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Probability, Random Variables, and Stochastic Processes

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Library of Congress Catalog Card Number 64-22956

PRINTED IN JAPAN

Preface

Several years ago I reached the conclusion that the theory of probability should no longer be treated as adjunct to statistics or noise or any other terminal topic, but should be included in the basic training of all engineers and physicists as a separate course. I made then a number of observations concerning the teaching of such a course, and it occurs to me that the following excerpts from my early notes might give you some insight into the factors that guided me in the planning of this book:

"Most students, brought up with a deterministic outlook of physics, find the subject unreliable, vague, difficult. The difficulties persist because of inadequate definition of the first principles, resulting in a constant confusion between assumptions and logical conclusions. Conceptual ambiguities can be removed only if the theory is developed axiomatically. They say that this approach would require measure theory, would reduce the subject to a branch of mathematics, would force the student to doubt his intuition leaving him without convincing alternatives, but I don't think so. I believe that most concepts needed in the applications can be explained with simple mathematics, that probability, like any other theory, should be viewed as a conceptual structure and its conclusions should rely not on intuition but on logic. The various concepts must, of course, be related to the physical world, but such motivating sections should be separated from the deductive part of the theory. Intuition will thus be strengthened, but not at the expense of logical rigor.

"There is an obvious lack of continuity between the elements of probability as presented in introductory courses, and the sophisticated concepts needed in today's applications. How can the average student, equipped only with the probability of cards and dice, understand prediction theory or harmonic analysis? The applied books give at most a brief discussion of background material; their objective is not the use of the applications to strengthen the student's understanding of basic concepts, but rather a detailed discussion of special topics.

"Random variables, transformations, expected values, conditional densities, characteristic functions cannot be mastered with mere exposure. These concepts must be clearly defined and must be developed, one at a time, with sufficient elaboration. Special topics should be used to illustrate the theory, but they must be so presented as to minimize peripheral, descriptive material and to concentrate on probabilistic content. Only then the student can learn a variety of applications with economy and perspective."

I realized that to teach a convincing course, a course that is not a mere presentation of results but a connected theory, I would have to reexamine not only the development of special topics, but also the proofs of many results and the method of introducing the first principles.

"The theory must be mathematical (deductive) in form but without the generality or rigor of mathematics. The philosophical meaning of probability must somehow be discussed. This is necessary to remove the mystery associated with probability and to convince the student of the need for an axiomatic approach and a clear distinction between assumptions and logical conclusions. The axiomatic foundation should not be a mere appendix but should be recognized throughout the theory.

"Random variables must be defined as functions with domain an abstract set of experimental outcomes and not as points on the real line. Only then infinitely dimensional spaces are avoided and the extension to stochastic processes is simplified.

"The inadequacy of averages as definitions and the value of an underlying space is most obvious in the treatment of stochastic processes. Time averages must be introduced as stochastic integrals, and their relationship to the statistical parameters of the process must be established only in the form of ergodicity.

"The emphasis on second-order moments and spectra, utilizing the student's familiarity with systems and transform techniques, is justified by the current needs.

"Mean-square estimation (prediction and filtering), a topic of considerable importance, needs a basic reexamination. It is best understood if it is divorced from the details of integral equations or the calculus of variations, and is presented as an application of the orthogonality principle (linear regression), simply explained in terms of random variables.

"To preserve conceptual order, one must sacrifice continuity of special topics, introducing them as illustrations of the general theory."

These ideas formed the framework of a course that I taught at the Polytechnic Institute of Brooklyn. Encouraged by the students' reaction, I decided to make it into a book. I should point out that I did not view my task as an impersonal presentation of a complete theory, but rather as an effort to explain the essence of this theory to a particular group of students. The book is written neither for the handbook-oriented students nor for the sophisticated few who can learn the subject from advanced mathematical texts. It is written for the majority of engineers and physicists who have sufficient maturity to appreciate and follow a logical presentation, but, because of their limited mathematical background, would find a book such as Doob's too difficult for a beginning text.

Although I have included many useful results, some of them new, my hope is that the book will be judged not for completeness but for organization and clarity. In this context I would like to anticipate a criticism and explain my approach. Some readers will find the proofs of many important theorems lacking in rigor. I emphasize that it was not out of negligence, but after considerable thought, that I decided to give, in several instances, only plausibility arguments. I realize too well that "a proof is a proof or it is not." However, a rigorous proof must be preceded by a clarification of the new idea and by a plausible explanation of its validity. I felt that, for the purposes of this book, the emphasis should be placed on explanation, facility, and economy. I hope that this approach will give you not only a working knowledge, but also an incentive for a deeper study of this fascinating subject.

Although I have tried to develop a personal point of view in practically every topic, I recognize that I owe much to other authors. In particular, the books "Stochastic Processes" by J. L. Doob and "Théorie des Functions Aléatoires" by A. Blanc-Lapierre and R. Fortet influenced greatly my planning of the chapters on stochastic processes.

Finally, it is my pleasant duty to express my sincere gratitude to Mischa Schwartz for his encouragement and valuable comments, to Ray Pickholtz for his many ideas and constructive suggestions, and to all my colleagues and students who guided my efforts and shared my enthusiasm in this challenging project.

Athanasios Papoulis

Contents

I

PROBABILITY AND RANDOM VARIABLES

1

The Meaning of Probability

Scientific theories deal with concepts, never with reality. All theoretical results are derived from certain axioms by deductive logic. In physical sciences the theories are so formulated as to correspond in some useful sense to the real world, whatever that may mean. However, this correspondence is approximate, and the physical justification of all theoretical conclusions is based on some form of inductive reasoning.

The student accepts readily this separation between the *conceptual world* (model) and the *physical world* for the so-called deterministic phenomena, but in probabilistic descriptions he confuses these two worlds. He has been taught that the universe evolves according to deterministic laws that specify exactly its future, and a probabilistic description is necessary only because of our ignorance. This deep-rooted skepticism in the validity of probabilistic results can be overcome only by a proper interpretation of the meaning of probability.

In the following discussion we attempt to show that probability is also a deductive science and must be developed axiomatically. It is of course true that the correspondence between analytical results and the real world is imprecise and cannot be "proved"; however, this is characteristic not only of probabilistic results but of all scientific conclusions.

1-1. Preliminary remarks

The theory of probability deals with averages of mass phenomena occurring sequentially or simultaneously: electron emission, telephone calls, radar detection, quality control, system failure, games of chance, statistical mechanics, turbulence, noise, birth and death rates, heredity.

It has been *observed* that in these and other fields certain averages approach a constant value as the number of observations increases. Furthermore, this limiting value remains the same if the averages are evaluated over any sub-sequence specified before the experiment is performed. Thus, in the coin-tossing game, the percentage of heads approaches 0.5 or some other constant, and the same average is obtained if one considers every fourth, say, tossing. (No betting system can beat the roulette.)

The purpose of the theory is to describe and predict such averages, and this is done by associating probabilities with various events. The probability $P(\mathcal{A})$ of an event $\mathcal{A}$ in a clearly specified experiment $\mathfrak{S}$ could be interpreted in the following sense:

> If the experiment is repeated n times and the event $\mathcal{A}$ occurs n_a times, then, with a *high degree of certainty*, the relative frequency n_a/n of the occurrence of $\mathcal{A}$ is *close* to $P(\mathcal{A})$,
>
> $$P(\mathcal{A}) \simeq \frac{n_a}{n} \tag{1-1}$$
>
> provided n is *sufficiently large.*

This interpretation is obviously imprecise; however, it cannot be essentially improved. One could modify it by giving, for example, a probabilistic content to the "high degree of certainty," but such modifications will only postpone the inevitable fact that probability, like any physical theory, is related to physical phenomena only in inexact terms. Nevertheless, the theory is an exact discipline developed logically from clearly defined axioms, and when it is applied to real problems, it *works.*

In any probabilistic investigation of a physical phenomenon one must distinguish three separate steps:

Step 1 (physical). We determine by a process that is not and cannot be made exact the probabilities $P(\mathcal{A})$ of certain events $\mathcal{A}$ (probabilistic data).

This step could be based on (1-1): $P(\mathcal{A})$ is equated to the experimentally determined ratio n_a/n (relative-frequency approach). For example, if a loaded die is rolled 1,000 times and five shows 203 times, then the probability of five equals about 0.2. In some cases, $P(\mathcal{A})$ is found a priori by pure reasoning without any experimentation (classical approach to be soon discussed). Given a "fair" die, one "reasons" that, because of its symmetry, the probability of five equals $\frac{1}{6}$.

Step 2 (conceptual). We assume that probabilities satisfy certain axioms, and by deductive reasoning we determine from the probabilities $P(\mathcal{A})$ of certain events $\mathcal{A}$ the probabilities $P(\mathcal{B})$ of other events $\mathcal{B}$.

For example, in the rolling of a fair die we deduce that the probability of the event "even" equals $\frac{3}{6}$. Our statement is of the following form:

$$\text{If} \qquad P(1) = P(2) = \cdots = P(6) = \tfrac{1}{6} \qquad \text{then} \qquad P(\text{even}) = \tfrac{3}{6}$$

Step 3 (physical). We make a physical *prediction* using the numbers $P(\mathcal{B})$ determined in step 2.

This step is again imprecise and could be based on (1-1). If, for example, we roll a fair die 1,000 times, we expect that an even number will show in about one-half of these rollings.

The theory of probability deals only with step 2; i.e., from certain assumed probabilities it tells us how to derive other probabilities. One might say that such derivations are mere tautologies because the results are contained in the assumptions. This is true in the same sense that the intricate equations of motion of a satellite are included in Newton's laws. But no one denies the value of the science of mechanics.

We could not emphasize too strongly the need for separating clearly the above three steps in the solution of a problem. The student must distinguish between the data that are either assumed or determined by inexact reasoning and the results that are obtained logically.

Steps 1 and 3 are the subject of investigation of statistics. However, even in statistics all results are given in terms of probabilities, with the difference that final experimental testing is applied to events whose probability is *almost* 1. In this case the relative-frequency interpretation takes the following form:

> If the probability of an event is *almost* 1, then, with a *high degree of certainty*, this event will occur in a single trial.

Even so, the difficulty of assigning probabilities to real events is not eliminated, and one might wonder how to proceed in a physical problem. To determine the probability of heads, should he toss a coin one hundred or one thousand times? Suppose that, after one thousand tossings, the average number of heads settled to the value 0.48. How can he make a *prediction* on the basis of this observation? Out of what logical necessity must he deduce that, at the next thousand tossings, the average will be about 0.48? This question can be answered only by some form of inductive reasoning. However, such reasoning is used, not only in probabilistic statements, but in all conclusions drawn from experience, even in the so-called deterministic sciences. Consider, for example, the development of classical mechanics. It was *observed* that bodies fell according to certain rules, and on the basis of this observation Newton's laws were formulated and used *successfully* to predict future events. If one wants to "prove" that the future will evolve in the predicted manner, he will have to invoke a metaphysical cause like "regularity in nature." The physicist bases his conclusions on inductive reasoning, and he is content that his predictions are correct. This point of view gives him also the flexibility to abandon any theory if subsequent evidence contradicts it. If he finds it useful to describe certain phenomena probabilistically, he does so without the need for a deterministic explanation.

To conclude, we repeat that the probability $P(\mathcal{A})$ of an event $\mathcal{A}$ must be interpreted as a number assigned to this event, as mass is assigned to a body. In the development of the theory one should not worry about the "physical meaning" of $P(\mathcal{A})$. This is what is done in all theories.

Consider, for example, circuit analysis. One assumes that a resistor is a two-terminal device whose voltage is proportional to the current

$$R = \frac{v(t)}{i(t)} \tag{1-2}$$

But a physical resistor does not obey (1-2). It is a complicated concept without obvious terminals, with distributed inductance and capacitance; it generates thermal noise; and only within certain errors in certain frequency ranges and with many other qualifications can a relationship of the form (1-2) be claimed. Nevertheless, in the development of the theory one ignores all these uncertainties. He assumes that a resistor has a value R satisfying (1-2), and with the help of certain laws, he develops circuit analysis. It would, indeed, be very confusing if at each stage of this development he were concerned with the *true* meaning of R.

Like circuit analysis, or electromagnetic theory, or any other scientific discipline, probability must be presented axiomatically. These theories would, of course, be of no value to physics unless they could help us solve real problems. We should be able to assign specific, if only approximate, values to real resistors or probabilities to certain events (step 1); we should also be able to give physical meaning to the quantities that were derived from the theory (step 3). This link between idealized concepts and the physical world is essential, but must be separated from the purely logical structure of each theory (step 2).

1-2. The various definitions of probability

We shall now introduce and analyze the various definitions of probability, and at the end of the chapter we shall comment on their role in our investigation.

There are essentially four ways of defining probability:

A. Axiomatic (measure)

B. Relative frequency (Von Mises)

C. A priori definition as a ratio of favorable to total number of alternatives (classical)

D. Probability as a measure of belief (inductive reasoning)

A. Axiomatic Definition

For the understanding of this important definition we shall need the following concepts:† In a given experiment we denote by $\mathcal{S}$ the certain event, i.e., the event that occurs in every trial. Given two events $\mathcal{A}$ and $\mathcal{B}$, we denote by $\mathcal{A} + \mathcal{B}$ the event that occurs when $\mathcal{A}$ or $\mathcal{B}$ or both occur. We say that $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive if the occurrence of one at a given trial excludes the occurrence of the other.

As an illustration of these simple concepts we mention the single-die experiment. The certain event $\mathcal{S}$ is the event that occurs whenever any one of the six faces shows. If $\mathcal{A}$ is the event "even" and $\mathcal{B}$ the event "less than 3," then $\mathcal{A} + \mathcal{B}$ is the event "1 or 2 or 4 or 6." If $\mathcal{A}$ is the event "even" and $\mathcal{B}$ the event "odd," then $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive.

In the axiomatic development of probability one proceeds as follows:

The probability of an event $\mathcal{A}$ is a number $P(\mathcal{A})$ assigned to this event. This number obeys the following three postulates but is otherwise unspecified:

I. $P(\mathcal{A})$ is positive:

$$P(\mathcal{A}) \geq 0 \tag{1-3}$$

II. The probability of the certain event equals 1:

$$P(\mathcal{S}) = 1 \tag{1-4}$$

III. If $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive, then

$$P(\mathcal{A} + \mathcal{B}) = P(\mathcal{A}) + P(\mathcal{B}) \tag{1-5}$$

The resulting theory is based on these postulates and on nothing else.‡ In our opinion, this is the clearest way to introduce probability,

† For details see Chap. 2.

‡ We need also the postulate (2-11) of infinite additivity.

even in introductory courses. It emphasizes the deductive character of the theory, removing all conceptual ambiguities; it provides a solid background for the understanding of the applications; and it secures at least a beginning for a deeper study of probability, eliminating the need to learn a new language.

It is true that most mathematical treatments (based on Kolmogoroff's classic work†) require knowledge of measure theory, and it is perhaps for this reason that the axiomatic approach is not used in applied courses. However, the belief that advanced mathematics is needed to understand the basic ideas is not justified. The elements of the theory can be developed using only simple set operations. And this is sufficient for most applications.

B. *Relative-frequency Definition*

The following definition is popular among engineers and physicists: The experiment under consideration is repeated n times. If the event $\mathcal{A}$ occurs $n_{\mathcal{A}}$ times, then its probability $P(\mathcal{A})$ is defined as the *limit* of the relative frequency $n_{\mathcal{A}}/n$ of the occurrence of $\mathcal{A}$:

$$P(\mathcal{A}) = \lim_{n\to\infty} \frac{n_{\mathcal{A}}}{n} \tag{1-6}$$

For example, if a coin, fair or not, is tossed n times and heads shows n_h times, then the probability of heads equals the limit of n_h/n.

A beginner's reaction to this definition might be very favorable (this author's was, a quarter of a century ago). Since probabilities are used to make statements about relative frequencies, it is natural to define them as the limit of these frequencies. The problems associated with a priori definitions are eliminated, and the theory is based, one would think, on solid experimental ground.

However, although the relative-frequency concept is essential in applying the theory to the physical world (steps 1 and 3), one can raise strong objections to its use as the basis of a deductive theory (step 2). Without elaboration we remark that in a physical experiment the number n might be large but is always finite; therefore $n_{\mathcal{A}}/n$ cannot be equated, even approximately, to a limit. Definition (1-6) is thus a *hypothesis* about the existence of this limit, and not an experimental quantity. The problem of relating the *assumed* limit $P(\mathcal{A})$ to the experimental ratio $n_{\mathcal{A}}/n$ is not eliminated.

Using the hypothesis that the limit of $n_{\mathcal{A}}/n$ exists, one could develop a deductive theory; however, once the axiomatic character of this theory

† A. Kolmogoroff, Grundbegriffe der Wahrscheinlichkeitsrechnung, *Ergeb. Mat. und ihrer Grenzg.*, vol. 2, no. 3, 1933.

is recognized, one is easily convinced that definition A is preferable. It then follows from the postulates that "almost certainly" $P(\mathcal{A})$ equals the limit of $n_{\mathcal{A}}/n$ (law of large numbers).

We shall venture a comparison with circuit theory. It is possible to define an ideal resistor of value R as a limit,

$$R = \lim_{n\to\infty} \frac{e(t)}{i_n(t)}$$

where $e(t)$ is a voltage source, and $i_n(t)$ are the resulting currents of a sequence of real resistors that tend in some sense to an ideal two-terminal element. This definition would show the relationship between the ideal element and the real resistors, but the resulting theory would be complicated.

The relative-frequency approach was developed by Von Mises about fifty years ago.† At that time the prevailing point of view was still the classical and his work was a healthy reaction to the a priori concept of probability. It removed from this concept its metaphysical character, and it demonstrated that it works only because it makes implicit use of relative frequencies masked in our long experience. In the author's opinion Von Mises' weakness is his claim that relative frequencies should form the basis of a deductive theory; that, in fact, such a theory can be based not on assumptions, but on measurements. Although the relative-frequency concept is of great value in determining probabilistic data and in giving physical content to computed probabilities, its value as the basis for an axiomatic theory is questionable. It became evident after Kolmogoroff's work that definition A is superior. This definition is not incompatible with Von Mises' approach, provided each is given its proper meaning.

C. *Classical Definition*

Up until the twentieth century the theory of probability was based on the classical definition. Although its importance as the basis for a deductive theory has diminished, it is still used to determine probabilistic data and as a working hypothesis. In the following discussion we shall attempt to evaluate this important definition.

According to this definition, the probability $P(\mathcal{A})$ of an event $\mathcal{A}$ is found a priori without actual experimentation. This is done by counting the total number N of the possible outcomes (alternatives) of the experiment $\mathfrak{E}$. If in $N_{\mathcal{A}}$ of these outcomes the event $\mathcal{A}$ occurs, then $P(\mathcal{A})$ is

† Richard Von Mises, "Wahrscheinlichkeit, Statistik und Wahrheit," Springer-Verlag OHG, Vienna, 1936.

given by

$$P(\mathcal{A}) = \frac{N_{\mathcal{A}}}{N} \tag{1-7}$$

Calling the above $N_{\mathcal{A}}$ outcomes "favorable to $\mathcal{A}$," we can phrase (1-7) as follows:

> $P(\mathcal{A})$ equals the ratio of the favorable to the total number of outcomes.

In the single-die experiment the possible outcomes are 1, 2, 3, 4, 5, 6. For the event "even," the favorable outcomes are 2, 4, 6; hence $P(\text{even}) = 3/6$.

We remark that the numbers in (1-7) are radically different from the numbers in (1-1). n and $n_{\mathcal{A}}$ are obtained by performing our experiment n times, and their meaning is unambiguous; N and $N_{\mathcal{A}}$, on the other hand, are determined without experimentation, and their significance is not always clear. To show this ambiguity and the need for improving the definition, we shall consider the following problem: We want the probability p that the sum of the appearing numbers when two dice are rolled equals seven. We could count as outcomes the 11 possible sums. Of these only 1, namely, the sum seven, is favorable to our event; hence $p = 1/11$. This is obviously wrong. We could count as outcomes all possible pairs of numbers, not distinguishing between the first and second die; our outcomes are now 21. Of these, only 3, namely, (3,4), (5,2) and (6,1) are favorable; hence $p = 3/21$. This is also wrong. We now "reason" that the above solutions are wrong because the possible outcomes are not *equally likely*. Thus, in the second solution, the outcome (1,1) is less likely than (1,2). To solve the problem correctly, we must count all pairs, distinguishing between the first and second die. We then have 36 possibilities, of which 6 are favorable to our event, namely, the pairs (3,4), (4,3), (5,2), (2,5), (6,1), (1,6); hence $p = 6/36$.

> We thus arrive at the improved version of the classical definition: The probability of an event $\mathcal{A}$ equals the ratio of the favorable to the total number of outcomes, provided they are *equally likely*.

Criticism of the classical definition. The equally likely condition does not eliminate the difficulties associated with the above notion. We shall comment below on the principal objections to this theory:

1. The definition is circular. Indeed, after some thought, one concludes that *equally likely* means *equally probable*. Thus, in the definition, use is made of the concept to be defined. This objection is not just academic. Because of the difficulty in deciding what are equally likely alternatives, serious problems often arise in determining N and $N_{\mathcal{A}}$.

2. It can be used only for a limited class of problems. For example, in the die experiment, it is applicable only if the six faces have the same probability. If the die is not fair and the probability of five equals $\frac{1}{4}$, say, there is no direct way of deriving this number from (1-7).

3. The classical definition, although presented as a priori logical necessity, makes implicit use of the relative-frequency interpretation of probability. The probability of five in a fair die equals $\frac{1}{6}$, not only because of its symmetry, but also because it was observed in the long history of rolling dice that n_a/n is close to $\frac{1}{6}$. The next illustration shows more clearly the need for experimental evidence. We want the probability p that a newborn baby is a boy. It is generally assumed that $p = \frac{1}{2}$; however, one cannot reason that there are "obviously" only two equally likely alternatives. In the first place, it is only approximately true that $p = \frac{1}{2}$. Furthermore, without access to records we should not know that the two alternatives are equally likely regardless of the sex history of the baby's family, the season or place of its birth, and many other conceivable factors. It is only after extensive accumulation of statistics that all these factors become irrelevant and the two alternatives are accepted as equally likely.

4. In many problems the possible number of outcomes is infinite. To determine probabilities of various events from (1-7) one must introduce some measure of infinity like length or area. In the next classic example we shall show the resulting difficulties.

Example 1-1 (*Bertrand Paradox*). In a circle C of radius r we draw at "random" a cord AB. What is the probability that the length l of this cord is greater than the length $r\sqrt{3}$ of the side of an inscribed equilateral triangle?

First Solution. The center M of the cord AB is a "random" point. If it lies inside the circle C_1 of radius $r/2$ (Fig. 1-1a), then $l > r\sqrt{3}$. It is reasonable, therefore, to say that the unknown probability p equals the ratio of the area of the circle

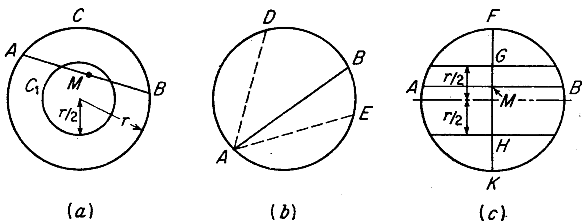

Fig. 1-1

C_1 (favorable outcomes) to the area of the circle C (all possible outcomes). Thus

$$p = \frac{\pi r^2/4}{\pi r^2} = \frac{1}{4}$$

Second Solution. Since the end A of the cord AB is a "random" point, we can assume that it is fixed. The possible outcomes are thus reduced, but the favorable ones are also reduced proportionately and the ratio remains the same. We shall have $l > r\sqrt{3}$ if B lies on the arc DBE (Fig. 1-1*b*) whose length equals $2\pi r/3$. The probability p is thus equal to the ratio of the length $2\pi r/3$ of DBE (favorable outcomes) to the length $2\pi r$ of the circle (all possible outcomes):

$$p = \frac{2\pi r/3}{2\pi r} = \frac{1}{3}$$

Third Solution. Since the direction of AB is "random," we can assume it fixed, perpendicular to the diameter FK shown in Fig. 1-1*c*. Again favorable and possible outcomes are reduced proportionately. We shall have $l > r\sqrt{3}$ if the center M of AB lies between G and H; therefore p equals the ratio of the length r of GH (favorable outcomes) to the length $2r$ of FK (all possible outcomes). Thus

$$p = \frac{r}{2r} = \frac{1}{2}$$

One might remark that the above solutions correspond to three different experiments. This is true; in the second solution the experiment was the random rotation of a straight line about a fixed point A, whereas in the third we had a parallel motion. In any case, the example shows the difficulties associated with the classical definition. We also see that it is meaningless to talk about the probability of an event; the underlying experiment must also be clearly specified.

Use of the classical definition. The a priori definition is a valuable tool in the solution of physical problems (step 1). We discuss below its various uses:

1. In many applications, the assumption that there are N equally likely alternatives is well established through long experience; one can then apply (1-7) without questioning its validity. Thus "the probability of selecting a white ball at random from a pile of m black and n white balls equals $n/(m + n)$" or "if a particle is placed at random in one of n boxes, the probability that it will be in the kth box equals $1/n$." This reasoning is also used for infinitely many alternatives. "If a call occurs at random in the $(0,T)$ interval, the probability that it will occur in the (t_1,t_2) interval equals $(t_2 - t_1)/T$."

Such applications are correct and valuable; however, the burden of justifying their correctness is placed on the validity of the term *random.* It is clear from the last example that one does not *prove* that "the probability of a call in the (t_1,t_2) interval equals $(t_2 - t_1)/T$" from the assumed *randomness* of the call; the two statements are merely equivalent. The assumed randomness can be approximately established only from records of many calls and does not follow from a priori reasoning.

2. In many problems it is impractical to determine probabilistic data $P(\mathcal{A})$ by repeating the experiment by which they are defined a sufficiently large number of times. In such cases one has no choice but to assume the data. This is often done by postulating that certain alternatives are

equally likely; $P(\mathcal{A})$ is then found from (1-7). In other words, one uses the classical definition as a *working hypothesis* and derives certain conclusions that can be tested experimentally. If these conclusions agree with the experiment, the hypothesis is accepted; if not, it is rejected. We shall illustrate this with an important application from statistical mechanics.

Example 1-2. We place at random n particles in m boxes. Assuming $m > n$, we want to find the probability p that the particles will be found in n preselected boxes, one in each box.

With $n = 2$ and $m = 6$, if we identify the two particles with two dice, the six boxes with the six face, and the $n = 2$ preselected boxes with 3 and 4, say, we see that p equals the probability of rolling three-four. Since we are interested here only in the underlying assumptions and not in the details of the solution, we shall merely state the results (for the proof see Prob. 3-8).

First Solution (Maxwell-Boltzmann). We consider as equally likely alternatives all possible ways of inserting n particles in m boxes, distinguishing the identity of each particle. It can be shown that the unknown probability is given by

$$p = \frac{n!}{m^n}$$

In the case of the two dice, $p = 2!/6^2$. This can be easily verified; the possible outcomes are 36, namely,

$$(1,1),\ (1,2),\ (2,1),\ \ldots,\ (5,6),\ (6,5),\ (6,6)$$

Of these only (3,4) and (4,3) are favorable; hence $p = 2/36$.

Second Solution (Bose-Einstein). We now assume that the particles are not distinguishable from each other. In this case all permutations of the particles count as one outcome as long as the number of particles in each box remains the same. With this assumption we obtain

$$p = \frac{(m - 1)!n!}{(n + m - 1)!}$$

In the case of the two dice, $p = 5!2!/7! = 1/21$. Indeed, now (i,j) and (j,i) count as one outcome, so that the total number of outcomes is 21. Of these, only (3,4) is favorable; hence $p = 1/21$.

Third Solution (Fermi-Dirac). As in the second solution, we do not distinguish among the particles; however, we now assume that, in each box, at most one particle can be placed. In this case

$$p = \frac{n!(m - n)!}{m!}$$

For the dice, $p = 2!4!/6! = 1/15$. Indeed, now we must exclude all rollings that show equal faces; i.e., (1,1), (2,2), . . . , (6,6) are assumed as impossible. The possible outcomes are thus 15 and $p = 1/15$.

Reasoning a priori, one might argue that from these solutions only the first is "correct." The fact is that in the absence of experimental evidence it makes no sense to reach such a conclusion. The physicist merely derives from the above *hypotheses* measurable conclusions and accepts the one that agrees with the experimental results. Actually, in the formulation of the proper hypothesis, he was guided by the available evidence. It required great scientific insight to unify the diverse evidence into a simple hypothesis. This was not done by merely counting obviously equally likely alternatives.

3. Sometimes one uses a priori reasoning, not to evaluate probabilities from (1-7), but to establish the *independence* of certain events (see also Sec. 3-1). Suppose that the probability of an event $\mathcal{A}_1$ in an experiment $\mathfrak{E}_1$ equals $P(\mathcal{A}_1)$ and the probability of an event $\mathcal{A}_2$ in an experiment $\mathfrak{E}_2$ equals $P(\mathcal{A}_2)$. We now construct a new experiment $\mathfrak{E}$ consisting of the performance of both experiments $\mathfrak{E}_1$ and $\mathfrak{E}_2$, and in this experiment we want the probability $P(\mathcal{A}_1 \text{ and } \mathcal{A}_2)$ that both events $\mathcal{A}_1$ and $\mathcal{A}_2$ will occur. In general, this probability cannot be found from mere knowledge of $P(\mathcal{A}_1)$ and $P(\mathcal{A}_2)$. If, however, we assume that the two experiments $\mathfrak{E}_1$ and $\mathfrak{E}_2$ are independent, then

$$P(\mathcal{A}_1 \text{ and } \mathcal{A}_2) = P(\mathcal{A}_1)P(\mathcal{A}_2) \tag{1-8}$$

In many applications this independence is established a priori by reasoning that the outcomes of $\mathfrak{E}_1$ do not influence the outcomes of $\mathfrak{E}_2$; (1-8) is thus justified without recourse to (1-1) or any other experimentation.

4. We finally remark that the classical definition can be used as basis of a deductive theory if one *assumes* that there are N equally likely alternatives and that the probability $P(\mathcal{A})$ of an event $\mathcal{A}$ is given by the ratio (1-7). In this theory, postulates (1-3) to (1-5) become theorems. Indeed, N and $N_{\mathcal{A}}$ are positive numbers; hence $P(\mathcal{A}) \geq 0$. Furthermore, $N_{\mathcal{S}} = N$ because all alternatives are favorable to the certain event; hence $P(\mathcal{S}) = 1$. Finally, if $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive, then $N_{\mathcal{A}+\mathcal{B}} = N_{\mathcal{A}} + N_{\mathcal{B}}$ because no alternatives can be favorable to $\mathcal{A}$ and $\mathcal{B}$; therefore

$$P(\mathcal{A} + \mathcal{B}) = \frac{N_{\mathcal{A}+\mathcal{B}}}{N} = \frac{N_{\mathcal{A}}}{N} + \frac{N_{\mathcal{B}}}{N} = P(\mathcal{A}) + P(\mathcal{B})$$

However, as we show on page 31, the above is nothing more than a very special case of definition A. One might dispute this since we have not used the axioms at all. The fact is that by accepting (1-7) as the definition of $P(\mathcal{A})$, we have implicitly assumed that the axioms are valid.

D. *Probability as a Measure of Belief*

Probability is often used as a measure of belief that something might or might not be true; for example, "it is probable that X is guilty." This interpretation of probability is different from the theory of probability as used in physical sciences. The reason is that it is a statement about a single event, and not about averages of mass phenomena. Furthermore, it is subjective. Indeed, X is either guilty or not guilty; the probability that we associate with his guilt is merely a measure of our own knowledge and reasoning ability.

This concept of probability is a form of *inductive reasoning.* It is important, not only in everyday applications, but also in the early stages of a scientific investigation leading eventually to a deductive theory. In fact, even in mathematics, the proofs of many theorems and the solutions of most problems are based on exploratory, inductive reasoning that in no way resembles the economical, deductive proofs given in mathematics texts.

From our point of view, this form of probability has no place in the development of an exact theory (step 2). In obtaining probabilistic data (step 1), however, one uses such reasoning as is sometimes done in statistics (Bayes' theory). In such cases, one reasons that subjective knowledge is better than no knowledge at all; but this is done only if the values $P(\mathcal{A})$ of the estimated probabilities are not critical in the determination of the final results. In fact, one can show that, as experimental evidence accumulates, the influence of the a priori numbers $P(\mathcal{A})$ on the computed results (a posteriori probabilities) becomes negligible (see Sec. 3-5).

1-3. Determinism versus probability

As we have already pointed out, many students are skeptical about the physical validity of a probabilistic law. They are used to the idea that a physical law describes the deterministic evolution of nature and a probabilistic interpretation is necessary only because of our ignorance. The controversy of determinism and causality versus randomness and probability has been the topic of extensive discussions. In our opinion, the difference lies not in the nature of this or that phenomenon, but in the quantities in which the observer is interested. If he is interested in the outcome of *one* experiment, then his statement is deterministic; if he is interested in certain averages of a large number n of experiments, then his statement is probabilistic. In either case no categorical assertion is possible. In the first case, the uncertainty of his conclusions takes the form *within certain errors and in certain ranges of the relevant parameters;* in the second case, *with a high degree of certainty if n is large enough.* We shall illustrate this dual interpretation with a simple example.

Example 1-3. A particle leaves the origin 0 under the influence of the force of gravity. Its initial velocity v forms an angle θ with the horizontal axis $0x$, and its path crosses $0x$ at the point B, as in Fig. 1-2. We want the distance $d \doteq 0B$.

From Newton's law one easily finds that

$$d = \frac{v^2}{g} \sin 2\theta \tag{1-9}$$

It seems at first that (1-9) is a perfect deterministic result of a causal law and that a probabilistic interpretation has no place here. But is it really so? After a closer examination we conclude that the above solution corresponds to an idealized problem in which we have neglected air friction, variation of g, inexactness in the initial position and velocity of the particle, and other factors. Let us ignore all unknown parameters except the uncertainty in the value of the angle θ. Clearly, since θ cannot be determined exactly, (1-9) should be given the following interpretation: *if the initial angle equals θ within an error δ, then $d = v^2 \sin 2\theta/g$ within a constant ε.*

Suppose now that we place little holes along the $0x$ axis and we want to find the reentry hole. If v is large, then, because of the uncertainty in the value of θ, we are in no position to make a deterministic statement about this hole. We can, however, ask a different question: if many particles with the same initial velocity (within

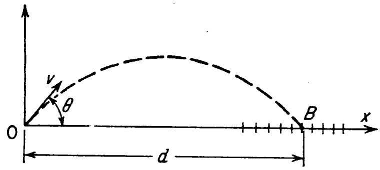

Fig. 1-2

certain errors) are ejected from 0, what percentage will fall in the nth hole? This question has no longer a causal answer. Thus the *same* physical phenomenon can be subjected to either a deterministic or a probabilistic analysis.

The reader might object to the above. He might argue that angle θ has *obviously a definite value*, even though we do not know it. If we knew θ, we should be in a position to predict *exactly* the reentry hole. The phenomenon is thus inherently deterministic, and probabilistic considerations are necessary only because of our ignorance. Our answer is that the physicist is not concerned with what *is* true, but only with what he can measure. Such explanations are therefore outside the sphere of his scientific interests.

A final comment: One sometimes associates probabilistic phenomena with discontinuities between cause and effect. A slight variation in the angle of tossing might change the outcome from heads to tails. This is, perhaps, best demonstrated by the familiar experiment of a ball falling through a pinboard. However, such discontinuities depend again on what we consider as effect in a given experiment. Suppose, for example, that a computer converts an analog input to a digital output and it records this output up to the tenth decimal place. If we can measure the input only with a 10^{-5} accuracy and we consider as effect the last decimal of the output, we shall observe a discontinuity between cause and effect. In fact, so far as we can tell, identical inputs (causes) will result in distinctly different outputs.

Concluding Remarks

In this book we shall present a deductive theory (step 2) based entirely on the axiomatic definition A. The other definitions, properly interpreted, could also be used; however, the classical is too limited and the relative frequency too involved.

In assigning probabilistic data (step 1) to various events in a specific problem we shall often make use of the classical definition.

Finally, as a motivation and a reminder to the reader of the connection between the conceptual and the physical world, we shall give a "relative-frequency interpretation" of various axioms, definitions, theorems, and their physical counterparts, based on (1-1). We emphasize that this portion of the book (in small indented print) does not obey the rules of deductive reasoning on which the theory is based.

2

The Axioms of Probability

A central concept in the theory of probability is the *event*. Events are combined in various ways to form other events. These operations are best described by what is known as *set*† *theory*. In this chapter we introduce the elements of this theory and use it to develop, simply but carefully, the foundations of probability theory.

2-1. Set theory

A set is a collection of objects. Examples of sets: "heads and tails"; the numbers "1, 2, 3, 4"; "a car, an apple, and a pencil." The objects of the set we shall call its *elements*. Thus the set of the third example consists of the three elements "car," "apple," "pencil." A *subset* $\mathcal{B}$ of a set $\mathcal{A}$ is another set whose elements are also elements of $\mathcal{A}$ (belong to $\mathcal{A}$). In the following discussion we shall consider not arbitrary sets, but only sets whose elements are taken from a largest set $\mathcal{S}$, which we shall call *space* (sometimes called also *universe*). In other words, all our sets will be subsets of the space $\mathcal{S}$.‡ For example, in the

† N. J. Fine, "Fundamentals of College Mathematics," prelim. ed., The Ronald Press Company, New York, 1962.

‡ With the exception of the *complement*, all operations of this section can be defined for arbitrary sets that are not necessarily subsets of a given space.

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rolling of a die one can consider as space the set consisting of its six faces $f_1, f_2, \ldots, f_6$. In the manufacture of resistors we can take as space all possible values of R.

We shall use script letters $\mathcal{A}, \mathcal{B}, \mathcal{C}, \ldots$ for all our sets. The elements will, in general, be identified by the Greek letter ζ; or sometimes they will be specified by words, e.g., "heads" or "tails," or by suitable symbols h or t. The elements that are contained in a specific set $\mathcal{A}$ can either be listed explicitly or can be described by their properties. This will be done by placing the elements or the describing property in braces. llaves Thus

$$\mathcal{A} = \{\zeta_1, \zeta_2, \ldots, \zeta_n\} \tag{2-1}$$

will mean that the set $\mathcal{A}$ consists of the elements $\zeta_1, \ldots, \zeta_n$. Clearly, Claramente: the order in which the elements are written is immaterial.

$$\mathcal{B} = \{\text{all positive integers}\} \tag{2-2}$$

will mean the set consisting of the numbers 1, 2, 3, ... Similarly, if $\mathcal{S}$ is the set of positive numbers, then

$$\mathcal{A} = \{\zeta^2 - 1 = 0\}$$

will mean the set† $\mathcal{A} = \{1\}$, because the only positive number satisfying the equation $\zeta^2 - 1 = 0$ is the number 1.

If ζ_i is an element of $\mathcal{A}$, this will be denoted by

$$\zeta_i \in \mathcal{A}$$

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We shall also say that ζ_i *belongs* to $\mathcal{A}$. The notation

$$\zeta_i \notin \mathcal{A}$$

will mean that ζ_i is not an element of $\mathcal{A}$.

A set might contain only one element,

$$\mathcal{A} = \{\zeta_1\}$$

entre

In this case one should be careful to distinguish between the element ζ_1 and the set $\{\zeta_1\}$. This might seem artificial, but we have a good reason to do so. Later on, for example, we shall assign numbers (probabilities) to various sets, and not to elements, and we shall define functions (random variables) only on the elements.

† The above is sometimes written in the form

$$\mathcal{A} = \{\zeta | \zeta^2 - 1 = 0\}$$

that is, "$\mathcal{A}$ consists of all positive integers ζ such that $\zeta^2 - 1 = 0$." In the notation (2-2) one must specify the underlying space. Thus, if the space consists of all real numbers, then

$$\mathcal{A} = \{\zeta^2 - 1 = 0\} = \{+1, -1\}$$

We shall also find it convenient to talk about the set that contains no elements. This set will be called *null set* or *empty set* and will be denoted† by 0. We do so because we do not want to exclude the possibility that a set specified by certain properties might contain no elements. For example, the set {all readers of this book who live in France} might very well be empty. Another reason is that, in performing certain operations on sets, we might end up with a set without elements.

Example 2-1. Our space consists of the six faces of a die:

$$\mathcal{S} = \{f_1,f_2,f_3,f_4,f_5,f_6\}$$

It is easy to see that $\mathcal{S}$ has $2^6 = 64$ subsets, namely,

$$0,\ \{f_1\},\ \ldots,\ \{f_6\},\ \{f_1,f_2\},\ \ldots,\ \{f_5,f_6\},\ \{f_1,f_2,f_3\},\ \ldots,\ \mathcal{S}$$

If the space $\mathcal{S}$ consists of a finite number n of elements, then the total number of its subsets is 2^n. We leave the proof as an exercise.

Example 2-2. Our space $\mathcal{S}$ is the set of points in the square of Fig. 2-1. It can also be considered as the collection of all ordered pairs of numbers (x,y), where

$$0 \le x \le T \qquad 0 \le y \le T$$

Subsets of $\mathcal{S}$ are all plane sets included in the above square. The shaded area is a subset $\mathcal{A}$ consisting of all points (x,y) such that $-b \le x - y \le a$. Clearly, we cannot express $\mathcal{A}$ in the form (2-1), but only by the property of its elements:

$$\mathcal{A} = \{-b \le x - y \le a\}$$

The above is an abbreviation of "$\mathcal{A}$ consists of all pairs (x,y) such that $0 \le x \le T$, $0 \le y \le T$, and $-b \le x - y \le a$."

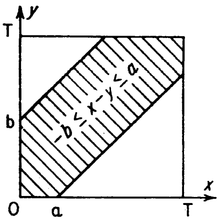

Fig. 2-1

An element of a set is any kind of object. It might even be a collection (set) of other objects. We can thus talk about a set (or better, a *class* or a *family*) of sets. In Example 2-1 one could consider sets whose elements are some of the 64 listed sets. Such abstractions will be rarely used here; however, we shall often introduce elements each of which consists of more than one object. This is done in the next example.

Example 2-3. A coin is tossed twice. A particular sequence of outcomes can be considered as an object. Thus "heads-heads" is the element (one element) *hh*. There are four such elements forming the space

$$\mathcal{S} = \{hh,ht,th,tt\}$$

With these elements we can form $2^4 = 16$ sets. We list below a number of these, where the first representation is as in (2-2) and the second as in (2-1):

$\mathcal{A}$ = {heads at the first tossing} = $\{hh,ht\}$
$\mathcal{B}$ = {only one head came up} = $\{ht,th\}$
$\mathcal{C}$ = {heads came up at least once} = $\{hh,ht,th\}$

† In many texts the empty set is denoted by ∅.

Set Operations

We have already said that a set $\mathcal{B}$ is a subset of $\mathcal{A}$ (is included in $\mathcal{A}$) if every element of $\mathcal{B}$ is also an element of $\mathcal{A}$. This is written in the form

$$\mathcal{B} \subset \mathcal{A} \qquad \text{or} \qquad \mathcal{A} \supset \mathcal{B}$$

In the following discussion we shall find it convenient to represent various sets by closed plane figures (Venn diagrams), and their elements by the enclosed points. The space $\mathcal{S}$ will be represented by a square. Many relationships involving sets will be "proved" with the help of appropriate Venn diagrams. Although geometric considerations are not formal proofs, they can guide us in developing such proofs.

In Fig. 2-2 we have shown three sets, $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, where $\mathcal{B} \subset \mathcal{A}$, $\mathcal{C} \subset \mathcal{B}$. It is clear that $\mathcal{C}$ is a subset of $\mathcal{A}$. Thus (transitivity)

$$\text{If} \quad \mathcal{C} \subset \mathcal{B} \quad \text{and} \quad \mathcal{B} \subset \mathcal{A} \quad \text{then} \quad \mathcal{C} \subset \mathcal{A}$$

It is easy to see that

$$\mathcal{A} \subset \mathcal{A} \qquad 0 \subset \mathcal{A} \qquad \mathcal{A} \subset \mathcal{S}$$

for any set $\mathcal{A}$.

$\mathcal{C} \subset \mathcal{B} \subset \mathcal{A}$

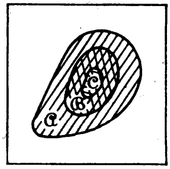

Fig. 2-2

Equality. We say that the set $\mathcal{A}$ equals the set $\mathcal{B}$ *if and only if* (abbreviation: iff) every element of $\mathcal{A}$ is an element of $\mathcal{B}$ and every element of $\mathcal{B}$ is an element of $\mathcal{A}$:

$$\mathcal{A} = \mathcal{B} \qquad \text{iff} \qquad \mathcal{A} \subset \mathcal{B} \qquad \text{and} \qquad \mathcal{B} \subset \mathcal{A}$$

The above definition might appear clumsy; however, frequently it is the most direct (and primitive) way of establishing equality of sets.

Sums. The sum or union

$$\mathcal{A} + \mathcal{B}$$

of two sets is a set whose elements are all the elements of $\mathcal{A}$ or of $\mathcal{B}$ or of both (Fig. 2-3).

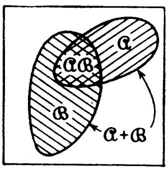

Fig. 2-3

In Example 2-3 the set $\mathcal{C}$ is the sum of $\mathcal{A}$ and $\mathcal{B}$. It is easy to see that (associative law)

$$(\mathcal{A} + \mathcal{B}) + \mathcal{C} = \mathcal{A} + (\mathcal{B} + \mathcal{C})$$

Hence the above sum can be written without parentheses

$$\mathcal{A} + \mathcal{B} + \mathcal{C}$$

Clearly, $\mathcal{A} + \mathcal{B} = \mathcal{B} + \mathcal{A}$ (commutative law), and

$$\mathcal{A} + \mathcal{A} = \mathcal{A} \qquad \mathcal{A} + 0 = \mathcal{A} \qquad \mathcal{A} + \mathcal{S} = \mathcal{S}$$

and if $\mathcal{B} \subset \mathcal{A}$, then $\mathcal{A} + \mathcal{B} = \mathcal{A}$.

Products. The product† or intersection

$$\mathcal{A}\mathcal{B}$$

of two sets is a set consisting of all elements that are common to set $\mathcal{A}$ and set $\mathcal{B}$ (Fig. 2-3).

Again,

$$(\mathcal{A}\mathcal{B})\mathcal{C} = \mathcal{A}(\mathcal{B}\mathcal{C}) = \mathcal{A}\mathcal{B}\mathcal{C}$$

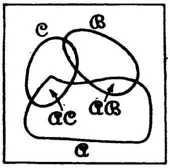

Fig. 2-4

Clearly, $\mathcal{A}\mathcal{B} = \mathcal{B}\mathcal{A}$, and

$$\mathcal{A}\mathcal{A} = \mathcal{A} \qquad \mathcal{A}0 = 0 \qquad \mathcal{A}\mathcal{S} = \mathcal{A}$$

and if $\mathcal{B} \subset \mathcal{A}$, then $\mathcal{A}\mathcal{B} = \mathcal{B}$.

It is easy to see (Fig. 2-4) that (distributive law)

$$\mathcal{A}(\mathcal{B} + \mathcal{C}) = (\mathcal{A}\mathcal{B}) + (\mathcal{A}\mathcal{C})$$

If the sets $\mathcal{A}$ and $\mathcal{B}$ are described by the properties of their elements as in (2-2), then their product $\mathcal{A}\mathcal{B}$ will often be specified by including these properties in braces. For example, if

$$\mathcal{S} = \{1,2,3,4,5,6\}$$
$$\mathcal{A} = \{\text{even}\} = \{2,4,6\} \qquad \mathcal{B} = \{\text{less than } 5\} = \{1,2,3,4\}$$

then $\mathcal{A}\mathcal{B} = \{2,4\}$. Thus

$$\mathcal{A}\mathcal{B} = \{\text{even}\}\{\text{less than } 5\} = \{\text{even, less than } 5\}$$

Two sets $\mathcal{A}$ and $\mathcal{B}$ are called *mutually exclusive* (or disjoint) if they have no common elements, i.e., if

$$\mathcal{A}\mathcal{B} = 0$$

The sets

$$\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3, \ldots$$

are called mutually exclusive if any two of them have no common elements, i.e., if

$$\mathcal{A}_i\mathcal{A}_j = 0 \qquad \text{for every } i, j \qquad i \neq j$$

† In mathematics one uses mostly the expressions "union" and "intersection" instead of "sum" and "product," and these operations are written in the form

$$\mathcal{A} \cup \mathcal{B} \qquad \text{and} \qquad \mathcal{A} \cap \mathcal{B}$$

We have adopted $\mathcal{A} + \mathcal{B}$ and $\mathcal{A}\mathcal{B}$ to conform with the convention in applied books and to avoid new symbolism.

Complement. The complement $\bar{\mathcal{A}}$ of a set $\mathcal{A}$ is defined as the set consisting of all elements of $\mathcal{S}$ that are not in $\mathcal{A}$ (Fig. 2-5).

Thus, with $\mathcal{C} = \{hh,ht,th\}$, the set of Example 2-3, its complement $\bar{\mathcal{C}}$ is the set $\{tt\}$. Clearly,

$$\bar{0} = \mathcal{S} \qquad \bar{\mathcal{S}} = 0 \qquad \bar{\bar{\mathcal{A}}} = \mathcal{A} \qquad \mathcal{A} + \bar{\mathcal{A}} = \mathcal{S} \qquad \mathcal{A}\bar{\mathcal{A}} = 0$$

and if $\mathcal{B} \subset \mathcal{A}$, then $\bar{\mathcal{B}} \supset \bar{\mathcal{A}}$. Finally, if $\mathcal{A} = \mathcal{B}$, then $\bar{\mathcal{A}} = \bar{\mathcal{B}}$.

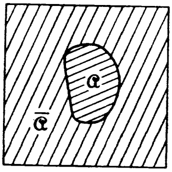

Fig. 2-5

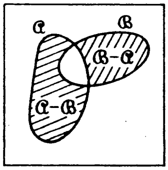

Fig. 2-6

Difference. The difference $\mathcal{A} - \mathcal{B}$ is a set consisting of the elements of $\mathcal{A}$ that are not in $\mathcal{B}$ (Fig. 2-6).

Thus

$$\mathcal{A} - \mathcal{B} = \mathcal{A}\bar{\mathcal{B}} = \mathcal{A} - \mathcal{A}\mathcal{B}$$

Notice that $(\mathcal{A} - \mathcal{B}) + \mathcal{B}$ is not equal to $\mathcal{A}$. We also observe that

$$(\mathcal{A} + \mathcal{A}) - \mathcal{A} = 0 \qquad \text{but} \qquad \mathcal{A} + (\mathcal{A} - \mathcal{A}) = \mathcal{A}$$

so that, although in sums and products one can omit the parentheses, this is not the case if a difference is involved. Finally,

$$\mathcal{A} - 0 = \mathcal{A} \qquad \mathcal{A} - \mathcal{S} = 0 \qquad \mathcal{S} - \mathcal{A} = \bar{\mathcal{A}}$$

for any set $\mathcal{A}$.

Example 2-4. With

$$\mathcal{A} = \{1,2,3\} \qquad \mathcal{B} = \{2,3,4\}$$

we have

$$\mathcal{A} + \mathcal{B} = \{1,2,3,4\} \qquad \mathcal{A}\mathcal{B} = \{2,3\} \qquad \mathcal{A} - \mathcal{B} = \{1\}$$

and if

$$\mathcal{S} = \{1,2,3,4,5\}$$

then

$$\bar{\mathcal{A}} = \{4,5\} \qquad \bar{\mathcal{B}} = \{1,5\}$$

De Morgan law. It is easy to see that (Fig. 2-7)

$$\overline{(\mathcal{A} + \mathcal{B})} = \bar{\mathcal{A}}\bar{\mathcal{B}} \qquad \overline{(\mathcal{A}\mathcal{B})} = \bar{\mathcal{A}} + \bar{\mathcal{B}}$$

Combining the above operations, one can show that:

> If in an identity we replace sums by products, products by sums, and sets by their complements, then the identity is preserved.

Thus from

$$\mathcal{A}(\mathcal{B} + \mathcal{C}) = (\mathcal{A}\mathcal{B}) + (\mathcal{A}\mathcal{C}) \quad \text{follows that} \quad \bar{\mathcal{A}} + (\bar{\mathcal{B}}\bar{\mathcal{C}}) = (\bar{\mathcal{A}} + \bar{\mathcal{B}})(\bar{\mathcal{A}} + \bar{\mathcal{C}})$$

Indeed, complementing the first identity, we have

$$\overline{\mathcal{A}(\mathcal{B} + \mathcal{C})} = \overline{(\mathcal{A}\mathcal{B}) + (\mathcal{A}\mathcal{C})}$$

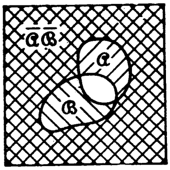

Fig. 2-7

But

$$\overline{\mathcal{A}(\mathcal{B} + \mathcal{C})} = \bar{\mathcal{A}} + \overline{(\mathcal{B} + \mathcal{C})} = \bar{\mathcal{A}} + \bar{\mathcal{B}}\bar{\mathcal{C}}$$

Furthermore,

$$\overline{(\mathcal{A}\mathcal{B}) + \mathcal{A}\mathcal{C})} = \overline{\mathcal{A}\mathcal{B}}\;\overline{\mathcal{A}\mathcal{C}} = (\bar{\mathcal{A}} + \bar{\mathcal{B}})(\bar{\mathcal{A}} + \bar{\mathcal{C}})$$

and our conclusion follows.

We finally remark that if $\mathcal{A} \subset \mathcal{B}$, then $\bar{\mathcal{A}} \supset \bar{\mathcal{B}}$. Hence De Morgan's law holds also for inclusions, provided $\subset$ is replaced by $\supset$.

Duality Principle. Since $\bar{\mathcal{S}} = 0$, $\bar{0} = \mathcal{S}$, and the complement $\bar{\mathcal{A}}$ of an arbitrary set $\mathcal{A}$ is also a set, we conclude from the above law that:

> If in an identity we replace sums by products, products by sums, $\mathcal{S}$ by 0, and 0 by $\mathcal{S}$, then the identity is preserved.

Thus from

$$\mathcal{A}(\mathcal{B} + \mathcal{C}) = (\mathcal{A}\mathcal{B}) + (\mathcal{A}\mathcal{C}) \quad \text{follows that} \quad \mathcal{A} + (\mathcal{B}\mathcal{C}) = (\mathcal{A} + \mathcal{B})(\mathcal{A} + \mathcal{C})$$

and from

$$\mathcal{S} + \mathcal{A} = \mathcal{S} \qquad \text{follows that} \qquad 0\mathcal{A} = 0$$

2-2. Probability space

We shall now identify the elements ζ of a space $\mathcal{S}$ with experimental *outcomes.* We shall call the subsets of $\mathcal{S}$ *events,* the space $\mathcal{S}$ *certain event,* and the empty set 0 *impossible event.* These concepts are not determined uniquely from the underlying physical experiment, and they should be carefully specified.

We shall try to explain the possible ambiguity, using as example the rolling of a die.

Mr. X will say that the outcomes of this experiment are obviously the six faces $f_1, \ldots, f_6$, forming the space

$$\mathcal{S} = \{f_1,f_2,f_3,f_4,f_5,f_6\}$$

Events are the 64 subsets of $\mathcal{S}$, for example, the set {even} or {odd}.

Mr. Y might want to bet only on "even" or "odd." He can then argue that for him the outcomes are only two, namely, "even" and "odd," forming the space

$$\mathcal{S} = \{\text{even, odd}\}$$

Events are now only four: 0, {even}, {odd}, $\mathcal{S}$. The event {even} consists of the single element "even." Mr. X will argue that this is not so. The "true" outcomes are six, no matter what we bet on, and the event {even} consists of the three elements f_2, f_4, f_6.

Mr. Z now says that he can consider as outcomes the six faces together with the coordinates of the center of gravity of the die. After all, one might want to bet on which side of the table the die will land. The "true" outcomes, he says, are not six but infinitely many, and {six} is an event containing not just one but many elements.

From the above we see that we must agree what to consider as experimental outcomes; they are not determined from the description of the experiment. From now on when we talk about an experiment $\mathfrak{E}$ we shall understand that its outcomes are specified.

Trials. We shall now relate the abstract concepts of "outcomes" and "events" to real experiments and trials. By "trial" we shall mean a single performance of a well-defined experiment. At this trial we observe a single outcome ζ_i (whatever we agreed to consider as outcome). We say that the event $\mathcal{A}$ occurred at this trial if it contains the element ζ_i. Thus the event $\mathcal{A}$ will occur in all trials whose outcomes are elements of $\mathcal{A}$. The event $\{\zeta_i\}$ containing the single element ζ_i will occur only if the outcome of the trial is ζ_i. Suppose, for example, that we roll a die once and six comes up. The outcome of this trial is the element f_6 (not the event $\{f_6\}$). The event $\{f_6\}$ occurs at this trial, but so does the event {even} and 30 other events (list them).

Clearly, the certain event $\mathcal{S}$ occurs at every trial because it contains every outcome ζ_i. The impossible event 0 never occurs. The event $\mathcal{A} + \mathcal{B}$ occurs when $\mathcal{A}$ or $\mathcal{B}$ or both occur because $\mathcal{A} + \mathcal{B}$ contains all elements that are in $\mathcal{A}$ or in $\mathcal{B}$ or in both. The event $\mathcal{A}\mathcal{B}$ occurs when both events $\mathcal{A}$ and $\mathcal{B}$ occur. If $\mathcal{A}\mathcal{B} = 0$, i.e., if the events $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive, then not both events can occur at the same trial. If $\mathcal{A} \subset \mathcal{B}$ and $\mathcal{A}$ occurs, then $\mathcal{B}$ also occurs. Finally, if at a trial the event $\mathcal{A}$ occurs, then its complement $\bar{\mathcal{A}}$ does not occur.

We should emphasize that in the theory of probability (step 2) one is not concerned with occurrences of events in one or more trials in the preceding sense, but only with probabilities of events. The concept of repeated trials is used only to create from a given experiment *other experiments.* We shall clarify this important point in Chap. 3.

The Axioms

As we have already observed, by an experiment $\mathfrak{E}$ we shall mean a set $\mathcal{S}$ of elements or outcomes ζ. Certain subsets of $\mathcal{S}$ are called events (the reason why not all subsets of $\mathcal{S}$ can be considered as events will be soon given). The space $\mathcal{S}$ is the certain event, and the empty set 0 the impossible event. If an event $\{\zeta_i\}$ consists of a single element ζ_i, it will be called *elementary event*.† Two events $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive if they have no common elements, i.e., if $\mathcal{A}\mathcal{B} = 0$.

We now assign to each event $\mathcal{A}$ a number $P(\mathcal{A})$ which we call "the probability of the event $\mathcal{A}$." This number‡ is so chosen as to satisfy the following three conditions (axioms):

$$\text{I} \qquad P(\mathcal{A}) \geq 0 \tag{2-3}$$

$$\text{II} \qquad P(\mathcal{S}) = 1 \tag{2-4}$$

$$\text{III If} \quad \mathcal{A}\mathcal{B} = 0 \quad \text{then} \quad P(\mathcal{A} + \mathcal{B}) = P(\mathcal{A}) + P(\mathcal{B}) \tag{2-5}$$

Corollaries. From the above axioms we easily conclude that

$$P(0) = 0 \tag{2-6}$$

$$P(\mathcal{A}) = 1 - P(\bar{\mathcal{A}}) \leq 1 \tag{2-7}$$

If $\mathcal{A}$ and $\mathcal{B}$ are not mutually exclusive, then, in general,

$$P(\mathcal{A} + \mathcal{B}) \neq P(\mathcal{A}) + P(\mathcal{B})$$

In this case

$$P(\mathcal{A} + \mathcal{B}) = P(\mathcal{A}) + P(\mathcal{B}) - P(\mathcal{A}\mathcal{B}) \leq P(\mathcal{A}) + P(\mathcal{B}) \tag{2-8}$$

Finally, if $\mathcal{B} \subset \mathcal{A}$, then

$$P(\mathcal{A}) = P(\mathcal{B}) + P(\mathcal{A}\bar{\mathcal{B}}) \geq P(\mathcal{B}) \tag{2-9}$$

Proof. Since $\mathcal{A} + 0 = \mathcal{A}$ and $\mathcal{A}0 = 0$, we have, from (2-5),

$$P(\mathcal{A}) = P(\mathcal{A} + 0) = P(\mathcal{A}) + P(0)$$

and (2-6) follows. Similarly, from

$$\mathcal{A} + \bar{\mathcal{A}} = \mathcal{S} \qquad \mathcal{A}\bar{\mathcal{A}} = 0$$

we conclude that

$$P(\mathcal{A}) + P(\bar{\mathcal{A}}) = P(\mathcal{S}) = 1$$

† In many texts the elements ζ_i are called elementary events. We did not adopt this convention because the element ζ_i is different from the event $\{\zeta_i\}$. Furthermore, the set $\{\zeta_i\}$ is not necessarily an event.

‡ If the event $\mathcal{A}$ is expressed with braces as in (2-1) or (2-2), then instead of $P(\{\quad\})$, we shall write $P\{\quad\}$. Thus $P\{\text{even}\}$ will mean the probability of the event $\{\text{even}\}$.

To show (2-8) we write $\mathcal{A} + \mathcal{B}$ and $\mathcal{B}$ as sums of two mutually exclusive events

$$\mathcal{A} + \mathcal{B} = \mathcal{A} + (\bar{\mathcal{A}}\mathcal{B}) \qquad \mathcal{B} = (\mathcal{A}\mathcal{B}) + (\bar{\mathcal{A}}\mathcal{B})$$

and apply (2-5):

$$P(\mathcal{A} + \mathcal{B}) = P(\mathcal{A}) + P(\bar{\mathcal{A}}\mathcal{B}) \qquad P(\mathcal{B}) = P(\mathcal{A}\mathcal{B}) + P(\bar{\mathcal{A}}\mathcal{B})$$

Eliminating $P(\bar{\mathcal{A}}\mathcal{B})$, we obtain (2-8). Finally, if $\mathcal{B} \subset \mathcal{A}$, then

$$\mathcal{A} = \mathcal{B} + (\mathcal{A}\bar{\mathcal{B}})$$

and since $\mathcal{B}$ and $\mathcal{A}\bar{\mathcal{B}}$ are mutually exclusive, (2-9) follows from (2-5).

Frequency Interpretation. The above conditions I, II, and III form the axioms of probability. These axioms are assumed, and it is meaningless to try to prove them. However, the axioms must be chosen wisely so that the resulting theory will give a satisfactory representation of the physical world. If this is so, then probabilities as used in real problems must also satisfy (more or less) the above axioms. Using the frequency interpretation (1-1):

$$P(\mathcal{A}) \simeq \frac{n_{\mathcal{A}}}{n}$$

we shall show that they do. Indeed, (2-3) is obviously true. Since the certain event $\mathcal{S}$ occurs at every trial, we have $n_{\mathcal{S}} = n$; hence

$$P(\mathcal{S}) \simeq \frac{n_{\mathcal{S}}}{n} = 1$$

Finally, if $\mathcal{A}$ and $\mathcal{B}$ are mutually exclusive, then

$$n_{\mathcal{A}+\mathcal{B}} = n_{\mathcal{A}} + n_{\mathcal{B}}$$

as it is easy to see; hence

$$P(\mathcal{A} + \mathcal{B}) \simeq \frac{n_{\mathcal{A}+\mathcal{B}}}{n} = \frac{n_{\mathcal{A}}}{n} + \frac{n_{\mathcal{B}}}{n} \simeq P(\mathcal{A}) + P(\mathcal{B})$$

and the axioms are thus justified.

Fields. Events are collections of outcomes (subsets of $\mathcal{S}$) on which we have assigned probabilities. However, not all such collections can be considered as events†; why? One reason is that we might not want to assign probabilities to every collection of outcomes. In the die experiment, for example, we might want to assign probabilities only to the four sets

$$0, \{\text{even}\}, \{\text{odd}\}, \mathcal{S}$$

† The distinction between subsets of $\mathcal{S}$ and events and the resulting concept of a field is given for completeness. In the application, all "sensible" collections of outcomes will be events.

Another reason is of deep mathematical nature. If $\mathcal{S}$ has infinitely many elements (if, for example, $\mathcal{S}$ is the set of all real numbers), it might be impossible to assign probabilities to all its subsets so as to satisfy (2-11). Since we intend to accept (2-11), our theory can deal only with certain subsets of $\mathcal{S}$. Thus we assign probabilities only to certain subsets of $\mathcal{S}$, and we call them events. The class of events will be denoted by $\mathfrak{F}$.

We shall demand that the empty set 0 and the space $\mathcal{S}$ be events (impossible event and certain event, respectively). Furthermore, the outcome of any set operation involving events should also be an event. Thus, if $\mathcal{A}$ and $\mathcal{B}$ are events, then $\mathcal{A} + \mathcal{B}$, $\mathcal{A}\mathcal{B}$, and $\mathcal{A} - \mathcal{B}$ should be events. If this is the case, then the class $\mathfrak{F}$ is called a field. The following definition gives a minimum set of conditions for $\mathfrak{F}$ to be a field:

Definition. A field $\mathfrak{F}$ is a nonempty class of sets such that:

1. If $\mathcal{A} \in \mathfrak{F}$, then $\bar{\mathcal{A}} \in \mathfrak{F}$.
2. If $\mathcal{A} \in \mathfrak{F}$ and $\mathcal{B} \in \mathfrak{F}$, then $\mathcal{A} + \mathcal{B} \in \mathfrak{F}$.

We shall show that $\mathfrak{F}$, so defined, has the following properties:

$$\text{If} \quad \mathcal{A} \in \mathfrak{F} \quad \text{and} \quad \mathcal{B} \in \mathfrak{F} \quad \text{then} \quad \mathcal{A}\mathcal{B} \in \mathfrak{F} \quad \text{and} \quad \mathcal{A} - \mathcal{B} \in \mathfrak{F}$$

Furthermore,

$$0 \in \mathfrak{F} \quad \text{and} \quad \mathcal{S} \in \mathfrak{F}$$

Proof. If $\mathcal{A} \in \mathfrak{F}$ and $\mathcal{B} \in \mathfrak{F}$, then (property 1) $\bar{\mathcal{A}} \in \mathfrak{F}$ and $\bar{\mathcal{B}} \in \mathfrak{F}$. Hence (property 2) $\bar{\mathcal{A}} + \bar{\mathcal{B}} \in \mathfrak{F}$; therefore $\overline{(\bar{\mathcal{A}} + \bar{\mathcal{B}})} \in \mathfrak{F}$. But

$$\overline{(\bar{\mathcal{A}} + \bar{\mathcal{B}})} = \mathcal{A}\mathcal{B}$$

hence $\mathcal{A}\mathcal{B} \in \mathfrak{F}$.

Similarly, if $\mathcal{A} \in \mathfrak{F}$ and $\mathcal{B} \in \mathfrak{F}$, then $\bar{\mathcal{B}} \in \mathfrak{F}$, and, as we have just proved, $\mathcal{A}\bar{\mathcal{B}} \in \mathfrak{F}$. But $\mathcal{A}\bar{\mathcal{B}} = \mathcal{A} - \mathcal{B}$; hence $\mathcal{A} - \mathcal{B} \in \mathfrak{F}$. Finally, since $\mathfrak{F}$ is nonempty, it contains at least a set $\mathcal{A}$. Thus $\mathcal{A} \in \mathfrak{F}$; hence $\bar{\mathcal{A}} \in \mathfrak{F}$; therefore $\mathcal{A} + \bar{\mathcal{A}} = \mathcal{S} \in \mathfrak{F}$ and $\mathcal{A}\bar{\mathcal{A}} = 0 \in \mathfrak{F}$, and the proof is complete.

From the above, we conclude that the outcome of any set operation involving finitely many sets in a field $\mathfrak{F}$ is also a set in $\mathfrak{F}$.

Example 2-5. The space $\mathcal{S}$ has six elements:

$$\mathcal{S} = \{1,2,3,4,5,6\}$$

The class

$$0,\ \mathcal{S},\ \{1,3,5\},\ \{2,4,6\},\ \{1\}$$

is not a field because, although the sets $\{1\}$ and $\{2,4,6\}$ belong to the field, their sum

$$\{1\} + \{2,4,6\} = \{1,2,4,6\}$$

does not belong to it. However, the first four sets of the above class form a field. At the risk of belaboring the obvious, we mention that the set $\{1,2,4,6\}$ does not belong to the above class, although it is a subset of $\mathcal{S}$ and $\mathcal{S}$ belongs to this class. Every set must be viewed as a separate entity.

Borel Field. If the sets $\mathcal{A}_1, \ldots, \mathcal{A}_n$ belong to a field $\mathfrak{F}$, then so do the sets

$$\mathcal{A}_1 + \mathcal{A}_2 + \cdots + \mathcal{A}_n \qquad \mathcal{A}_1\mathcal{A}_2 \cdots \mathcal{A}_n$$

This follows readily from the definition. Indeed, the events $\mathcal{A}_1 + \mathcal{A}_2$ and $\mathcal{A}_3$ are in $\mathfrak{F}$; hence $(\mathcal{A}_1 + \mathcal{A}_2) + \mathcal{A}_3$ is in $\mathfrak{F}$, and so on. However, the above does not necessarily hold for infinite sums and products. If a field has the property that, if the sets $\mathcal{A}_1, \ldots, \mathcal{A}_n, \ldots$ belong to it, then so do the sets

$$\mathcal{A}_1 + \cdots + \mathcal{A}_n + \cdots \qquad \mathcal{A}_1 \cdots \mathcal{A}_n \cdots$$

then the field is called Borel field.

Clearly, the class of *all* subsets of $\mathcal{S}$ is a Borel field. Given an arbitrary class C of subsets of $\mathcal{S}$, we can add to it other subsets of $\mathcal{S}$ so as to form a field (if necessary by including all subsets of $\mathcal{S}$). We mention without proof that there exists a smallest Borel field containing C.

The Real Line. A very important example of a space is the set of all real numbers x. Its subsets are collections of such numbers, e.g., the interval $x_2 \leq x \leq x_1$ or the point $x = x_1$. One can show that (measure theory) it is not possible to consider any set of real numbers as an event, because it is not possible to assign probabilities to all sets so as to satisfy (2-11). When we talk about events consisting of real numbers, we shall mean all sets that belong to the smallest Borel field $\mathfrak{F}$ containing all intervals $x \leq x_1$ for any x_1. It is easy to see that this field contains all open or closed intervals $x_1 \leq x \leq x_2$, $x_1 < x < x_2$, all points $x = x_1$, and any countable union or intersection of intervals or points; i.e., it contains every set on the real line that is of interest in applications. One might wonder if we have not already included *all* sets of real numbers in $\mathfrak{F}$. Actually, one can construct certain pathological sets (such constructions are not simple) that are not countable unions or intersections of intervals. Sets of this kind have no probabilities, but are of no importance in applications, and we can forget them.

By events on the plane we shall mean sets of points that can be expressed as unions or intersections (countable) of rectangles. This includes any plane figure that is of interest. Finally, events in an n-dimensional cartesian space will be defined similarly.

Events. From now on we shall assume that events are certain (possibly all) subsets of $\mathcal{S}$ forming a Borel field $\mathfrak{F}$.

We do so because, when we consider unions, intersections, sequences, and limits of events, we want to be sure that the resulting sets are also events, i.e., that probabilities are assigned to them.

Infinite Additivity. From (2-5) we easily conclude that, if the events $\mathcal{A}_1, \ldots, \mathcal{A}_n$ are mutually exclusive, then

$$P(\mathcal{A}_1 + \mathcal{A}_2 + \cdots + \mathcal{A}_n) = P(\mathcal{A}_1) + P(\mathcal{A}_2) + \cdots + P(\mathcal{A}_n) \tag{2-10}$$

The extension to an infinite number of events does not follow from (2-5), but is an added axiom:

III*a*. If

$$\mathcal{A}_i\mathcal{A}_j = 0 \qquad i \neq j \qquad i, j = 1, \ldots, n, \ldots$$

then

$$P(\mathcal{A}_1 + \cdots + \mathcal{A}_n + \cdots) = P(\mathcal{A}_1) + \cdots + P(\mathcal{A}_n) + \cdots \tag{2-11}$$

Definition of an experiment. We are now in a position to give a complete axiomatic definition of an experiment. An experiment $\mathfrak{S}$ is:

1. A set $\mathcal{S}$ of elements or outcome ζ; this set is called space or certain event.
2. A Borel field $\mathfrak{F}$ consisting of certain subsets of $\mathcal{S}$ called events.
3. A number $P(\mathcal{A})$ assigned to every event $\mathcal{A}$; this number is called probability of the event $\mathcal{A}$, and it satisfies axioms I, II, III, and III*a*.

Thus $\mathfrak{S}$ is specified by the above three concepts, $\mathcal{S}$, $\mathfrak{F}$, and P:

$$\mathfrak{S}: (\mathcal{S}, \mathfrak{F}, P)$$

Example 2-6. Our experiment is the single tossing of a coin generating the outcomes *h*, *t*. The certain event $\mathcal{S}$ consists of these two outcomes,

$$\mathcal{S} = \{h,t\}$$

Events are the four sets

$$0, \{h\}, \{t\}, \mathcal{S}$$

If

$$P\{h\} = p \qquad P\{t\} = q$$

we must have

$$p + q = 1$$

because the events $\{h\}$ and $\{t\}$ are mutually exclusive and their sum equals $\mathcal{S}$; therefore [see (2-4) and (2-5)] $P(\mathcal{S}) = 1 = P\{h\} + P\{t\}$.

Construction of probability spaces. In an experiment $\mathfrak{S}$ one must know the probabilities of all events. In many problems these probabilities are determined uniquely from the axioms and the probabilities of certain events. We shall discuss two important applications of this method.

Finitely Many Outcomes. Suppose that $\mathcal{S}$ consists of finitely many elements $\zeta_1, \zeta_2, \ldots, \zeta_N$ and that all subsets of $\mathcal{S}$ are events. With $p_1, \ldots, p_N$ N numbers such that

$$p_1 + \cdots + p_N = 1 \qquad p_i \geq 0 \tag{2-12}$$

we assume that the probability of the elementary event $\{\zeta_i\}$ containing the single element ζ_i equals p_i:

$$P\{\zeta_i\} = p_i \tag{2-13}$$

If $\mathcal{A}$ is an arbitrary set containing the elements $\zeta_{k_1}, \ldots, \zeta_{k_r}$, we have

$$\mathcal{A} = \{\zeta_{k_1}, \ldots, \zeta_{k_r}\} = \{\zeta_{k_1}\} + \cdots + \{\zeta_{k_r}\}$$

Hence [see (2-10)]

$$P(\mathcal{A}) = P\{\zeta_{k_1}\} + \cdots + P\{\zeta_{k_r}\} = p_{k_1} + \cdots + p_{k_r} \tag{2-14}$$

We can thus determine the probability of any event in terms of the probabilities of the elementary events. This is also true if the elements in $\mathcal{S}$ are denumerably infinite.

Classical Definition. We now assume that the numbers p_i are all equal, i.e., that

$$P\{\zeta_i\} = \frac{1}{N} \tag{2-15}$$

It follows from (2-14) that if an event $\mathcal{A}$ contains k elements, then

$$P(\mathcal{A}) = \frac{k}{N} \tag{2-16}$$

This very special but important case is equivalent to the classical definition (1-7). This equivalence is valid, however, only if (2-15) is interpreted as an assumption and not as a logical necessity.

Example 2-7. Our experiment is the tossing of a coin three times. A *single outcome* is a certain succession of heads or tails. The possible outcomes are eight, namely,

hhh, hht, hth, htt, thh, tht, tth, ttt

(Notice that our space is not $\{h,t\}$.) Assuming that all elementary events have the same probability, we conclude from (2-15) that the probability of each elementary event equals $\frac{1}{8}$. Thus $P\{hhh\} = \frac{1}{8}$; i.e., the probability that we get three heads in a row equals $\frac{1}{8}$. Since the event {heads at the first two tossings} consists of the two outcomes *hhh, hht,* we conclude from (2-16) that its probability equals $\frac{2}{8}$.

Probabilities on the Real Line. An important probability space is the set of numbers on the entire real axis or on a portion of it. The experimental outcomes are real numbers, usually time instances, and events are sums and products of intervals as defined on page 29. Examples of experiments generating such spaces are many. We mention electron emission, arrival time, time of death, time of failure of a system, telephone calls. We shall assume that $\mathcal{S}$ is the set of positive real numbers (instead of ζ we now use t for an outcome). The probabilities of various events can be determined if one knows the probability of the event $\{0 \leq t \leq t_1\}$ for any t_1. This can be defined as follows: With $\alpha(t)$ an integrable bounded function of time such that

$$\alpha(t) \geq 0 \qquad \int_0^\infty \alpha(t)\,dt = 1 \tag{2-17}$$

we assume that

$$P\{0 \leq t \leq t_1\} = \int_0^{t_1} \alpha(t)\,dt \tag{2-18}$$

To determine the probability of the event $\{t_1 \leq t \leq t_2\}$ we observe that

$$\{0 \leq t \leq t_2\} = \{0 \leq t \leq t_1\} + \{t_1 < t \leq t_2\} \tag{2-19}$$

But the last two events in (2-19) are obviously mutually exclusive; hence [see (2-5)]

$$P\{0 \leq t \leq t_2\} = P\{0 \leq t \leq t_1\} + P\{t_1 < t \leq t_2\}$$

Therefore [see (2-18)]

$$P\{t_1 < t \leq t_2\} = \int_{t_1}^{t_2} \alpha(t)\,dt \tag{2-20}$$

The probability of an event consisting of two or more intervals is given by a sum of integrals of the form (2-20). We shall now show that the probability of the elementary event $\{t = t_1\}$ equals zero:

$$P\{t = t_1\} = 0 \tag{2-21}$$

Indeed, with $\varepsilon > 0$ any number, we have from (2-20)

$$P\{t_1 - \varepsilon < t \leq t_1\} = \int_{t_1-\varepsilon}^{t_1} \alpha(t)\,dt$$

This can be made arbitrarily small if we choose ε small enough. But

$$\{t_1 - \varepsilon < t \leq t_1\} \supset \{t = t_1\}$$

Hence [see (2-9)]

$$P\{t_1 - \varepsilon < t \leq t_1\} \geq P\,\{t = t_1\}$$

and (2-21) follows. From (2-21) and (2-20) we easily conclude that

$$P\{t_1 < t < t_2\} = P\{t_1 < t \leq t_2\} = P\{t_1 \leq t \leq t_2\} \tag{2-22}$$

In certain applications the probability of the elementary event $\{t = t_1\}$ is not zero. In such cases probabilities cannot be determined as in (2-18) [unless we allow $\alpha(t)$ to include impulses].

Example 2-8. A telephone call occurs at "random" in the time interval $(0,T)$. Define an appropriate experiment.

The probability space is the interval $(0,T)$, and its outcomes, instances of time in this interval. Events are sums and products of intervals as on page 29. The probability of the event $\{0 \le t \le t_1\}$ is given by

$$P\{0 \le t \le t_1\} = \frac{t_1}{T}$$

Hence [see (2-20)]

$$P\{t_1 \le t \le t_2\} = \frac{t_2 - t_1}{T} \tag{2-23}$$

This is what one means by "random" calls.

Example 2-9. Our experiment is the selection of a substance and the observation of the time of emission of an electron after $t = 0$. The space is now the interval $t \ge 0$, and the probability of the event $\{t_1 \le t \le t_2\}$, that is, the event {the electron is emitted in the interval (t_1,t_2)}, can be written in the form (2-20)

$$P\{t_1 \le t \le t_2\} = \int_{t_1}^{t_2} \alpha(t)\, dt$$

where $\alpha(t)$ is a suitable function satisfying (2-17).

Equal Events with Probability 1. If the probability of an event $\mathcal{N}$ equals zero, it does not follow that $\mathcal{N}$ is the impossible event. Thus, in Example 2-8, we have $P\{t = t_1\} = 0$ [see (2-21)]; however, the event $\{t = t_1\}$ is not empty since it contains the element t_1.

If the event $\mathcal{N}$ is such that

$$P(\mathcal{N}) = 0$$

we say that "$\mathcal{N}$ equals the impossible event with probability 1." Similarly, if

$$P(\mathcal{A}) = 1$$

it does not follow that $\mathcal{A}$ equals the certain event; we merely say that it equals $\mathcal{S}$ with probability 1. More generally, if the events $\mathcal{A}$ and $\mathcal{B}$ are such that the set of elements $(\mathcal{A} + \mathcal{B}) - (\mathcal{A}\mathcal{B})$ that belong to $\mathcal{A}$ or $\mathcal{B}$ but not to both (Fig. 2-8) has zero probability, we then say that "$\mathcal{A}$ equals $\mathcal{B}$ with probability 1 (or almost everywhere)." It is clear, we hope, that "equality of events with probability 1" does not mean that the probability of these events equals 1.

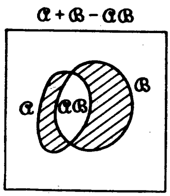

Fig. 2-8

2-3. Conditional probabilities and independent events

Given an event $\mathcal{M}$ with nonzero probability,

$$P(\mathcal{M}) > 0 \tag{2-24}$$

we define the "conditional probability of $\mathcal{A}$ assuming $\mathcal{M}$" by

$$P(\mathcal{A}|\mathcal{M}) = \frac{P(\mathcal{A}\mathcal{M})}{P(\mathcal{M})} \tag{2-25}$$

In words: $P(\mathcal{A}|\mathcal{M})$ equals the probability of the event $\mathcal{A}\mathcal{M}$ (the part of $\mathcal{A}$ included in $\mathcal{M}$) divided by the probability of $\mathcal{M}$. Clearly, if $\mathcal{A}$ and $\mathcal{M}$ have no common elements (mutually exclusively), then $P(\mathcal{A}|\mathcal{M}) = 0$. We also remark that if $\mathcal{A} \subset \mathcal{M}$, then $\mathcal{A}\mathcal{M} = \mathcal{A}$; hence

$$P(\mathcal{A}|\mathcal{M}) = \frac{P(\mathcal{A})}{P(\mathcal{M})} \geq P(\mathcal{A})$$

Finally, if $\mathcal{M} \subset \mathcal{A}$, then

$$P(\mathcal{A}|\mathcal{M}) = \frac{P(\mathcal{M})}{P(\mathcal{M})} = 1$$

Frequency Interpretation. The experiment $\mathfrak{S}$ is performed n times. With $n_{\mathcal{A}}$, $n_{\mathcal{M}}$, and $n_{\mathcal{A}\mathcal{M}}$ the number of occurrences of the events $\mathcal{A}$, $\mathcal{M}$, and $\mathcal{A}\mathcal{M}$, respectively, we have from (1-1)

$$P(\mathcal{A}) \simeq \frac{n_{\mathcal{A}}}{n} \qquad P(\mathcal{M}) \simeq \frac{n_{\mathcal{M}}}{n} \qquad P(\mathcal{A}\mathcal{M}) \simeq \frac{n_{\mathcal{A}\mathcal{M}}}{n}$$

Hence

$$P(\mathcal{A}|\mathcal{M}) = \frac{P(\mathcal{A}\mathcal{M})}{P(\mathcal{M})} \simeq \frac{n_{\mathcal{A}\mathcal{M}}/n}{n_{\mathcal{M}}/n} = \frac{n_{\mathcal{A}\mathcal{M}}}{n_{\mathcal{M}}}$$

Thus the frequency interpretation of the conditional probability takes the form

$$P(\mathcal{A}|\mathcal{M}) \simeq \frac{n_{\mathcal{A}\mathcal{M}}}{n_{\mathcal{M}}}$$

This result can be phrased as follows: If we discard all trials in which the event $\mathcal{M}$ did not occur and retain the sub-sequence of $n_{\mathcal{M}}$ trials in which it occurred, then $P(\mathcal{A}|\mathcal{M})$ equals the relative frequency of occurrence of the event $\mathcal{A}$ in that sub-sequence.

Fundamental remark. Given an event $\mathcal{M}$, the quantities $P(\mathcal{A}|\mathcal{M})$ are numbers assigned to all events of $\mathfrak{F}$. We maintain that these numbers are, indeed, probabilities; i.e., they satisfy axioms I, II, and III. If this is so, then conditional probabilities permit us to create from the original experiment specified by $\mathcal{S}$, $\mathfrak{F}$, $P(\mathcal{A})$ a *new* experiment $\mathcal{S}$, $\mathfrak{F}$, $P(\mathcal{A}|\mathcal{M})$. This experiment has the same outcomes ζ forming the space $\mathcal{S}$, the same events forming the field $\mathfrak{F}$, but new probabilities $P(\mathcal{A}|\mathcal{M})$ assigned to these events.

Proof. It suffices to prove that

$$P(\mathcal{A}|\mathcal{M}) \geq 0 \tag{2-26}$$
$$P(\mathcal{S}|\mathcal{M}) = 1 \tag{2-27}$$

and if

$$\mathcal{A}\mathcal{B} = 0 \qquad \text{then} \qquad P(\mathcal{A} + \mathcal{B}|\mathcal{M}) = P(\mathcal{A}|\mathcal{M}) + P(\mathcal{B}|\mathcal{M}) \tag{2-28}$$

The first equation is obvious. To show (2-27) we observe that $\mathcal{S}\mathcal{M} = \mathcal{M}$; hence

$$P(\mathcal{S}|\mathcal{M}) = \frac{P(\mathcal{S}\mathcal{M})}{P(\mathcal{M})} = 1$$

From $\mathcal{AB} = 0$ we conclude that (see Fig. 2-9) the events $\mathcal{AM}$ and $\mathcal{BM}$ are also mutually exclusive; therefore

$$P[(\mathcal{A} + \mathcal{B})\mathcal{M}] = P[(\mathcal{AM}) + (\mathcal{BM})] = P(\mathcal{AM}) + P(\mathcal{BM})$$

Thus

$$P(\mathcal{A} + \mathcal{B}|\mathcal{M}) = \frac{P[(\mathcal{A} + \mathcal{B})\mathcal{M}]}{P(\mathcal{M})} = \frac{P(\mathcal{AM})}{P(\mathcal{M})} + \frac{P(\mathcal{BM})}{P(\mathcal{M})}$$
$$= P(\mathcal{A}|\mathcal{M}) + P(\mathcal{B}|\mathcal{M})$$

and (2-28) is proved. The implications of the above remark will be

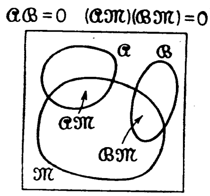

Fig. 2-9

appreciated later on, when we introduce conditional distributions and densities.

Example 2-10. Our space consists of the six faces of a die as in Example 2-1. We assume that

$$P\{f_i\} = 1/6 \qquad i = 1, \ldots, 6$$

(This is what we shall mean by *fair die.*) With

$$\mathcal{M} = \{\text{even}\} = \{f_2, f_4, f_6\}$$

and $\mathcal{A} = \{f_2\}$, we have $\mathcal{AM} = \{f_2\}$. Since $P(\mathcal{M}) = 3/6$, $P(\mathcal{AM}) = 1/6$, we obtain from (2-25)

$$P\{f_2|\text{even}\} = \frac{P(\mathcal{AM})}{P(\mathcal{M})} = \frac{1}{3}$$

Thus the probability of the event $\{f_2\}$, assuming that an even number came up, equals $1/3$. (This equals the relative frequency of occurrence of {two} in the subsequence of rollings whose outcome is even.)

Example 2-11. Our space is the real-time axis $(0, \infty)$. A single outcome is a particular instance $t = t_i$, for example, the emission time of an electron. The probability that the electron is emitted in the time interval (t_1, t_2) is given by [see (2-20)]

$$P\{t_1 \le t \le t_2\} = \int_{t_1}^{t_2} \alpha(t)\, dt$$

where

$$\alpha(t) \ge 0 \qquad \int_0^\infty \alpha(t)\, dt = 1$$

We shall now determine the conditional probability of the event $\{t_1 \leq t \leq t_2\}$, assuming that the electron was not emitted prior to $t = t_0$. Denoting this last event by

$$\mathfrak{M} = \{t \geq t_0\}$$

we observe that, if $t_0 \leq t_1$, then

$$\{t_1 \leq t \leq t_2\}\{t \geq t_0\} = \{t_1 \leq t \leq t_2\}$$

But

$$P\{t \geq t_0\} = \int_{t_1}^{\infty} \alpha(t)\, dt$$

Hence [see (2-25)] for $t_1 \geq t_0$

$$P\{t_1 \leq t \leq t_2 | t \geq t_0\} = \frac{\int_{t_1}^{t_2} \alpha(t)\, dt}{\int_{t_0}^{\infty} \alpha(t)\, dt} \tag{2-29}$$

Special Case. It can be shown that (see Prob. 2-14), if the conditional probability of emission of an electron in the interval $(t_0, t_0 + t)$, assuming that it was not emitted prior to t_0, equals the unconditional probability of its emission in the interval $(0,t)$, then

$$\alpha(t) = ce^{-ct} \tag{2-30}$$

Example 2-12. The probability that a person will die in the time interval (t_1,t_2) is given by

$$P\{t_1 \leq t \leq t_2\} = \int_{t_1}^{t_2} \alpha(t)\, dt$$

The function $\alpha(t)$ is determined from long records; we shall assume that it equals the function

$$\alpha(t) = \begin{cases} At^2(100 - t)^2 & 0 \leq t \leq 100 \\ 0 & \text{elsewhere} \end{cases}$$

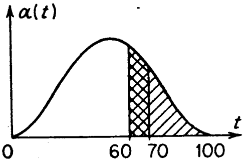

Fig. 2-10

shown in Fig. 2-10. The constant A is determined from (2-17); the result of the integration gives

$$A = 3 \times 10^{-9}$$

We shall now determine the probability p_1 that the person will die between the ages of 60 and 70 and the probability p_2 that he will die between those ages assuming that he lived to be 60.

Clearly, $p_1 = P\{60 \leq t \leq 70\}$; hence

$$p_1 = 3 \times 10^{-9} \int_{60}^{70} t^2(100 - t)^2\, dt = 0.154$$

p_2 equals the conditional probability

$$P\{60 \leq t \leq 70 | t \geq 60\}$$

Hence [see (2-29)]

$$p_2 = \frac{3 \times 10^{-9} \int_{60}^{70} t^2(100 - t)^2\, dt}{3 \times 10^{-9} \int_{60}^{100} t^2(100 - t)^2\, dt} = 0.486$$

Example 2-13. A box contains three white and two red balls:

$$w_1,\ w_2,\ w_3,\ r_1,\ r_2$$

We remove from it "at random" two balls in succession. What is the probability that the first ball is white and the second red?

First Solution. An element of our space is a pair of different balls. The total number of such pairs is 20, namely,

$$\begin{gathered} w_1w_2,\ w_1w_3,\ w_1r_1,\ w_1r_2,\ w_2w_1,\ w_2w_3,\ w_2r_1,\ w_2r_2,\ w_3w_1,\ w_3w_2 \\ w_3r_1,\ w_3r_2,\ r_1w_1,\ r_1w_2,\ r_1w_3,\ r_1r_2,\ r_2w_1,\ r_2w_2,\ r_2w_3,\ r_2r_1 \end{gathered}$$

Hence the probability of each elementary event equals $\frac{1}{20}$. The event {white first, red second} consists of the six elements

$$w_1r_1,\ w_1r_2,\ w_2r_1,\ w_2r_2,\ w_3r_1,\ w_3r_2$$

$\mathcal{F}\{f(t)\} = \int_{-\infty}^{\infty} e^{-j\omega t} f(t)\, dt$

Therefore [see (2-16)] its probability equals $\frac{6}{20}$.

Second Solution. It is easy to see that

$$\begin{aligned} P\{\text{white first}\} &= \tfrac{3}{5} \\ P\{\text{red second}|\text{white first}\} &= \tfrac{2}{4} \end{aligned}$$

Hence [see (2-25)]

$$P\{\text{white first, red second}\} = \tfrac{3}{5} \times \tfrac{2}{4} = \tfrac{6}{20}$$

Total Probability

We are given n mutually exclusive events

$$\mathcal{A}_1,\ \mathcal{A}_2,\ \ldots,\ \mathcal{A}_n$$

whose sum equals the certain event $\mathcal{S}$ (Fig. 2-11)

$$\mathcal{A}_i\mathcal{A}_j = 0 \qquad i \neq j = 1, 2, \ldots, n \tag{2-31}$$

$$\mathcal{A}_1 + \mathcal{A}_2 + \cdots + \mathcal{A}_n = \mathcal{S} \tag{2-32}$$

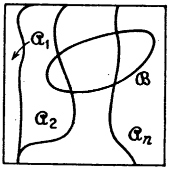

Fig. 2-11

With $\mathcal{B}$ an arbitrary event, we shall show that†

$$P(\mathcal{B}) = P(\mathcal{B}|\mathcal{A}_1)P(\mathcal{A}_1) + \cdots + P(\mathcal{B}|\mathcal{A}_n)P(\mathcal{A}_n) \tag{2-33}$$

Proof. Since $\mathcal{B} = \mathcal{B}\mathcal{S}$, we obtain from (2-32)

$$\mathcal{B} = \mathcal{B}(\mathcal{A}_1 + \cdots + \mathcal{A}_n) = \mathcal{B}\mathcal{A}_1 + \cdots + \mathcal{B}\mathcal{A}_n$$

From the fact that $\mathcal{A}_i$ has no common elements with $\mathcal{A}_j$, it easily follows that $\mathcal{B}\mathcal{A}_i$ has no common elements with $\mathcal{B}\mathcal{A}_j$. Therefore [see (2-10)]

$$P(\mathcal{B}) = P(\mathcal{B}\mathcal{A}_1) + \cdots + P(\mathcal{B}\mathcal{A}_n) \tag{2-34}$$

But [see (2-25)]

$$P(\mathcal{B}\mathcal{A}_i) = P(\mathcal{B}|\mathcal{A}_i)P(\mathcal{A}_i)$$

Inserting the above into (2-34), we obtain (2-33).

† Throughout the book, whenever we write $P(\mathcal{A}|\mathcal{B})$, we shall assume $P(\mathcal{B}) \neq 0$ even if this is not explicitly stated.

Equation (2-33), known as "theorem on total probability," is used to evaluate $P(\mathcal{B})$ in terms of $P(\mathcal{B}|\mathcal{A}_i)$ and $P(\mathcal{A}_i)$. It is valid even if the sum of the sets $\mathcal{A}_i$ does not equal $\mathcal{S}$, provided

$$\mathcal{A}_1 + \mathcal{A}_2 + \cdots + \mathcal{A}_n \supset \mathcal{B}$$

Example 2-14. A box contains 2,000 components, of which 5% are defective. A second box contains 500 components, of which 40% are defective. Two other boxes contain 1,000 components each, with 10% defective components. We select at *random* one of the above boxes and remove from it at *random* a single component. What is the probability that this component is defective?

We have 4,000 good (g) components and 500 defective (d) components arranged as follows:

Box 1: 1,900g, 100d	Box 2: 300g, 200d
Box 3: 900g, 100d	Box 4: 900g, 100d

Thus our probability space has 4,500 elements. We denote by $\mathcal{B}_i$ the event consisting of all elements in the ith box and by $\mathcal{D}$ the event consisting of all 500 defective elements. We want to determine $P(\mathcal{D})$. Clearly,

$$\mathcal{B}_1 + \mathcal{B}_2 + \mathcal{B}_3 + \mathcal{B}_4 = \mathcal{S} \qquad \mathcal{B}_i\mathcal{B}_j = 0 \qquad i \neq j$$

By random selection of a box one means that

$$P(\mathcal{B}_1) = P(\mathcal{B}_2) = P(\mathcal{B}_3) = P(\mathcal{B}_4) = \tfrac{1}{4} \tag{2-35}$$

Once a box is selected, the probability that the removed element is defective equals the ratio of the number of defective to the total number of elements in that box. In our language this means that

$$\begin{aligned} P(\mathcal{D}|\mathcal{B}_1) &= \frac{100}{2{,}000} = 0.05 \qquad & P(\mathcal{D}|\mathcal{B}_2) &= \frac{200}{500} = 0.4 \\ P(\mathcal{D}|\mathcal{B}_3) &= \frac{100}{1{,}000} = 0.1 \qquad & P(\mathcal{D}|\mathcal{B}_4) &= \frac{100}{1{,}000} = 0.1 \end{aligned} \tag{2-36}$$

Inserting the above quantities into (2-33), we obtain

$$P(\mathcal{D}) = 0.05 \times \tfrac{1}{4} + 0.4 \times \tfrac{1}{4} + 0.1 \times \tfrac{1}{4} + 0.1 \times \tfrac{1}{4} = 0.1625$$

We shall now comment on the significance of the above solution. From the point of view of probability theory, Eqs. (2-35) and (2-36) are not proved, they are merely *assumed*. Based on this assumption, we deduced that the probability of selecting a defective part equals 0.1625. Although it is "reasonable" to accept the validity of (2-35) and (2-36) in a real problem, such reasoning belongs to step 1 (page 4) of a probabilistic investigation. The classical definition, on the other hand, tells us that (2-35), for example, is correct because "there are four equally likely choices; hence, the probability of selecting a particular box must equal ¼." As we have pointed out, such conclusions are outside the scope of our analysis.

Bayes' theorem. Using Eq. (2-33), we shall show that

$$P(\mathcal{A}_i|\mathcal{B}) = \frac{P(\mathcal{B}|\mathcal{A}_i)P(\mathcal{A}_i)}{P(\mathcal{B}|\mathcal{A}_1)P(\mathcal{A}_1) + \cdots + P(\mathcal{B}|\mathcal{A}_n)P(\mathcal{A}_n)} \tag{2-37}$$

where the events $\mathcal{A}_i$ satisfy (2-31) and (2-32) and $\mathcal{B}$ is arbitrary. This formula permits us to evaluate the so-called a posteriori probabilities $P(\mathcal{A}_i|\mathcal{B})$ of the events $\mathcal{A}_i$ in terms of the (a priori) probabilities $P(\mathcal{A}_i)$ and the conditional probabilities $P(\mathcal{B}|\mathcal{A}_i)$.

Proof. From (2-25) we have

$$P(\mathcal{A}_i\mathcal{B}) = P(\mathcal{A}_i|\mathcal{B})P(\mathcal{B}) = P(\mathcal{B}|\mathcal{A}_i)P(\mathcal{A}_i)$$

Hence

$$P(\mathcal{A}_i|\mathcal{B}) = \frac{P(\mathcal{B}|\mathcal{A}_i)P(\mathcal{A}_i)}{P(\mathcal{B})} \tag{2-38}$$

Inserting $P(\mathcal{B})$ as given by (2-33) into (2-38), we obtain the desired result (2-37).

Example 2-14 (Continued). We examine the selected component in Example 2-14, and we find it defective. What is the probability that it was taken from box 2?

We want the conditional probability $P(\mathcal{B}_2|\mathfrak{D})$ of the event $\mathcal{B}_2$ (second box), assuming $\mathfrak{D}$ (defective component). Since

$$P(\mathfrak{D}) = 0.1625 \qquad P(\mathfrak{D}|\mathcal{B}_2) = 0.4 \qquad P(\mathcal{B}_2) = \tfrac{1}{4}$$

we obtain from (2-38)

$$P(\mathcal{B}_2|\mathfrak{D}) = \frac{0.4 \times \frac{1}{4}}{0.1625} \simeq 0.615$$

Thus the a posteriori (conditional) probability of having selected box 2, assuming that the selected part is defective, equals 0.615.

The frequency interpretation of the a posteriori probability $P(\mathcal{B}_2|\mathfrak{D})$ is the following: We repeat the experiment (selection of a box and removal of a particle) n times. In $n_{\mathfrak{D}}$ of these trials the selected part is defective. In the above $n_{\mathfrak{D}}$ trials we count the number of times the selected part was taken from box 2. If this number is $n_{\mathcal{B}_2\mathfrak{D}}$, then

$$P(\mathcal{B}_2|\mathfrak{D}) \simeq \frac{n_{\mathcal{B}_2\mathfrak{D}}}{n_{\mathfrak{D}}}$$

Comment. If in Bayes' theorem (2-37) the a priori probabilities $P(\mathcal{A}_i)$ are of the same order of magnitude and

$$P(\mathcal{B}|\mathcal{A}_1) \gg \sum_{i=2}^{n} P(\mathcal{B}|\mathcal{A}_i)$$

then

$$P(\mathcal{A}_1|\mathcal{B}) = \frac{P(\mathcal{B}|\mathcal{A}_1)P(\mathcal{A}_1)}{P(\mathcal{B}|\mathcal{A}_1)P(\mathcal{A}_1) + \cdots} \simeq 1 \tag{2-39}$$

This result is used extensively in statistics (see page 80).

Example 2-15. A box contains 999 white (w) cards and 1 red (r) card. Another box contains 999 red cards and 1 white card. We select a box at random and remove a card. This card is white. What is the probability that it was taken from the first box?

With $\mathcal{B}_1$ and $\mathcal{B}_2$ the elements in the first and second box, respectively, and $\mathcal{W}$ and $\mathcal{R}$ all the white and red cards, we have, as in Example 2-14,

$$P(\mathcal{B}_1) = \tfrac{1}{2} \qquad P(\mathcal{B}_2) = \tfrac{1}{2}$$
$$P(\mathcal{W}|\mathcal{B}_1) = 0.999 \qquad P(\mathcal{W}|\mathcal{B}_2) = 0.001$$

Therefore [see (2-37)]

$$P(\mathcal{B}_1|\mathcal{W}) = \frac{0.999 \times \frac{1}{2}}{0.999 \times \frac{1}{2} + 0.001 \times \frac{1}{2}} \simeq 1$$

From the last two examples we see that, in order to solve a problem, we must first carefully translate its wording into the language of probability theory, i.e., into events, probabilities, conditional probabilities. For the beginner this step is not always easy and requires practice. However, once this is done, the solution usually involves simple mathematical operations. We recognize that certain problems can be easily solved by intuition, without the need to specify, in our sense, the underlying experiment (see, for instance, the second solution in Example 2-13). However, intuition without supporting theory is inadequate for more sophisticated applications. Furthermore, the student is often unable to identify such simple concepts as independence or mutual exclusiveness. In fact he might not know the difference between a correct and a wrong solution. A precise definition eliminates such doubts and refines intuition.

Independent Events

The concept of independence is basic. One could say that because of it the theory of probability is developed as a separate discipline and is not considered merely as a topic in measure theory. The reader might find the following definition arbitrary. He might even wonder if, in a general probability space, there exist events satisfying (2-40). In the next chapter we shall show the reason for introducing this concept and the way to construct independent events through repeated trials.

Definition. Two events $\mathcal{A}$ and $\mathcal{B}$ are called independent if

$$P(\mathcal{A}\mathcal{B}) = P(\mathcal{A})P(\mathcal{B}) \tag{2-40}$$

From the definition and (2-25) we conclude that, if the events $\mathcal{A}$ and $\mathcal{B}$ are independent, then

$$P(\mathcal{A}|\mathcal{B}) = P(\mathcal{A}) \qquad P(\mathcal{B}|\mathcal{A}) = P(\mathcal{B}) \tag{2-41}$$

Frequency Interpretation. With $n_{\mathcal{A}}$, $n_{\mathcal{B}}$, and $n_{\mathcal{A}\mathcal{B}}$ the number of occurrences of the events $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{A}\mathcal{B}$, respectively, in a sequence of n trials, we have

$$P(\mathcal{A}) \simeq \frac{n_{\mathcal{A}}}{n} \qquad P(\mathcal{B}) \simeq \frac{n_{\mathcal{B}}}{n} \qquad P(\mathcal{A}\mathcal{B}) \simeq \frac{n_{\mathcal{A}\mathcal{B}}}{n}$$

If the events $\mathcal{A}$ and $\mathcal{B}$ are independent, then

$$P(\mathcal{A}) = \frac{P(\mathcal{A}\mathcal{B})}{P(\mathcal{B})}$$

Therefore

$$\frac{n_{\mathcal{A}}}{n} \simeq \frac{n_{\mathcal{A}\mathcal{B}}/n}{n_{\mathcal{B}}/n} = \frac{n_{\mathcal{A}\mathcal{B}}}{n_{\mathcal{B}}}$$

Thus: The relative frequency $n_{\mathcal{A}}/n$ of the occurrence of the event $\mathcal{A}$ in the original sequence of n trials equals the relative frequency $n_{\mathcal{A}\mathcal{B}}/n_{\mathcal{B}}$ of the occurrence of $\mathcal{A}$ in the sub-sequence of $n_{\mathcal{B}}$ trials in which $\mathcal{B}$ occurred.

Independence of n Events. The notion of independence will now be extended to n events. We shall first assume that $n = 3$. The events $\mathcal{A}_1$, $\mathcal{A}_2$, $\mathcal{A}_3$ are called independent if

$$P(\mathcal{A}_1\mathcal{A}_2) = P(\mathcal{A}_1)P(\mathcal{A}_2) \qquad P(\mathcal{A}_1\mathcal{A}_3) = P(\mathcal{A}_1)P(\mathcal{A}_3)$$
$$P(\mathcal{A}_2\mathcal{A}_3) = P(\mathcal{A}_2)P(\mathcal{A}_3) \tag{2-42}$$

and

$$P(\mathcal{A}_1\mathcal{A}_2\mathcal{A}_3) = P(\mathcal{A}_1)P(\mathcal{A}_2)P(\mathcal{A}_3) \tag{2-43}$$

In general, we say that the events $\mathcal{A}_1, \ldots, \mathcal{A}_n, \ldots$ are independent if, with $k_1, k_2, \ldots, k_r$ any set of integers we have

$$P(\mathcal{A}_{k_1}\mathcal{A}_{k_2} \cdots \mathcal{A}_{k_r}) = P(\mathcal{A}_{k_1})P(\mathcal{A}_{k_2}) \cdots P(\mathcal{A}_{k_r}) \tag{2-44}$$

The total number of equations of the form (2-44) that are required to establish the independence of n events equals (see Prob. 2-10)

$$2^n - (n + 1)$$

We emphasize that if the above events are independent in pairs, i.e., if

$$P(\mathcal{A}_i\mathcal{A}_j) = P(\mathcal{A}_i)P(\mathcal{A}_j)$$

for every i and j $(i \neq j)$, it does not follow that they are independent. This is illustrated in the next example.

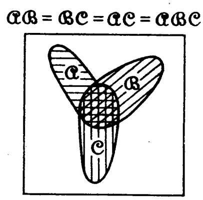

Fig. 2-12

Example 2-16. The events $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ of Fig. 2-12 are such that

$$P(\mathcal{A}) = P(\mathcal{B}) = P(\mathcal{C}) = 1/5$$

and

(*a*) $$P(\mathcal{A}\mathcal{B}) = P(\mathcal{A}\mathcal{C}) = P(\mathcal{B}\mathcal{C}) = P(\mathcal{A}\mathcal{B}\mathcal{C}) = 1/25$$

These events are obviously independent in pairs, but they are not independent because

$$P(\mathcal{A}\mathcal{B}\mathcal{C}) \neq P(\mathcal{A})P(\mathcal{B})P(\mathcal{C})$$

(*b*) $$P(\mathcal{A}\mathcal{B}) = P(\mathcal{A}\mathcal{C}) = P(\mathcal{B}\mathcal{C}) = P(\mathcal{A}\mathcal{B}\mathcal{C}) = 1/125$$

We now have

$$P(\mathcal{A}\mathcal{B}\mathcal{C}) = P(\mathcal{A})P(\mathcal{B})P(\mathcal{C})$$

However, $P(\mathcal{A}\mathcal{B}) \neq P(\mathcal{A})P(\mathcal{B})$.

Thus three events are independent if they satisfy Eqs. (2-42) *and* (2-43).

Properties of Independent Events. If the events $\mathcal{A}$ and $\mathcal{B}$ are independent, then

$$P(\mathcal{A} + \mathcal{B}) = P(\mathcal{A}) + P(\mathcal{B}) - P(\mathcal{A})P(\mathcal{B}) \tag{2-45}$$

This follows from (2-8) and (2-40).

If the events $\mathcal{A}$ and $\mathcal{B}$ are independent, then their complements $\bar{\mathcal{A}}$ and $\bar{\mathcal{B}}$ are also independent. Indeed, from

$$\bar{\mathcal{A}}\bar{\mathcal{B}} = (\overline{\mathcal{A} + \mathcal{B}}) \qquad P(\overline{\mathcal{A} + \mathcal{B}}) = 1 - P(\mathcal{A} + \mathcal{B})$$

[see De Morgan's law and (2-7)] we obtain

$$\begin{aligned} P(\bar{\mathcal{A}}\bar{\mathcal{B}}) = 1 - P(\mathcal{A} + \mathcal{B}) = 1 - P(\mathcal{A}) - P(\mathcal{B}) &+ P(\mathcal{A})P(\mathcal{B}) \\ &= [1 - P(\mathcal{A})][1 - P(\mathcal{B})] \end{aligned}$$

But

$$1 - P(\mathcal{A}) = P(\bar{\mathcal{A}}) \qquad 1 - P(\mathcal{B}) = P(\bar{\mathcal{B}})$$

Therefore

$$P(\bar{\mathcal{A}}\bar{\mathcal{B}}) = P(\bar{\mathcal{A}})P(\bar{\mathcal{B}})$$

One shows, similarly, that $\mathcal{A}$ and $\bar{\mathcal{B}}$ are independent.

If the events $\mathcal{A}_1$, $\mathcal{A}_2$, $\mathcal{A}_3$ are independent, then $\mathcal{A}_1$ is independent of $\mathcal{A}_2\mathcal{A}_3$ because

$$P(\mathcal{A}_1\mathcal{A}_2\mathcal{A}_3) = P(\mathcal{A}_1)P(\mathcal{A}_2)P(\mathcal{A}_3) = P(\mathcal{A}_1)P(\mathcal{A}_2\mathcal{A}_3)$$

It is also independent of $\mathcal{A}_2 + \mathcal{A}_3$ because

$$\begin{aligned} P[\mathcal{A}_1(\mathcal{A}_2 + \mathcal{A}_3)] &= P[(\mathcal{A}_1\mathcal{A}_2) + (\mathcal{A}_1\mathcal{A}_3)] \\ &= P(\mathcal{A}_1\mathcal{A}_2) + P(\mathcal{A}_1\mathcal{A}_3) - P(\mathcal{A}_1\mathcal{A}_2\mathcal{A}_3) \end{aligned}$$

The last equality followed from (2-8) and the obvious fact that

$$(\mathcal{A}_1\mathcal{A}_2)(\mathcal{A}_1\mathcal{A}_3) = \mathcal{A}_1\mathcal{A}_2\mathcal{A}_3$$

From the assumed independence we have

$$\begin{aligned} P(\mathcal{A}_1\mathcal{A}_2) &+ P(\mathcal{A}_1\mathcal{A}_3) - P(\mathcal{A}_1\mathcal{A}_2\mathcal{A}_3) \\ &= P(\mathcal{A}_1)P(\mathcal{A}_2) + P(\mathcal{A}_1)P(\mathcal{A}_3) - P(\mathcal{A}_1)P(\mathcal{A}_2)P(\mathcal{A}_3) \\ &= P(\mathcal{A}_1)[P(\mathcal{A}_2) + P(\mathcal{A}_3) - P(\mathcal{A}_2)P(\mathcal{A}_3)] \end{aligned}$$

Therefore [see (2-45)]

$$P[\mathcal{A}_1(\mathcal{A}_2 + \mathcal{A}_3)] = P(\mathcal{A}_1)P(\mathcal{A}_2 + \mathcal{A}_3)$$

The above is not true if the events $\mathcal{A}_1$, $\mathcal{A}_2$, $\mathcal{A}_3$ are independent only in pairs (see Example 2-16 and Fig. 2-12).

One can similarly show that if the events $\mathcal{A}_1, \ldots, \mathcal{A}_n$ are independent, then any one of them is independent of any event formed by sums, products, and complements of the others.

Example 2-17. Our experiment is the tossing of a coin twice. The certain event $\mathcal{S}$ consists of the four elements *hh, ht, th,* and *tt*. Assuming that the probabilities of the elementary events are given by

$$P\{hh\} = p^2 \qquad P\{ht\} = pq \qquad P\{th\} = pq \qquad P\{tt\} = q^2$$

where

$$p + q = 1$$

we shall show that the events

$$\mathcal{H}_1 = \{\text{heads at first tossing}\} = \{hh,ht\}$$
$$\mathcal{H}_2 = \{\text{heads at second tossing}\} = \{hh,th\}$$

are independent.

Since $\mathcal{H}_1$ has the two elements hh and ht, we conclude from (2-14) that

$$P\{\mathcal{H}_1\} = p^2 + pq = p$$

Similarly,

$$P\{\mathcal{H}_2\} = p^2 + pq = p$$

But $\mathcal{H}_1\mathcal{H}_2 = \{hh\}$; hence

$$P\{\mathcal{H}_1\mathcal{H}_2\} = p^2 = P\{\mathcal{H}_1\}P\{\mathcal{H}_2\}$$

Therefore the events $\mathcal{H}_1$, $\mathcal{H}_2$ are independent.

Example 2-18. Train X arrives at the station at random in the time interval $(0,T)$, stopping for a minutes. Train Y arrives independently in the same interval, stopping for b minutes.

1. Find the probability p_1 that X will arrive before Y.
2. Find the probability p_2 that the trains will meet.
3. Assuming that they met, find the probability p_3 that X arrived before Y.

We shall first determine the underlying probability space. An outcome of our experiment is the arrival of trains X and Y at specific time instances x and y. Thus

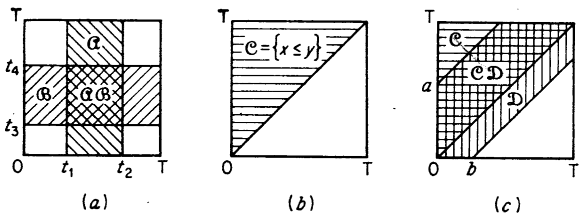

Fig. 2-13

the space $\mathcal{S}$ consists of all pairs of numbers (x,y) such that $0 \le x \le T$, $0 \le y \le T$ or, equivalently, of the points in the square of Fig. 2-13 (see also Example 2-2). The event

$$\{X \text{ arrives in the interval } (t_1,t_2)\} = \{t_1 \le x \le t_2\} = \mathcal{A}$$

is a vertical strip in Fig. 2-13a, and its probability is given by

$$P\{t_1 \le x \le t_2\} = \frac{t_2 - t_1}{T}$$

(this is the meaning of random arrival). The event

$$\{Y \text{ arrives in the interval } (t_3,t_4)\} = \{t_3 \le y \le t_4\} = \mathcal{B}$$

is a horizontal strip, and

$$P\{t_3 \le y \le t_4\} = \frac{t_4 - t_3}{T}$$

The product (intersection)

$$\{t_1 \le x \le t_2\}\{t_3 \le y \le t_4\} = \{t_1 \le x \le t_2,\ t_3 \le y \le t_4\}$$

of these two events is the double-shaded rectangle in the same figure. From the assumed independence of the arrival times we conclude that

$$P\{t_1 \le x \le t_2,\ t_3 \le y \le t_4\} = \frac{(t_2 - t_1)(t_4 - t_3)}{T^2} \tag{2-46}$$

Thus: The probability of a rectangle equals its area divided by T^2. An arbitrary event $\mathfrak{M}$ (two-dimensional Borel set) can be written as a limit of sums of nonoverlapping intervals; therefore, if its area equals $D_{\mathfrak{M}}$, we conclude from (2-46) and axiom III*a* that

$$P(\mathfrak{M}) = \frac{D_{\mathfrak{M}}}{T^2} \tag{2-47}$$

Our experiment is thus completely specified.

The event

$$\{X \text{ arrives before } Y\} = \{x \le y\} = \mathfrak{C}$$

is shown in Fig. 2-13*b*. Its area equals $T^2/2$; hence

$$p_1 = P\{x \le y\} = \tfrac{1}{2} \tag{2-48}$$

The area of the event

$$\{X \text{ meets } Y\} = \{-a \le x - y \le b\} = \mathfrak{D}$$

equals $(a + b)T - (a^2 + b^2)/2$, as it is easy to see from Fig. 2-13*b*; hence

$$p_2 = P\{-a \le x - y \le b\} = \frac{a + b}{T} - \frac{a^2 + b^2}{2T^2} \tag{2-49}$$

Finally since,

$$\{x \le y,\ -a \le x - y \le b\} = \{-a \le x - y \le 0\}$$

(double-shaded area in Fig. 2-13*c*), we conclude from (2-49) with $b = 0$ that

$$P\{x \le y,\ -a \le x - y \le b\} = \frac{a}{T} - \frac{a^2}{2T^2}$$

Therefore [see (2-25)]

$$p_3 = P\{x \le y \mid -a \le x - y \le b\} = \frac{2aT - a^2}{2(a + b)T - (a^2 + b^2)}$$

2-4. Summary

We shall now summarize the essential concepts of this chapter. In the theory of probability we deal with a *single* experiment $\mathfrak{E}$. The outcomes or elements of this experiment are well-defined objects ζ forming a set $\mathcal{S}$ called space or certain event. Events are various subsets $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, . . . of $\mathcal{S}$. Two events are mutually exclusive if they have no common elements. To each event we assign a probability P, that is, a number satisfying axioms I, II, and III. Given an event $\mathfrak{M}$ such that

$P(\mathfrak{M}) \neq 0$, the conditional probability of $\mathfrak{A}$, assuming $\mathfrak{M}$ is defined by

$$P(\mathfrak{A}|\mathfrak{M}) = \frac{P(\mathfrak{A}\mathfrak{M})}{P(\mathfrak{M})}$$

where $\mathfrak{A}\mathfrak{M}$ is the event consisting of the elements that are common to $\mathfrak{A}$ and $\mathfrak{M}$. Two events $\mathfrak{A}$ and $\mathfrak{B}$ are called independent if

$$P(\mathfrak{A}\mathfrak{B}) = P(\mathfrak{A})P(\mathfrak{B})$$

In the application of probability theory to physical problems one must clearly identify actual experimental outcomes with the elements ζ of $\mathcal{S}$. This identification is essential particularly in problems involving repeated trials. Numerical probabilities of various events are determined either as relative frequencies or through the classical definition; however, the underlying reasoning is outside the scope of probability theory.

Problems

2-1. $\mathcal{S}$ is the set of all real numbers x. With

$$\mathfrak{A} = \{2 \leq x \leq 5\} \qquad \mathfrak{B} = \{3 \leq x \leq 6\}$$

find $\mathfrak{A} + \mathfrak{B}$, $\mathfrak{A}\mathfrak{B}$, $\bar{\mathfrak{A}}$, $\mathfrak{A} - \mathfrak{B}$.

2-2. The set $\mathcal{S}$ contains a countable number of elements. It is known that any subset of $\mathcal{S}$ consisting of a single element is an event. Show that all subsets of $\mathcal{S}$ are events. Is the above necessarily true if the number of elements of $\mathcal{S}$ is not countable?

2-3. Show that if $\mathfrak{A}\mathfrak{B} = 0$, then $P(\mathfrak{A}) \leq P(\bar{\mathfrak{B}})$.

2-4. Show that, for any $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$,

$$P(\mathfrak{A} + \mathfrak{B} + \mathfrak{C}) = P(\mathfrak{A}) + P(\mathfrak{B}) + P(\mathfrak{C}) - P(\mathfrak{A}\mathfrak{B}) - P(\mathfrak{A}\mathfrak{C}) - P(\mathfrak{B}\mathfrak{C}) + P(\mathfrak{A}\mathfrak{B}\mathfrak{C})$$

Generalize.

2-5. (a) Show that, if $P(\mathfrak{A}) = P(\mathfrak{B}) = 1$, then $P(\mathfrak{A}\mathfrak{B}) = 1$. (b) If $\mathfrak{A} = \mathfrak{B}$ with probability 1, then $P(\mathfrak{A}) = P(\mathfrak{B}) = P(\mathfrak{A}\mathfrak{B})$.

2-6. The space $\mathcal{S}$ has six elements $\mathcal{S} = \{1,2,3,4,5,6\}$. Find the smallest field $\mathfrak{F}$ containing the sets $\{1\}$, $\{1,3,5\}$, $\{2,4,6\}$. *Hint.* $\mathfrak{F}$ consists of eight sets.

2-7. Is it possible that two events are independent and mutually exclusive?

2-8. Show that

$$P(\mathfrak{A}_1\mathfrak{A}_2 \cdots \mathfrak{A}_n) = P(\mathfrak{A}_1)P(\mathfrak{A}_2|\mathfrak{A}_1)P(\mathfrak{A}_3|\mathfrak{A}_1\mathfrak{A}_2) \cdots P(\mathfrak{A}_n|\mathfrak{A}_1\mathfrak{A}_2 \cdots \mathfrak{A}_{n-1})$$

2-9. (a) Show that if the events $\mathfrak{M}_1, \mathfrak{M}_2, \ldots, \mathfrak{M}_n$ are mutually exclusive and $\mathfrak{M} = \mathfrak{M}_1 + \mathfrak{M}_2 + \cdots + \mathfrak{M}_n$, then

$$P(\mathfrak{A}|\mathfrak{M}) = P(\mathfrak{A}|\mathfrak{M}_1)\frac{P(\mathfrak{M}_1)}{P(\mathfrak{M})} + \cdots + P(\mathfrak{A}|\mathfrak{M}_n)\frac{P(\mathfrak{M}_n)}{P(\mathfrak{M})}$$

(b) From the above follows that if

$$P(\mathfrak{A}|\mathfrak{M}_i) = P(\mathfrak{B}|\mathfrak{M}_i) \qquad i = 1, 2, \ldots, n$$

then

$$P(\mathcal{A}|\mathcal{M}) = P(\mathcal{B}|\mathcal{M})$$

Is this conclusion true if the events $\mathcal{M}_i$ are not mutually exclusive?

• **2-10.** Prove that, to establish the independence of n events, one needs $2^n - (n + 1)$ equations.

2-11. Show that if the events $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ are independent, then

$$P(\mathcal{B}_1\mathcal{B}_2 \cdots \mathcal{B}_n) = P(\mathcal{B}_1)P(\mathcal{B}_2) \cdots P(\mathcal{B}_n)$$

where

$$\mathcal{B}_i = \mathcal{A}_i \quad \text{or} \quad \bar{\mathcal{A}}_i \quad \text{or} \quad \mathcal{S}$$

2-12. The probability that two components 1 and 2 are good equals 0.9 and 0.8 respectively. Assuming independence, we conclude that the probability for both components to be good equals 0.9×0.8. Define an appropriate experiment and explain "independence" in the sense of (2-40).

Outline. The space $\mathcal{S}$ consists of four outcomes with

$$P\{g_1g_2\} = 0.72 \qquad P\{g_1b_2\} = 0.18 \qquad P\{b_1g_2\} = 0.08 \qquad P\{b_1b_2\} = 0.02$$

2-13. A telephone call occurs at random in the interval $(0,T)$. Find the probability $P\{t_0 \le t \le t_1 | t \ge t_0\}$ that it will occur in the interval (t_0,t_1), assuming that it did not occur before t_0.

• **2-14.** An electron is emitted from a substance after $t = 0$. The probability of its emission in the interval (t_1,t_2) is given by

$$P\{t_1 \le t \le t_2\} = \int_{t_1}^{t_2} \alpha(t)\,dt$$

Show that if (see page 36)

$$P\{0 \le t \le t_1\} = P\{t_0 \le t \le t_0 + t_1 | t \ge t_0\}$$

for every t_0 and t_1, then

$$\alpha(t) = ce^{-ct}$$

• **2-15.** Two shipments of parts are received. The first shipment contains 1,000 parts with 10% defective. The second shipment contains 2,000 parts with 5% defective. One shipment is selected at random: two parts are tested and found good. Find the probability (a posteriori) that the tested parts were selected from the first shipment.

2-16. A train and Mr. X arrive at the station at random and independently between 9 A.M. and 10 A.M. The train makes a 10-minute stop. Mr. X wants to board the train, but he waits only t_0 minutes and then leaves. Determine t_0 so that the probability that Mr. X will board the train equals 0.5.

3

Repeated Trials

As we have already noted, the theory of probability deals with the outcomes of a *single* experiment and with events formed from these outcomes. In the applications one often deals with two or more experiments or with the repeated performance of the same experiment. In order to analyze such problems, one must conceive a combined experiment whose outcomes are multiples of objects. The underlying set theoretical space is the so-called cartesian product formed with the elements of two or more sets. In the following discussion we shall define this product and apply it to repeated trials.

Most of the material of this chapter can be omitted at first reading without essential loss of continuity. The principal results will be rederived in subsequent chapters in terms of random variables.

3-1. Combined experiments

Suppose that we perform the following two experiments: Experiment $\mathfrak{G}_1$ is the rolling of a fair die; its space $\mathcal{S}_1$ consists of six elements

$$\mathcal{S}_1 = \{f_1, f_2, \ldots, f_6\}$$

and the probability of each elementary event equals $1/6$:

$$P_1\{f_i\} = 1/6 \qquad i = 1, 2, \ldots, 6$$

Experiment $\mathfrak{E}_2$ is the tossing of a fair coin; its space $\mathcal{S}_2$ consists of the elements "heads" and "tails":

$$\mathcal{S}_2 = \{h,t\}$$

and the probability of each elementary event equals $\frac{1}{2}$:

$$P_2\{h\} = P_2\{t\} = \tfrac{1}{2}$$

We now perform both experiments, and we want the probability that we get "two" on the die and "heads" on the coin. Assuming that the outcome of the coin tossing does not depend on what face of the die came up, as is obviously the case, we conclude that the events "two" and "heads" are independent and the unknown probability equals

$$\tfrac{1}{6} \cdot \tfrac{1}{2} = \tfrac{1}{12}$$

This conclusion is correct; however, it does not agree with our definition of independence as given in Sec. 2-3. According to this definition, two independent events were subsets of the *same* space $\mathcal{S}$; therefore, in order to fit the above conclusion into our language, we must introduce a new space $\mathcal{S}$ containing "two" and "heads" as subsets. This can be done as follows:

The two experiments $\mathfrak{E}_1$ and $\mathfrak{E}_2$ are considered as a new experiment $\mathfrak{E}$ (rolling of a die and tossing of a coin). A single outcome of this experiment is a pair of objects

$$(\zeta_1,\zeta_2)$$

where ζ_1 is one of the six faces of the die, and ζ_2 is heads or tails.† The resulting space $\mathcal{S}$ consists of the 12 elements (f_1,h), (f_2,h), (f_3,h), (f_4,h), (f_5,h), (f_6,h), (f_1,t), (f_2,t), (f_3,t), (f_4,t), (f_5,t), (f_6,t). Subsets of $\mathcal{S}$ are sets formed with these elements. "Two" is no longer an elementary event, but a subset of $\mathcal{S}$ consisting of two elements

$$\{(f_2,h),\ (f_2,t)\} = \{\text{two}\}$$

Similarly, "heads" is an event with six elements

$$\{(f_1,h),\ (f_2,h),\ (f_3,h),\ (f_4,h),\ (f_5,h),\ (f_6,h)\} = \{\text{heads}\}$$

To complete the specification of this experiment one must assign probabilities P to various events of the new space $\mathcal{S}$. Clearly, the event $\{\text{two}\}$ occurs if in the die rolling "two" shows, no matter what comes up in the coin tossing; hence

$$P\{\text{two}\} = P_1\{\text{two}\} = \tfrac{1}{6}$$

† Up to now, by ζ_i we meant a *single* element of a set $\mathcal{S}$. In order to avoid double subscripts, we shall in the following denote by ζ_i *any* element of a set $\mathcal{S}_i$. Whether ζ_i is one particular element or any one of many will be understood from the context.

Similarly,

$$P\{\text{heads}\} = P_2\{\text{heads}\} = \tfrac{1}{2}$$

Assuming independence of the events {two} and {heads} in the sense of Sec. 2-3, we conclude that, since the set product

$$\{\text{two}\}\{\text{heads}\}$$

has the single element (f_2,h), the probability of {two on the die and heads on the coin} equals

$$P\{\text{two}\}P\{\text{heads}\} = \tfrac{1}{6} \cdot \tfrac{1}{2}$$

in agreement with our heuristic conclusion. We turn now to the general case.

Very important

Cartesian product space. Given two sets $\mathcal{S}_1$ and $\mathcal{S}_2$ consisting of the elements ζ_1 and ζ_2, we take a single element ζ_1 from $\mathcal{S}_1$ and a single element ζ_2 from $\mathcal{S}_2$ (see footnote, page 48, for the dual meaning of ζ_1 and ζ_2):

$$\zeta_1 \in \mathcal{S}_1 \qquad \zeta_2 \in \mathcal{S}_2$$

The resulting ordered pair

$$(\zeta_1,\zeta_2)$$

can be considered as a single object. If we form all such pairs with every element of $\mathcal{S}_1$ and $\mathcal{S}_2$, we obtain a new collection of objects (ζ_1,ζ_2), that is, a new set $\mathcal{S}$. This set we call the cartesian product of $\mathcal{S}_1$ and $\mathcal{S}_2$ and shall write it in the form

$$\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2$$

Example 3-1. With

$$\mathcal{S}_1 = \{\text{car, apple, bird}\} \qquad \mathcal{S}_2 = \{\text{heads, tails}\}$$

we have

$$\mathcal{S} = \{(\text{car, heads}), (\text{car, tails}), (\text{apple, heads}), (\text{apple, tails}), (\text{bird, heads}), (\text{bird, tails})\}$$

It is easy to see that if the sets $\mathcal{S}_1$ and $\mathcal{S}_2$ have finitely many elements m and n, respectively, then $\mathcal{S}$ has mn elements.

The cartesian product can be defined even if the sets $\mathcal{S}_1$ and $\mathcal{S}_2$ are identical.

Example 3-2. With

$$\mathcal{S}_1 = \{\text{heads, tails}\} \qquad \mathcal{S}_2 = \mathcal{S}_1$$

we have

$$\mathcal{S} = \{(\text{heads, heads}), (\text{heads, tails}), (\text{tails, heads}), (\text{tails, tails})\}$$

In the preceding example the element (heads, tails) is different from the element (tails, heads). Thus the elements of a cartesian product are *ordered* pairs of objects.

If $\mathcal{A}$ is a subset of $\mathcal{S}_1$ and $\mathcal{B}$ is a subset of $\mathcal{S}_2$, then the set

$$\mathcal{C} = \mathcal{A} \times \mathcal{B}$$

consisting of all pairs (ζ_1,ζ_2) such that

$$\zeta_1 \in \mathcal{A} \qquad \zeta_2 \in \mathcal{B}$$

is a subset of $\mathcal{S}$. It is easy to see that $\mathcal{A} \times \mathcal{B}$ can be written as a product of the sets $\mathcal{A} \times \mathcal{S}_2$ and $\mathcal{S}_1 \times \mathcal{B}$:

$$\mathcal{A} \times \mathcal{B} = (\mathcal{A} \times \mathcal{S}_2)(\mathcal{S}_1 \times \mathcal{B}) \tag{3-1}$$

Example 3-3. If $\mathcal{S}_1$ is the set of all real numbers x, and $\mathcal{S}_2$ the set of all real numbers y,

$$\mathcal{S}_1 = \{\text{all real } x\} \qquad \mathcal{S}_2 = \{\text{all real } y\}$$

then

$$\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2$$

is the set of all pairs of numbers (x,y) or, equivalently, the set of all points in the xy plane. The set

$$\{x_1 \leq x \leq x_2\} \times \{y_1 \leq y \leq y_2\}$$

is a rectangle, and the sets

$$\{x_1 \leq x \leq x_2\} \times \mathcal{S}_2 \qquad \mathcal{S}_1 \times \{y_1 \leq y \leq y_2\}$$

vertical and horizontal strips, respectively. Finally,

$$\{x = x_1\} \times \mathcal{S}_2$$

is a vertical line.

The probability space of two experiments. We are given two experiments $\mathfrak{S}_1$ and $\mathfrak{S}_2$ with outcomes ζ_1 and ζ_2 forming the spaces $\mathcal{S}_1$ and $\mathcal{S}_2$. The probabilities of various events of these two experiments we denote by $P_1(\ )$ and $P_2(\ )$, respectively. The performance of both experiments can be considered as a new experiment $\mathfrak{S}$ whose outcomes are all pairs of objects

$$(\zeta_1,\zeta_2)$$

forming the space

$$\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2$$

This combined experiment $\mathfrak{S}$ we shall write in the form

$$\mathfrak{S} = \mathfrak{S}_1 \times \mathfrak{S}_2$$

In this experiment, we consider as events all subsets of $\mathcal{S}$ of the form

$$\mathcal{A} \times \mathcal{B} \tag{3-2}$$

where $\mathcal{A}$ and $\mathcal{B}$ are events of the experiments $\mathfrak{S}_1$ and $\mathfrak{S}_2$, respectively. And since events must form a field, all sums and products of sets of the form (3-2) are also events.

To complete the specification of the experiment $\mathfrak{S}$, we must define probabilities of its various events. We first observe that if $\mathcal{A}$ is an event of $\mathfrak{S}_1$ with probability $P_1(\mathcal{A})$, then the probability $P(\mathcal{A} \times \mathcal{S}_2)$ of the event $\mathcal{A} \times \mathcal{S}_2$ in the experiment $\mathfrak{S}$ must be such that

$$P(\mathcal{A} \times \mathcal{S}_2) = P_1(\mathcal{A}) \tag{3-3}$$

because $\mathcal{A} \times \mathcal{S}_2$ consists of all elements (ζ_1,ζ_2) such that

$$\zeta_1 \in \mathcal{A}$$

and ζ_2 is any element of $\mathcal{S}_2$; hence $\mathcal{A} \times \mathcal{S}_2$ occurs in the experiment $\mathfrak{S}$ if $\mathcal{A}$ occurs in the experiment $\mathfrak{S}_1$ no matter what the outcome of the experiment $\mathfrak{S}_2$. Therefore the probability of $\mathcal{A} \times \mathcal{S}_2$ in $\mathfrak{S}$ must equal the probability of $\mathcal{A}$ in $\mathfrak{S}_1$. One concludes, similarly, that

$$P(\mathcal{S}_1 \times \mathcal{B}) = P_2(\mathcal{B}) \tag{3-4}$$

The relationships (3-3) and (3-4) do not, of course, suffice to determine probabilities of all events of the combined experiment $\mathfrak{S}$. Thus the probability of the event $\mathcal{A} \times \mathcal{B}$, that is, of the occurrence of the event $\mathcal{A}$ in $\mathfrak{S}_1$ and of $\mathcal{B}$ in $\mathfrak{S}_2$, cannot be determined from $P_1(\mathcal{A})$ and $P_2(\mathcal{B})$. It must be given as additional information about the two experiments.

In many important cases, one assumes that

$$P(\mathcal{A} \times \mathcal{B}) = P_1(\mathcal{A})P_2(\mathcal{B})$$

If this is the case for any $\mathcal{A}$ and $\mathcal{B}$, then we say that the experiments $\mathfrak{S}_1$ and $\mathfrak{S}_2$ are independent. In the following discussion we shall consider product spaces resulting from independent experiments.

Independent Experiments. We assume that, in the experiment $\mathfrak{S}$, the events

$$(\mathcal{A} \times \mathcal{S}_2) \qquad \text{and} \qquad (\mathcal{S}_1 \times \mathcal{B})$$

are independent (in the sense of Sec. 2-3) for every $\mathcal{A}$ and $\mathcal{B}$. Since [see (3-1)]

$$(\mathcal{A} \times \mathcal{S}_2)(\mathcal{S}_1 \times \mathcal{B}) = \mathcal{A} \times \mathcal{B}$$

and

$$P(\mathcal{A} \times \mathcal{S}_2) = P_1(\mathcal{A}) \qquad P(\mathcal{S}_1 \times \mathcal{B}) = P_2(\mathcal{B})$$

we conclude [see (2-40)] that

$$P(\mathcal{A} \times \mathcal{B}) = P_1(\mathcal{A})P_2(\mathcal{B}) \tag{3-5}$$

Thus the probability of the event $\mathcal{A} \times \mathcal{B}$ of the combined experiment $\mathfrak{S} = \mathfrak{S}_1 \times \mathfrak{S}_2$ is determined from the probabilities of the events $\mathcal{A}$ and $\mathcal{B}$ in the experiments $\mathfrak{S}_1$ and $\mathfrak{S}_2$. And since all events in $\mathfrak{S}$ are sums and products of sets of the form $\mathcal{A} \times \mathcal{B}$, we conclude from (3-5) and the axioms that experiment $\mathfrak{S}$ is completely specified in terms of P_1 and P_2.

If $\{\zeta_1\}$ and $\{\zeta_2\}$ are two elementary events of $\mathfrak{E}_1$ and $\mathfrak{E}_2$, then

$$\{(\zeta_1,\zeta_2)\}$$

is an elementary event of $\mathfrak{E}$ and

$$P\{(\zeta_1,\zeta_2)\} = P_1\{\zeta_1\}P_2\{\zeta_2\} \tag{3-6}$$

From now on, in order to simplify the notation, we shall write an element of the product space $\mathcal{S}$, not as (ζ_1,ζ_2), but merely as $\zeta_1\zeta_2$.

Example 3-4. A box contains 10 white and 5 red balls; a second box contains 20 white and 20 red balls. A ball is drawn from each box. What is the probability that the first will be white and the second red?

The above problem can be considered as a combined experiment. Experiment $\mathfrak{E}_1$ is the drawing from the first box, and experiment $\mathfrak{E}_2$ the drawing from the second box. The space $\mathcal{S}_1$ has 15 elements, namely, the 10 white and the 5 red balls. The event

$$\mathcal{W}_1 = \{\text{white balls in box 1}\}$$

has 10 elements, and its probability is [see (2-16)]

$$P_1\{\mathcal{W}_1\} = {}^{10}\!/_{15}$$

The space $\mathcal{S}_2$ has 40 elements; the event

$$\mathcal{R}_2 = \{\text{red balls in box 2}\}$$

has 20 elements, and its probability is

$$P_2(\mathcal{R}_2) = {}^{20}\!/_{40}$$

The space $\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2$ of the experiment $\mathfrak{E} = \mathfrak{E}_1 \times \mathfrak{E}_2$ has

$$40 \times 15 = 600$$

elements, namely, all possible pairs of balls that one can draw. We want the probability of the event $\mathcal{W}_1 \times \mathcal{R}_2$ = {white from the first and red from the second box}. Assuming independence of the two experiments, we obtain from (3-5)

$$P(\mathcal{W}_1 \times \mathcal{R}_2) = P_1(\mathcal{W}_1)P_2(\mathcal{R}_2) = {}^1\!/_3$$

Comment. In certain problems, the space $\mathcal{S}_2$ of the experiment $\mathfrak{E}_2$ is not given in advance, but it depends on the outcome of the experiment $\mathfrak{E}_1$ (for example, consecutive drawings from a box without replacement). In such cases it is best to define the underlying space, not as a cartesian product, but directly as a set of ordered pairs (see Example 2-13).

The most important example of a cartesian product is the space resulting from the repeated trials of the same experiment. We give here a simple illustration and shall elaborate in the next section.

Example 3-5. We are given a coin with probability of heads or tails equal p and q, respectively, where

$$p + q = 1$$

We toss the coin twice; what is the probability space of the resulting experiment

$$\mathfrak{E} = \mathfrak{E}_1 \times \mathfrak{E}_2$$

In the above, $\mathfrak{E}_1$ is the experiment of the first tossing with

$$\mathcal{S}_1 = \{h,t\} \qquad P_1\{h\} = p \qquad P_1\{t\} = q$$

and $\mathfrak{E}_2 = \mathfrak{E}_1$ is the experiment of the second tossing.†

The space

$$\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2$$

consists of the four elements

$$hh,\ ht,\ th,\ tt$$

Assuming that the two tossings are independent, we have from (3-5)

$$P\{hh\} = P_1\{h\}P_2\{h\} = p^2$$

Similarly,

$$P\{ht\} = pq \qquad P\{th\} = qp \qquad P\{tt\} = q^2$$

With $\mathcal{H}_1$ the event {heads at the first tossing}, we have

$$\mathcal{H}_1 = \{hh,ht\}$$

Hence [see (2-14)]

$$P(\mathcal{H}_1) = P\{hh\} + P\{ht\} = p^2 + pq = p$$

This follows also from

$$\mathcal{H}_1 = \{h\} \times \mathcal{S}_2$$

and (3-5).

Generalization. Given n sets

$$\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_n$$

we define as their cartesian product

$$\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2 \times \cdots \times \mathcal{S}_n \tag{3-7}$$

the set whose elements are all ordered n-tuples

$$\zeta_1\zeta_2 \ldots \zeta_n$$

where ζ_i is an element of the set $\mathcal{S}_i$.

Given n experiments

$$\mathfrak{E}_i\colon (\mathcal{S}_i,\mathfrak{F}_i,P_i) \qquad i = 1, 2, \ldots, n$$

by

$$\mathfrak{E} = \mathfrak{E}_1 \times \mathfrak{E}_2 \times \cdots \times \mathfrak{E}_n \tag{3-8}$$

we shall mean the combined experiment resulting from the performance of all the given experiments. The space $\mathcal{S}$ of $\mathfrak{E}$ is the cartesian product

† We say that two experiments $\mathfrak{E}_1$ and $\mathfrak{E}_2$ are equal if $\mathcal{S}_1 = \mathcal{S}_2$, $\mathfrak{F}_1 = \mathfrak{F}_2$, $P_1 = P_2$.

(3-7) of the spaces $\mathcal{S}_i$, and its events are all sets of the form

$$\mathcal{A}_1 \times \mathcal{A}_2 \times \cdots \times \mathcal{A}_n \tag{3-9}$$

where $\mathcal{A}_i \subset \mathcal{S}_i$, and their sums and products.

If the experiments are independent, then the probability $P(\mathcal{A})$ of the event $\mathcal{A}_1 \times \mathcal{A}_2 \times \cdots \times \mathcal{A}_n$ is given by

$$P(\mathcal{A}_1 \times \mathcal{A}_2 \times \cdots \times \mathcal{A}_n) = P_1(\mathcal{A}_1)P_2(\mathcal{A}_2) \cdots P_n(\mathcal{A}_n) \tag{3-10}$$

where $P_i(\mathcal{A}_i)$ is the probability of the event $\mathcal{A}_i$ in the experiment $\mathfrak{G}_i$.

One can similarly define a combined experiment resulting from the performance of an infinite number of other experiments.

Example 3-6. A coin as in Example 3-5 is tossed n times. We shall determine the probability space of the resulting combined experiment

$$\mathfrak{G} = \mathfrak{G}_1 \times \mathfrak{G}_2 \times \cdots \times \mathfrak{G}_n$$

where

$$\mathfrak{G}_1 = \mathfrak{G}_2 = \cdots = \mathfrak{G}_n$$

The space

$$\mathcal{S} = \mathcal{S}_1 \times \mathcal{S}_2 \times \cdots \times \mathcal{S}_n$$

consists of 2^n elements, namely, all ordered n-tuples $\zeta_1\zeta_2 \ldots \zeta_n$, where $\zeta_i = h$ or t. Since [see (3-6)] $P\{\zeta_1 \ldots \zeta_n\} = P_1\{\zeta_1\}P_2\{\zeta_2\} \cdots P_n\{\zeta_n\}$ where

$$P_i\{\zeta_i\} = \begin{cases} p & \text{if } \zeta_i = h \\ q & \text{if } \zeta_i = t \end{cases}$$

is the probability of heads or tails in the ith experiment, we conclude that, if the elementary event

$$\{\zeta_1\zeta_2 \ldots \zeta_n\}$$

has k heads and n-k tails, then

$$P\{\zeta_1\zeta_2 \ldots \zeta_n\} = p^kq^{n-k} \tag{3-11}$$

Thus

$$P\{hh \ldots h\} = p^n \qquad P\{th \ldots h\} = p^{n-1}q \qquad \text{and so on}$$

Notice that (3-11) is not the probability of {k heads in n tossings}, but of the event {k heads in a specific order}. We also remark that the event

$$\mathcal{H}_1 = \{\text{heads at first tossing}\}$$

is not an elementary event of the experiment $\mathfrak{G}$, but an event consisting of 2^{n-1} outcomes

$$\zeta_1\zeta_2 \ldots \zeta_n$$

where

$$\zeta_1 = h \qquad \text{and} \qquad \zeta_i = h \text{ or } t \qquad \text{for } i > 1$$

Since

$$\mathcal{H}_1 = \{h\} \times \mathcal{S}_2 \times \cdots \times \mathcal{S}_n$$

we conclude from (3-10) that

$$P(\mathcal{H}_1) = P_1\{h\}P_2(\mathcal{S}_2) \cdots P_n(\mathcal{S}_n) = p$$

Similarly, with

$$\mathfrak{K}_i = \{\text{heads at } i\text{th tossing}\} \qquad \mathfrak{T}_i = \{\text{tails at } i\text{th tossing}\}$$

we have

$$P(\mathfrak{K}_i) = p \qquad P(\mathfrak{T}_i) = q$$

If

$$p = q = \tfrac{1}{2}$$

then

$$P\{\zeta_1\zeta_2 \ldots \zeta_n\} = 1/2^n \tag{3-12}$$

This conclusion can be reached also from (2-16) because there are 2^n elementary events with the same probability. Similarly, the event $\mathfrak{K}_1$ consists of 2^{n-1} elements, hence its probability equals $2^{n-1}/2^n = \frac{1}{2}$.

Sum of two or more spaces. Given two sets $\mathcal{S}_1$ and $\mathcal{S}_2$, we know that

$$\mathcal{S} = \mathcal{S}_1 + \mathcal{S}_2 \tag{3-13}$$

is the set consisting of all elements that are in the sets $\mathcal{S}_1$ or $\mathcal{S}_2$.

Suppose now that $\mathcal{S}_1$ is the set of outcomes of an experiment $\mathfrak{E}_1$ and $\mathcal{S}_2$ is the set of outcomes of another experiment $\mathfrak{E}_2$. We shall show that if $\mathcal{S}_1\mathcal{S}_2 = 0$, then the sum $\mathcal{S}$ can be considered as the space of another experiment $\mathfrak{E}$ written in the form

$$\mathfrak{E} = \mathfrak{E}_1 + \mathfrak{E}_2$$

and defined as follows:

We decide at random to perform either experiment $\mathfrak{E}_1$ or experiment $\mathfrak{E}_2$. After this decision is made, the selected experiment is then performed.

Clearly, the outcomes of $\mathfrak{E}$ are all the elements of $\mathcal{S}_1$ and of $\mathcal{S}_2$; that is, its space $\mathcal{S}$ is the sum (3-13). In the new experiment $\mathfrak{E}$, we consider as events all sets of the form

$$\mathcal{A} = \mathcal{A}_1 + \mathcal{A}_2 \tag{3-14}$$

where $\mathcal{A}_1$ and $\mathcal{A}_2$ are events of $\mathfrak{E}_1$ and $\mathfrak{E}_2$: $\mathcal{A}_1 \subset \mathcal{S}_1$, $\mathcal{A}_2 \subset \mathcal{S}_2$. The set $\mathcal{S}_1$ is no longer the certain event, but a subset of $\mathcal{S}$, and it occurs iff experiment $\mathfrak{E}_1$ is performed. Similarly for $\mathcal{S}_2$.

We shall now determine the probabilities $P(\mathcal{A})$ of the events $\mathcal{A}$ of the experiment $\mathfrak{E}$ in terms of the probabilities $P_1(\mathcal{A}_1)$ and $P_2(\mathcal{A}_2)$ of the two experiments $\mathfrak{E}_1$ and $\mathfrak{E}_2$. We maintain that, to determine P, it suffices to know P_1, P_2 and the probabilities

$$P(\mathcal{S}_1) \qquad \text{and} \qquad P(\mathcal{S}_2)$$

of the events $\mathcal{S}_1$ and $\mathcal{S}_2$, that is, the probabilities that we selected $\mathfrak{E}_1$ or $\mathfrak{E}_2$. Indeed, suppose that

$$\mathcal{A}_1 \subset \mathcal{S}_1$$

Then

$$P(\mathcal{A}_1|\mathcal{S}_1) = P_1(\mathcal{A}_1) \tag{3-15}$$

because the conditional probability of $\mathcal{A}_1$, assuming $\mathcal{S}_1$ ($\mathfrak{E}_1$ was selected), equals the probability of occurrence of $\mathcal{A}_1$ in the experiment $\mathfrak{E}_1$. From the above and (2-25) we conclude that

$$P(\mathcal{A}_1) = P(\mathcal{A}_1|\mathcal{S}_1)P(\mathcal{S}_1) = P_1(\mathcal{A}_1)P(\mathcal{S}_1) \tag{3-16}$$

Similarly if

$$\mathcal{A}_2 \subset \mathcal{S}_2$$

then

$$P(\mathcal{A}_2) = P(\mathcal{A}_2|\mathcal{S}_2)P(\mathcal{S}_2) = P_2(\mathcal{A}_2)P(\mathcal{S}_2) \tag{3-17}$$

Finally, to determine the probability $P(\mathcal{A})$ of an arbitrary event $\mathcal{A}$, we observe that

$$\mathcal{A} = \mathcal{A}\mathcal{S} = \mathcal{A}\mathcal{S}_1 + \mathcal{A}\mathcal{S}_2$$

because $\mathcal{S} = \mathcal{S}_1 + \mathcal{S}_2$. But the sets

$$\mathcal{A}\mathcal{S}_1 = \mathcal{A}_1 \qquad \mathcal{A}\mathcal{S}_2 = \mathcal{A}_2$$

are subsets of $\mathcal{S}_1$ and $\mathcal{S}_2$, respectively; therefore their probabilities can be found from (3-16) and (3-17). And since these events are mutually exclusive, we conclude from (2-5) that

$$P(\mathcal{A}) = P(\mathcal{A}_1) + P(\mathcal{A}_2) \tag{3-18}$$

Example 3-7. A box contains 10 white and 5 red balls; a second box contains 20 white and 20 red balls. We select "at random" one box and pick a ball. What is the probability that this ball is white?

Denoting by $\mathcal{S}_1$ and $\mathcal{S}_2$ the spaces of the experiments $\mathfrak{E}_1$ and $\mathfrak{E}_2$ as defined in Example 3-4, we observe that the space $\mathcal{S} = \mathcal{S}_1 + \mathcal{S}_2$ of the experiment $\mathfrak{E} = \mathfrak{E}_1 + \mathfrak{E}_2$ consists of 15 + 40 elements. We want $P(\mathcal{W})$, where $\mathcal{W}$ is the event consisting of all white balls. Clearly,

$$P(\mathcal{S}_1) = P(\mathcal{S}_2) = 1/2$$

(random selection of a box), and with $\mathcal{W}_1$ and $\mathcal{W}_2$ the white balls in the first and second boxes, we have

$$\mathcal{W} = \mathcal{W}_1 + \mathcal{W}_2$$

But

$$P_1(\mathcal{W}_1) = 10/15 \qquad P_2(\mathcal{W}_2) = 20/40$$
$$\mathcal{W}_1 \subset \mathcal{S}_1 \qquad \mathcal{W}_2 \subset \mathcal{S}_2$$

Hence [see (3-16) to (3-18)]

$$P(\mathcal{W}) = 10/15 \cdot 1/2 + 20/40 \cdot 1/2 = 7/12$$

Dual meaning of repeated trials. In the theory of probability the notion of repeated trials has two basically different interpretations. The first (physical) is the approximate relationship (1-1) between the probability $P(\mathcal{A})$ of an event $\mathcal{A}$ in an experiment $\mathfrak{E}$ and the relative frequency of its occurrence. The second (conceptual) is the creation of a new experiment

$$\mathfrak{E} \times \mathfrak{E} \times \cdots \times \mathfrak{E}$$

whose underlying space is the cartesian product

$$\mathcal{S} \times \mathcal{S} \times \cdots \times \mathcal{S}$$

where $\mathcal{S}$ is the space of $\mathfrak{S}$.

Consider, for example, the experiment with a fair coin. How do we interpret its repeated tossings?

First Interpretation (Physical). Our experiment $\mathfrak{S}$ is the *single* tossing of the coin. Its space $\mathcal{S}$ consists of the two elements "heads" and "tails." The probability of each elementary event equals $\frac{1}{2}$. A trial is the tossing of the coin *once.*

If we now toss the coin n times and n is sufficiently large, then

$$\frac{n_h}{n} \simeq \frac{1}{2}$$

where n_h is the number of heads. Thus the first meaning of repeated trials is the above imprecise statement relating probabilities with observed frequencies.

Second Interpretation (Conceptual). We consider as our experiment the tossing of the coin n times, where n is any number large or small. A single outcome of this experiment is a specific sequence of n heads and tails. The total number of outcomes equals 2^n, and the probability of each equals $\frac{1}{2^n}$. A trial is the tossing of the coin n times. The event $\mathcal{H}_1 = \{\text{heads at first tossing}\}$ consists of 2^{n-1} outcomes, namely, all sequences

$$hth \ldots h$$

starting with h, and its probability equals $2^{n-1}/2^n = \frac{1}{2}$. All statements are precise and in the form of probabilities.

If one wants to give a physical interpretation to these probabilities [in the sense of (1-1)], one must repeat the n tossing of the coin a large number of times. The probability of the various events of our experiment is then approximately equal to their relative frequency of occurrence.

Comment. On page 4 we commented that the probability $P(\mathcal{A})$ of a physical event $\mathcal{A}$ equals about $n_{\mathcal{A}}/n$ for large enough n and that we obtain approximately the same ratio if we consider only a sub-sequence of the performed experiments (no betting system can beat the roulette). This requirement is necessary only for the first interpretation of repeated trials. It would be true in the second interpretation only if we assumed independence of the consecutive trials.

3-2. Bernoulli trials

In this section we shall determine the probability that an event occurs k times in n independent trials of an experiment $\mathfrak{S}$. This basic problem is essentially the same as the problem of obtaining k heads in n tossings of a coin, and we shall start the discussion with the coin-tossing experiment.

We shall make use of the following well-known result from combinatorial analysis:

Given n objects

$$a_1, a_2, \ldots, a_n \tag{3-19}$$

we form all possible groups of k of these objects where a group is distinct from another if it differs by at least one object. Denoting by $N_n(k)$ the total number of these groups, we can easily show that

$$N_n(k) = \binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{1\cdot 2\cdots k} = \frac{n!}{k!(n-k)!} \tag{3-20}$$

As an illustration, suppose that $n = 4$ and $k = 2$; then

$$N_4(2) = \frac{4!}{2!2!} = 6$$

i.e., using the four objects a_1, a_2, a_3, a_4, we can form the six pairs

$$a_1a_2 \qquad a_1a_3 \qquad a_1a_4 \qquad a_2a_3 \qquad a_2a_4 \qquad a_3a_4$$

A problem related to the above is the following: We have k identical objects of a certain kind b and $n - k$ identical objects of another kind c. With these objects we form all possible sequences

$$d_1d_2 \ldots d_n \tag{3-21}$$

where d_i is b or c. We maintain that the total number of such sequences equals $N_n(k)$. Indeed, if we identify the n positions in (3-21) with the n objects (3-19) and the positions occupied by the objects b as one group of k objects, we see that the number of sequences of the form (3-21) equals the total number of ways of taking from n objects k at a time.

Example 3-8. A coin with

$$P\{h\} = p \qquad P\{t\} = q \qquad p + q = 1$$

is tossed n times. Find the probability that heads will show k times.

Denoting the unknown probability by $p_n(k)$, we shall show that

$$p_n(k) = P\{k \text{ heads in } n \text{ tossings}\} = \binom{n}{k} p^k q^{n-k} \tag{3-22}$$

Proof. Our experiment is the n tossing of the coin. A single outcome of this experiment is a particular sequence of heads and tails. The event

$$\{k \text{ heads in } n \text{ tossings}\}$$

consists of all sequences

$$\zeta_1\zeta_2 \ldots \zeta_n$$

containing k heads and $n - k$ tails. The total number of such sequences equals [see (3-20)]

$$N_n(k) = \binom{n}{k}$$

As we have shown in Example 3-6, the probability of a specific sequence of k heads and $n - k$ tails equals

$$p^k q^{n-k}$$

Therefore [see (2-14)]

$$P\{k \text{ heads in } n \text{ tossings}\} = N_n(k) p^k q^{n-k}$$

and (3-22) is thus proved.

Success or failure of an event $\mathcal{A}$ in n independent trials. We consider now our main problem. Suppose that the probability of an event $\mathcal{A}$ in an experiment $\mathfrak{S}$ equals $P(\mathcal{A})$. We perform this experiment n times; i.e., we perform the combined experiment

$$\mathfrak{S} \times \mathfrak{S} \times \cdots \times \mathfrak{S}$$

and we want to determine the probability $p_n(k)$ that the event $\mathcal{A}$ occurs exactly k times in any order.

With

$$P(\mathcal{A}) = p \qquad P(\bar{\mathcal{A}}) = q \qquad p + q = 1$$

we shall show that

$$p_n(k) = P\{\mathcal{A} \text{ occurs } k \text{ times}\} = \binom{n}{k} p^k q^{n-k} \tag{3-23}$$

as in the coin-tossing example.

Proof. The event $\{\mathcal{A}$ occurs k times in a specific order$\}$ can be written as a cartesian product

$$\mathcal{B}_1 \times \mathcal{B}_2 \times \cdots \times \mathcal{B}_n$$

where k of the events $\mathcal{B}_i$ equal $\mathcal{A}$, and the remaining $n - k$ equal $\bar{\mathcal{A}}$. As we know from (3-10), its probability is given by

$$P_1(\mathcal{B}_1)P_2(\mathcal{B}_2) \cdots P_n(\mathcal{B}_n) = p^k q^{n-k} \tag{3-24}$$

because

$$P_i(\mathcal{B}_i) = \begin{cases} p & \text{if } \mathcal{B}_i = \mathcal{A} \\ q & \text{if } \mathcal{B}_i = \bar{\mathcal{A}} \end{cases}$$

Thus

$$P\{\mathcal{A} \text{ occurs } k \text{ times in a specific order}\} = p^k q^{n-k} \tag{3-25}$$

Clearly, the event

$$\{\mathcal{A} \text{ occurs } k \text{ times in any order}\}$$

is the sum of the events

$$\{\mathcal{A} \text{ occurs } k \text{ times in a specific order}\}$$

These events are mutually exclusive, and their number equals

$$N_n(k) = \binom{n}{k}$$

as it is easy to see. Hence [see (2-10)]

$$P\{\mathcal{A} \text{ occurs } k \text{ times}\} = N_n(k)p^k q^{n-k}$$

and (3-23) is thus proved.

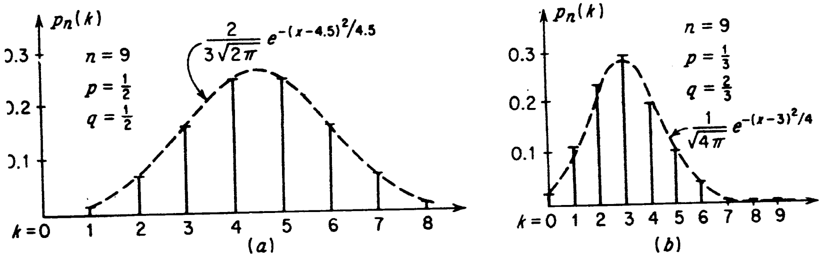

Fig. 3-1

In Fig. 3-1 we show the probabilities $p_n(k)$ for $k = 0, 1, \ldots, n$. In Fig. 3-1*a* we assume

$$n = 9 \qquad p = \tfrac{1}{2} \qquad q = \tfrac{1}{2}$$

and in Fig. 3-1*b*,

$$n = 9 \qquad p = \tfrac{1}{3} \qquad q = \tfrac{2}{3}$$

Example 3-9. A fair die is rolled five times. Find the probability $p_5(2)$ that "six" will come up twice.

With $\mathcal{A} = \{f_6\}$ the event "six" in the single rolling of the die, we have

$$P(\mathcal{A}) = \tfrac{1}{6} \qquad P(\bar{\mathcal{A}}) = \tfrac{5}{6} \qquad n = 5 \qquad k = 2$$

Therefore [see (3-23)]

$$p_5(2) = \frac{5!}{2!3!}(\tfrac{1}{6})^2(\tfrac{5}{6})^3$$

Example 3-10. A pair of fair dice is rolled four times. What is the probability that "seven" (sum of appearing faces) will not show at all?

We denote by $\mathfrak{G}$ the experiment of the single rolling of the two dice. Its space $\mathcal{S}$ consists of the 36 elements

$$f_i f_j \qquad i, j = 1, 2, \ldots, 6$$

The event $\mathcal{A} = \{\text{seven}\}$ consists of the six elements

$$f_1f_6,\ f_6f_1,\ f_2f_5,\ f_5f_2,\ f_3f_4,\ f_4f_3$$

Hence its probability equals $\tfrac{6}{36}$. Thus

$$P(\mathcal{A}) = \tfrac{1}{6} \qquad P(\bar{\mathcal{A}}) = \tfrac{5}{6}$$

The four rollings of the two dice form the combined experiment

$$\mathfrak{E} \times \mathfrak{E} \times \mathfrak{E} \times \mathfrak{E}$$

With

$$n = 4 \qquad \text{and} \qquad k = 0$$

we obtain from (3-23)

$$p_4(0) = (5/6)^4$$

Example 3-11. We place "at random" n points in the $(0,T)$ interval. What is the probability that k of these points will lie in the interval (t_1,t_2) (Fig. 3-2)?

This example can be considered as a problem in repeated trials. With $\mathfrak{E}$ the experiment of placing a *single* point in the $(0,T)$ interval and $\mathcal{A}$ the event {the point is placed in the (t_1,t_2) interval}, we have, as in Example 2-8,

$$P(\mathcal{A}) = \frac{t_2 - t_1}{T}$$

Our problem can be viewed as a combined experiment

$$\mathfrak{E} \times \mathfrak{E} \times \cdots \times \mathfrak{E} \qquad n \text{ times}$$

In this experiment, the event {k points in the (t_1,t_2) interval} equals the event {$\mathcal{A}$

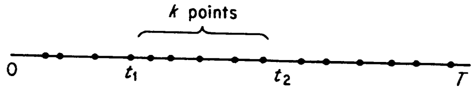

Fig. 3-2

occurs k times}; therefore its probability is given by (3-23):

$$P\{k \text{ points in the } (t_1,t_2) \text{ interval}\} = \binom{n}{k} p^k q^{n-k} \tag{3-26}$$

where

$$p = \frac{t_2 - t_1}{T} \qquad q = 1 - p$$

Example 3-12. A system contains n components. The probability that a particular component will fail in the interval $(0,t_1)$ is given by

$$\int_0^{t_1} \alpha(t)\, dt$$

where

$$\alpha(t) \geq 0 \qquad \int_0^{\infty} \alpha(t)\, dt = 1$$

What is the probability that exactly k components will fail prior to t_1?

With

$$\mathcal{A} = \{\text{one particular component fails in the } (0,t_1) \text{ interval}\}$$

we have

$$P(\mathcal{A}) = \int_0^{t_1} \alpha(t)\, dt$$

and as in Example 3-11,

$$P\{k \text{ failures prior to } t_1\} = \binom{n}{k} p^k q^{n-k} \qquad \text{where } p = \int_0^{t_1} \alpha(t)\, dt$$

Notice that the probability that only one (any one) component will fail before $t = t_1$ is given

$$\binom{n}{1} pq^{n-1}$$

and not by $P(\alpha)$.

Most Likely Number of Successes. For a fixed n, the quantity $p_n(k)$ in (3-23) depends on k. We maintain that it is maximum for k equal to about np. More precisely, if $(n+1)p$ is not an integer, then the value $k_{\max}$ of k for which $p_n(k)$ is maximum is given by

$$k_{\max} = [(n+1)p] \tag{3-27}$$

where by $[(n+1)p]$ we mean the largest integer that is smaller than $(n+1)p$. If $(n+1)p$ is an integer, then $p_n(k)$ is maximum for

$$k_{m_1} = (n+1)p \qquad \text{and} \qquad k_{m_2} = (n+1)p - 1 = np - q \tag{3-28}$$

For example, if

$$n = 10 \qquad p = 1/3$$

then

$$k_{\max} = [11/3] = 3$$

If

$$n = 11 \qquad p = 1/2$$

then

$$k_{m_1} = 12/2 = 6 \qquad \text{and} \qquad k_{m_2} = 5$$

The value of k that makes $p_n(k)$ maximum is called *most likely*.

Proof. The ratio of two consecutive values $p_n(k-1)$ and $p_n(k)$ is given by

$$\frac{p_n(k-1)}{p_n(k)} = \frac{\dfrac{n!}{(k-1)!(n-k+1)!}\, p^{k-1}q^{n-k+1}}{\dfrac{n!}{k!(n-k)!}\, p^k q^{n-k}} = \frac{kq}{(n-k+1)p}$$

If

$$\frac{kq}{(n-k+1)p} < 1$$

i.e., if

$$k < (n+1)p \tag{3-29}$$

then

$$p_n(k-1) < p_n(k)$$

Therefore $p_n(k)$ increases until k equals the largest integer that satisfies (3-29). If $(n+1)p$ is an integer, then

$$\frac{p_n(k-1)}{p_n(k)} = 1 \qquad \text{for } k = (n+1)p$$

Hence $p_n(k)$ is maximum for $k_{m_1} = (n+1)p$ and $k_{m_2} = (n+1)p - 1$.

Determination of $P\{k_1 \leq k \leq k_2\}$. In many problems one is interested not in the probability of exactly k successes of an event $\mathcal{A}$, but in the probability that the number of successes of $\mathcal{A}$ will lie between k_1 and k_2, that is, of the event $\{\mathcal{A}$ occurs k times, where k is any number between k_1 and $k_2\}$.

Denoting this event by

$$\{k_1 \leq k \leq k_2\}$$

we observe that

$$\{k_1 \leq k \leq k_2\} = \sum_{k=k_1}^{k_2} \{\mathcal{A} \text{ occurs } k \text{ times}\}$$

where the above is a sum of sets. But the events

$$\{\mathcal{A} \text{ occurs } k \text{ times}\} \qquad k = 0, 1, \ldots, n$$

are mutually exclusive; hence [see (2-10)]

$$P\{k_1 \leq k \leq k_2\} = \sum_{k=k_1}^{k_2} P\{\mathcal{A} \text{ occurs } k \text{ times}\}$$

From the above and (3-23) we finally obtain

$$P\{k_1 \leq k \leq k_2\} = \sum_{k=k_1}^{k_2} p_n(k) = \sum_{k=k_1}^{k_2} \binom{n}{k} p^k q^{n-k} \tag{3-30}$$

Example 3-13. An order of 10,000 parts is received. The probability that a part is defective equals 0.1. What is the probability that the total number of defective parts does not exceed 1,100?

Our experiment $\mathfrak{S}$ is the selection of a single part. The event

$$\mathcal{A} = \{\text{the part is defective}\}$$

has probability 0.1. We want the probability that, in 10,000 trials of the experiment $\mathfrak{S}$, the event $\mathcal{A}$ occurs at most 1,100 times. With

$$P(\mathcal{A}) = 0.1 \qquad n = 10{,}000 \qquad k_1 = 0 \qquad k_2 = 1{,}100$$

we obtain from (3-30)

$$P\{0 \leq k \leq 1{,}100\} = \sum_{k=0}^{1{,}100} \binom{10{,}000}{k} (0.1)^k (0.9)^{10{,}000-k}$$

3-3. Asymptotic theorems

In the preceding section we showed that, if the probability $P(\mathcal{A})$ of an event $\mathcal{A}$ in a given experiment equals p and the experiment is repeated n times, then the probability that $\mathcal{A}$ occurs k times in any order

Table 3-1

$$\operatorname{erf} x = \frac{1}{\sqrt{2\pi}} \int_0^x e^{-y^2/2}\, dy$$

x	erf x	x	erf x
0.05	0.01994	1.55	0.43943
0.10	0.03983	1.60	0.44520
0.15	0.05962	1.65	0.45053
0.20	0.07926	1.70	0.45543
0.25	0.08971	1.75	0.45994
0.30	0.11791	1.80	0.46407
0.35	0.13683	1.85	0.46784
0.40	0.15542	1.90	0.47128
0.45	0.17364	1.95	0.47441
0.50	0.19146	2.00	0.47725
0.55	0.20884	2.05	0.47982
0.60	0.22575	2.10	0.48214
0.65	0.24215	2.15	0.48422
0.70	0.25804	2.20	0.48610
0.75	0.27337	2.25	0.48778
0.80	0.28814	2.30	0.48928
0.85	0.30234	2.35	0.49061
0.90	0.31594	2.40	0.49180
0.95	0.32894	2.45	0.49286
1.00	0.34134	2.50	0.49379
1.05	0.35314	2.55	0.49461
1.10	0.36433	2.60	0.49534
1.15	0.37493	2.65	0.49597
1.20	0.38493	2.70	0.49653
1.25	0.39435	2.75	0.49702
1.30	0.40320	2.80	0.49744
1.35	0.41149	2.85	0.49781
1.40	0.41924	2.90	0.49813
1.45	0.42647	2.95	0.49841
1.50	0.43319	3.00	0.49865

is given by

$$p_n(k) = \binom{n}{k} p^k q^{n-k} \tag{3-23}$$

and the probability that the number of occurrences of $\mathcal{A}$ is between k_1 and k_2 by

$$P\{k_1 \le k \le k_2\} = \sum_{k=k_1}^{k_2} \binom{n}{k} p^k q^{n-k}$$

If n is large, then these expressions are difficult to evaluate. In this section we shall develop simple formulas for their numerical evaluation.

Gaussian and error functions. An important function for this discussion and for the general theory of probability is the *Gaussian* or *normal function* $G(x)$ given by

$$G(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

and its integral

$$\operatorname{erf} x = \frac{1}{\sqrt{2\pi}} \int_0^x e^{-y^2/2}\, dy \tag{3-31}$$

known as *error function.*† The values of erf x are shown in Table 3-1,

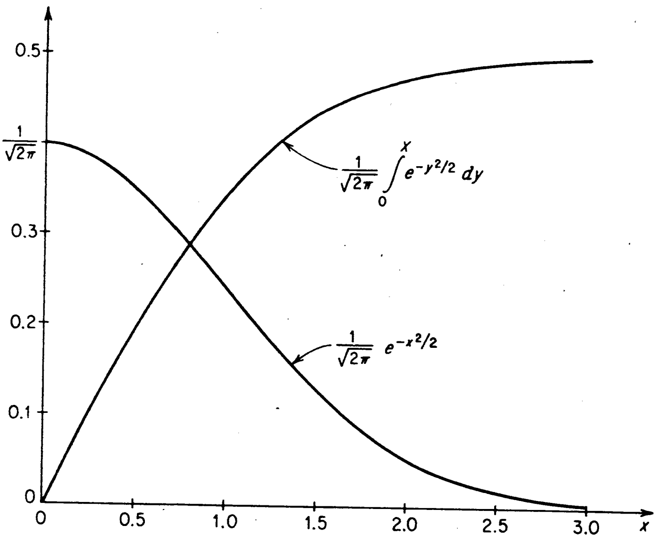

Fig. 3-3

and in Fig. 3-3 the functions $G(x)$ and erf x are plotted. Notice that

$$\operatorname{erf}(-x) = -\operatorname{erf} x$$

and

$$\operatorname{erf} \infty = \frac{1}{\sqrt{2\pi}} \int_0^\infty e^{-x^2/2}\, dx = \tfrac{1}{2} \tag{3-32}$$

as is well known. With a change of variable we easily conclude from

† The error function is defined in several ways; however, all definitions are essentially equivalent.

(3-31) that

$$\frac{1}{a\sqrt{2\pi}}\int_{x_1}^{x_2} e^{-(x-b)^2/2a^2}\,dx = \operatorname{erf}\frac{x_2-b}{a} - \operatorname{erf}\frac{x_1-b}{a} \tag{3-33}$$

where a and b are arbitrary constants.

DeMoivre-Laplace Theorem

We shall assume not only that n is large but also that

$$npq \gg 1 \tag{3-34}$$

It can be shown that, for values of k in the $\sqrt{npq}$ neighborhood of its most likely value np,

$$|k - np| \text{ of the order of } \sqrt{npq} \tag{3-35}$$

the quantity $p_n(k)$ in (3-23) is approximately given by

$$p_n(k) = \binom{n}{k} p^k q^{n-k} \simeq \frac{1}{\sqrt{2\pi npq}}\, e^{-(k-np)^2/2npq} \tag{3-36}$$

This important result, known as the DeMoivre-Laplace theorem, can be stated as a limit: The ratio of the two sides in (3-36) tends to 1 as n tends to infinity. The proof is somewhat lengthy and will be omitted.†

Thus, to evaluate the complicated expression (3-23), it suffices to determine the value of the function

$$\frac{1}{\sqrt{2\pi npq}}\, e^{-(x-np)^2/2npq} = \frac{1}{\sqrt{npq}}\, G\left(\frac{x-np}{\sqrt{npq}}\right) \tag{3-37}$$

for $x = k$. This function is merely the Gaussian curve $G(x)$ centered at $x = np$ with a linear scaling of the x axis (Fig. 3-4). Notice that, for

$$|k - np| \ll \sqrt{npq}$$

we have

$$\binom{n}{k} p^k q^{n-k} \simeq \frac{1}{\sqrt{2\pi npq}} \tag{3-38}$$

because the exponent in (3-36) is small compared to 1. In particular, the probability that the number of successes of the event $\mathcal{A}$ equals the most likely value $k_{\max}$ [see (3-27)] is given by $1/\sqrt{2\pi npq}$.

Example 3-14. A fair coin is tossed 10,000 times. Find the probability p_a that heads will show 5,000 times and the probability p_b that heads will show 5,100 times.

We have

$$n = 10{,}000 \qquad p = q = \tfrac{1}{2} \qquad npq = 2{,}500$$

† W. Feller, "An Introduction to Probability Theory and Its Applications," John Wiley & Sons, Inc., New York, 1957, p. 168.

From (3-38) we obtain

$$p_a \simeq \frac{1}{\sqrt{5{,}000\pi}}$$

and with

$$k = 5{,}100 \qquad np = 5{,}000 \qquad \frac{k - np}{\sqrt{npq}} = 2$$

(3-36) gives

$$p_b \simeq \frac{e^{-2}}{\sqrt{5{,}000\pi}}$$

Approximate evaluation of $P\{k_1 \le k \le k_2\}$. We shall now use (3-36) to evaluate the sum in (3-30). Assuming that k_1 and k_2 satisfy (3-35), we obtain from (3-36)

$$\sum_{k=k_1}^{k_2} \binom{n}{k} p^k q^{n-k} \simeq \frac{1}{\sqrt{2\pi npq}} \sum_{k=k_1}^{k_2} e^{-(k-np)^2/2npq} \tag{3-39}$$

This expression can be simplified. Since

$$npq \gg 1$$

by assumption, the curve (3-37) is approximately constant in a unit interval (see also Fig. 3-4). Hence its area in the interval

$$k \le x \le k + 1$$

is approximately equal to the ordinate

$$\frac{1}{\sqrt{2\pi npq}} e^{-(k-np)^2/2npq}$$

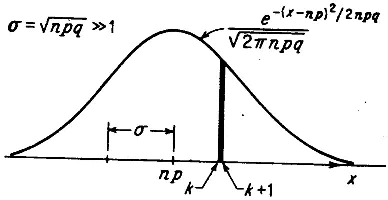

Fig. 3-4

Therefore the right-hand-side sum in (3-39) can be approximated by the integral

$$\frac{1}{\sqrt{2\pi npq}} \int_{k_1}^{k_2} e^{-(x-np)^2/2npq}\, dx$$

Using (3-33) with

$$a = \sqrt{npq} \qquad b = np$$

we obtain from (3-39)

$$\sum_{k=k_1}^{k_2} \binom{n}{k} p^k q^{n-k} \simeq \operatorname{erf} \frac{k_2 - np}{\sqrt{npq}} - \operatorname{erf} \frac{k_1 - np}{\sqrt{npq}} \tag{3-40}$$

Thus the determination of the probability that in n trials the number of occurrences of an event $\mathcal{A}$ is between k_1 and k_2 is reduced to the simple evaluation of the error function.

Example 3-15. A fair coin is tossed 10,000 times. What is the probability that the number of heads will be between 4,900 and 5,100? We have

$$n = 10{,}000 \qquad p = q = \tfrac{1}{2} \qquad k_1 = 4{,}900 \qquad k_2 = 5{,}100$$

Hence

$$\frac{k_2 - np}{\sqrt{npq}} = \frac{100}{50} = 2 \qquad \frac{k_1 - np}{\sqrt{npq}} = -2$$

Therefore [see (3-40)] the unknown probability equals

$$\operatorname{erf} 2 - \operatorname{erf}(-2) = 2 \operatorname{erf} 2 = 0.9545$$

as we see from Table 3-1.

Example 3-16. Over a period of 12 hours 100 calls are made at random. What is the probability that in a 6-hour interval (t_1,t_2) the number of calls is between 40 and 60?

Since the probability that a particular call will occur in the interval (t_1,t_2) is

$$\frac{t_2 - t_1}{T} = \frac{6}{12} = \frac{1}{2}$$

the probability that k calls will occur in this interval is given by [see also (3-23)]

$$\binom{100}{k}\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{100-k}$$

To solve our problem we must sum the above from 40 to 60. Using the approximation (3-40) we obtain

$$\sum_{k=40}^{60}\binom{100}{k}\left(\frac{1}{2}\right)^{100} \simeq \operatorname{erf} \tfrac{10}{5} - \operatorname{erf}\left(-\tfrac{10}{5}\right)$$

because

$$\frac{k_2 - np}{\sqrt{npq}} = \frac{60 - 50}{\sqrt{25}} = \frac{10}{5} \qquad \frac{k_1 - np}{\sqrt{npq}} = -\frac{10}{5}$$

Hence

$$P\{40 \le k \le 60\} \simeq 2 \operatorname{erf} 2 = 0.9545$$

Comment. For the validity of (3-40) it was assumed that the numbers k_1 and k_2 were in the $\sqrt{npq}$ vicinity of np. If this is not so, then the approximation (3-36), on which (3-40) was based, is not valid. However, if k is not in the above vicinity, the term

$$\binom{n}{k} p^k q^{n-k}$$

is small compared with $1/\sqrt{2\pi npq}$. Therefore, if the sum (3-40) contains terms near np, then the approximation is still satisfactory because the terms for which (3-36) does not hold are negligible. We can thus assume $k_1 = 0$ as long as $|k_2 - np|$ is of the order of $\sqrt{npq}$. Since for $k_1 = 0$ we have

$$\frac{k_1 - np}{\sqrt{npq}} = -\sqrt{\frac{np}{q}} \ll -1$$

and

$$\operatorname{erf} x \simeq -\tfrac{1}{2} \qquad \text{for } x \ll -1$$

we conclude that

$$\sum_{k=0}^{k_2} \binom{n}{k} p^k q^{n-k} \simeq \tfrac{1}{2} + \operatorname{erf} \frac{k_2 - np}{\sqrt{npq}} \tag{3-41}$$

Example 3-13 (continued). The final sum in Example 3-13, page 63, can be evaluated with the help of (3-41). Since

$$p = 0.1 \qquad n = 10{,}000 \qquad \text{and} \qquad k_2 = 1{,}100$$

we have

$$np = 1{,}000 \qquad npq = 900 \qquad \frac{k_2 - np}{\sqrt{npq}} = \frac{10}{3}$$

Therefore

$$\sum_{k=0}^{1,100} \binom{10{,}000}{k} (0.1)^k (0.9)^{10,000-k} \simeq \tfrac{1}{2} + \operatorname{erf} \tfrac{10}{3} \simeq 0.99936$$

Thus it is "almost certain" that the number of defective parts will not exceed 1,100.

We remark that the terms from 0 to 900, in the above sum, are negligible. In fact, the probability that the number of defective parts is between 900 and 1,100, given by [see (3-40)]

$$\sum_{k=900}^{1,100} \simeq \operatorname{erf} \tfrac{10}{3} - \operatorname{erf} \left(-\tfrac{10}{3}\right) \simeq 0.99912$$

is very close to 0.99936.

The law of large numbers. This important concept will be extensively discussed in Sec. 8-5 in terms of random variables. We introduce it here briefly as a corollary to the DeMoivre-Laplace theorem. We have shown that if an event $\mathfrak{A}$ occurs k times in n repetitions of an experiment $\mathfrak{S}$, then [see (3-40)]

$$P\{k_1 \le k \le k_2\} \simeq \operatorname{erf} \frac{k_2 - np}{\sqrt{npq}} - \operatorname{erf} \frac{k_1 - np}{\sqrt{npq}} \tag{3-42}$$

From the above we shall conclude that if n is large enough, then it is "almost certain" that the ratio k/n will be "close" to p. More precisely: Given

$$\varepsilon > 0 \qquad \text{and} \qquad \delta > 0$$

if we choose n_0 sufficiently large, then we shall have

$$P\left\{\left|\frac{k}{n} - p\right| \le \varepsilon\right\} \ge 1 - \delta \qquad \text{for every } n > n_0 \tag{3-43}$$

or, equivalently,

$$\lim P\left\{\left|\frac{k}{n} - p\right| \le \varepsilon\right\} = 1 \text{ with } n \to \infty \tag{3-44}$$

for any $\varepsilon > 0$.

Proof. With

$$k_1 = n(p - \varepsilon) \qquad k_2 = n(p + \varepsilon)$$

the inequalities

$$\left|\frac{k}{n} - p\right| \le \varepsilon \qquad \text{and} \qquad k_1 \le k \le k_2$$

are equivalent; hence [see (3-42)]

$$P\left\{\left|\frac{k}{n} - p\right| \le \varepsilon\right\} \simeq \operatorname{erf}\frac{\varepsilon n}{\sqrt{npq}} - \operatorname{erf}\frac{-\varepsilon n}{\sqrt{npq}} = 2\operatorname{erf}\varepsilon\sqrt{\frac{n}{pq}} \tag{3-45}$$

For a given $\varepsilon > 0$, we have

$$\varepsilon\sqrt{\frac{n}{pq}} \underset{n\to\infty}{\longrightarrow} \infty \qquad \text{hence} \qquad 2\operatorname{erf}\varepsilon\sqrt{\frac{n}{pq}} \underset{n\to\infty}{\longrightarrow} 1$$

because $\operatorname{erf}\infty = \frac{1}{2}$, and (3-44) is thus proved.

Example 3-17. We toss a fair coin n times. With

$$\left\{\left|\frac{k}{n} - \frac{1}{2}\right| \le 0.05\right\}$$

the event {the number of heads is between $0.45n$ and $0.55n$}, we want to find n so that the probability of the above event is not less than 0.997:

$$P\left\{\left|\frac{k}{n} - \frac{1}{2}\right| \le 0.05\right\} \ge 0.997$$

From Table 3-1 we see that

$$2\operatorname{erf} x \ge 0.997 \qquad \text{for } x > 3$$

and since

$$P\left\{\left|\frac{k}{n} - p\right| \le \varepsilon\right\} \simeq 2\operatorname{erf}\left(\varepsilon\sqrt{\frac{n}{pq}}\right)$$

we must choose n so that

$$\varepsilon\sqrt{\frac{n}{pq}} > 3 \qquad \text{that is} \qquad n > \frac{9pq}{\varepsilon^2} = 900$$

because

$$\varepsilon = 0.05 \qquad \text{and} \qquad p = q = \tfrac{1}{2}$$

Comment. The law of large numbers could be considered as a "link" between theory and application. Since the event

$$\left\{\left|\frac{k}{n} - p\right| \leq \varepsilon\right\}$$

has probability close to 1 if n is sufficiently large, we conclude that, in a single trial of the experiment

$$\mathfrak{E} \times \mathfrak{E} \times \cdots \times \mathfrak{E} \qquad n \text{ times}$$

we expect "with a high degree of certainty" that this event will occur, i.e., that the number k of successes of $\mathcal{A}$ will be such that

$$p - \varepsilon \leq \frac{k}{n} \leq p + \varepsilon \qquad \text{where } p = P(\mathcal{A})$$

If, therefore, we take ε sufficiently small, we shall have

$$P(\mathcal{A}) \simeq \frac{k}{n}$$

as in (1-1). The meaning of "high degree of certainty" is, however, still undefined. In Sec. 8-5 we shall comment further on the relationship between the law of large numbers and the relative-frequency interpretation of probability.

Poisson Theorem

We have shown that the probability of the event $\{\mathcal{A}$ occurs k times in n trials$\}$ is given by

$$\frac{n(n-1)\cdots(n-k+1)}{1\cdot 2\cdots k}\,p^k q^{n-k} \tag{3-46}$$

where $p = P(\mathcal{A})$ is the probability of the event $\mathcal{A}$ in a single trial. In the following we shall assume that

$$n \gg 1 \qquad p \ll 1 \tag{3-47}$$

If n is so large that

$$np \simeq npq \gg 1$$

then the expression (3-36) can be used for the approximate evaluation of (3-46). If, however, np is of the order of 1, then (3 36) is no longer valid. In this case the following estimate, known as the *Poisson theorem*, can be used:

$$\frac{n!}{k!(n-k)!}\,p^k q^{n-k} \simeq e^{-np}\,\frac{(np)^k}{k!} \tag{3-48}$$

for k of the order of np. This result can be stated as a limit: If

$$n \to \infty \qquad p \to 0 \qquad np \to a$$

then

$$\frac{n!}{k!(n-k)!} p^k q^{n-k} \to e^{-a} \frac{a^k}{k!} \tag{3-49}$$

We shall "justify" (3-48) using qualitative reasoning.† Since k is of the order of np and $p \ll 1$, we conclude that $k \ll n$. Therefore, in the numerator of (3-46), all factors can be approximated by n. Thus

$$n(n-1) \cdots (n-k+1) \simeq n \cdot n \cdots n = n^k$$

and since $p \ll 1$, we have

$$q = 1 - p \simeq e^{-p} \qquad q^{(n-k)} \simeq e^{-(n-k)p} \simeq e^{-np}$$

because $kp \ll 1$ by assumption. Inserting the above approximations into (3-46), we obtain (3-48).

Example 3-18. A system contains 1,000 components. Each component fails independently of the others, and the probability of its failure in one month equals 10^{-3} What is the probability that the system will function (i.e., no component will fail) at the end of a month?

This can be considered as a problem in repeated trials with

$$p = 10^{-3} \qquad k = 0 \qquad n = 10^3$$

From (3-23) we have

$$P\{k = 0\} = \binom{n}{0} p^k q^{n-k} = (1 - 10^{-3})^{1,000}$$

Since the conditions for the validity of (3-48) are satisfied, we obtain the following approximation for the above probability:

$$P\{k = 0\} \simeq e^{-np} = \frac{1}{e} = 0.368$$

Evaluation of $P\{k_1 \le k \le k_2\}$. Since the events

$$\{\mathfrak{A} \text{ occurs } k \text{ times}\} \qquad k = 0, 1, 2, \ldots$$

are mutually exclusive, we obtain from (3-48)

$$P\{k_1 \le k \le k_2\} \simeq e^{-np} \sum_{k=k_1}^{k_2} \frac{(np)^k}{k!} \tag{3-50}$$

where the summation must contain terms of the order of np.

Example 3-19. An order of 3,000 parts is received. The probability that a part is defective equals 10^{-3}. What is the probability that there will be more than five defective parts?

---

† For the proof see Feller, *loc. cit.*

We have $p = 10^{-3}$, $np = 3$, and

$$P\{k > 5\} + P\{k \le 5\} = 1$$

But

$$P\{k \le 5\} = e^{-3} \sum_{k=0}^{5} \frac{3^k}{k!} = 0.916$$

Hence

$$P\{k > 5\} = 1 - 0.916 = 0.084$$

Random points in time. An important application of Poisson's theorem is the approximate evaluation of (3-26) under the assumption that n is large and the interval $t_2 - t_1 = t_a$ is small compared with T. We repeat the problem: We place at random n points in the interval $(0,T)$. With

$$P\{k \text{ in } t_a\}$$

the probability that k of these points will lie in the interval t_a included in $(0,T)$, we have from (3-26)

$$P\{k \text{ in } t_a\} = \binom{n}{k} p^k q^{n-k} \qquad \text{where } p = \frac{t_a}{T} \tag{3-51}$$

If

$$n \gg 1 \qquad \text{and} \qquad \frac{t_a}{T} \ll 1 \tag{3-52}$$

then for k of the order of $n(t_2 - t_1)/T$ we obtain from (3-48)

$$P\{k \text{ in } t_a\} \simeq e^{-nt_a/T} \frac{(nt_a/T)^k}{k!} \tag{3-53}$$

and with

$$\frac{n}{T} = \lambda$$

$$P\{k \text{ in } t_a\} \simeq e^{-\lambda t_a} \frac{(\lambda t_a)^k}{k!} \qquad t_a = t_2 - t_1 \tag{3-54}$$

If

$$n \to \infty \qquad T \to \infty \qquad \frac{n}{T} \to \lambda$$

then (3-53) is no longer an approximation but an equality [see (3-49)].

Comment. Notice that if t_a is so small that $\lambda t_a \ll 1$, then $e^{-\lambda t_a} \simeq 1$; hence

$$P\{\text{one in } t_a\} \simeq e^{-\lambda t_a} \lambda t_a \simeq \lambda t_a$$

or, exactly,

$$\lim_{t_a \to 0} \frac{P\{\text{one in } t_a\}}{t_a} = \lambda$$

Furthermore, the probability of having more than one point in the interval t_a is of a higher order than t_a. It will presently be shown that if two intervals t_a and t_b are nonoverlapping, then the events

$$\{k_a \text{ points in } t_a\} \text{ and } \{k_b \text{ points in } t_b\}$$

are independent for $T \to \infty$. The above properties are often used to characterize random points in time and to derive (3-53) (see page 558).

Nonuniform Density. Suppose now that the probability that a specific point lies in the interval (t_1,t_2) is given, not by $(t_2 - t_1)/T$, but by

$$p = \int_{t_1}^{t_2} \alpha(t)\, dt \tag{3-55}$$

as in Example 3-12. If

$$\int_{t_1}^{t_2} \alpha(t)\, dt \ll 1$$

we conclude that the probability $P\{k \text{ in } (t_1,t_2)\}$ of finding k points in the interval (t_1,t_2) is given by (3-53) where $(t_2 - t_1)/T$ is replaced by p as in (3-55). With

$$n\alpha(t) = \lambda(t) \tag{3-56}$$

we thus obtain

$$P\{k \text{ in } (t_1,t_2)\} = e^{-\int_{t_1}^{t_2}\lambda(t)\,dt} \frac{\left[\int_{t_1}^{t_2} \lambda(t)\, dt\right]^k}{k!} \tag{3-57}$$

3-4. Generalized Bernoulli trials

The problem of repeated trials considered so far can be phrased as follows: Given two mutually exclusive events

$$\mathcal{A}_1 = \mathcal{A} \qquad \mathcal{A}_2 = \bar{\mathcal{A}} \qquad \mathcal{A}_1 + \mathcal{A}_2 = \mathcal{S}$$

the probability that $\mathcal{A}_1$ occurs $k_1 = k$ times and $\mathcal{A}_2$ occurs $k_2 = n - k$ times in n independent trials is given by (3-23). We shall now generalize this result.

We assume that in the experiment $\mathfrak{S}$ the events

$$\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_r$$

are mutually exclusive:

$$\mathcal{A}_i\mathcal{A}_j = 0 \qquad i \neq j = 1, 2, \ldots, r \tag{3-58}$$

and their sum equals the certain event $\mathcal{S}$ (Fig. 2-11):

$$\mathcal{A}_1 + \mathcal{A}_2 + \cdots + \mathcal{A}_r = \mathcal{S} \tag{3-59}$$

With

$$P(\mathcal{A}_1) = p_1, P(\mathcal{A}_2) = p_2, \ldots, P(\mathcal{A}_r) = p_r \tag{3-60}$$

their respective probabilities, we have

$$p_1 + p_2 + \cdots + p_r = 1 \tag{3-61}$$

In the combined experiment (n independent trials)

$$\mathfrak{E} \times \mathfrak{E} \times \cdots \times \mathfrak{E} \qquad n \text{ times}$$

we denote by

$$p_n(k_1, k_2, \ldots, k_r)$$

the probability of the event

$$\{\mathcal{A}_1 \text{ occurs } k_1 \text{ times}, \ldots, \mathcal{A}_r \text{ occurs } k_r \text{ times}\}$$

Reasoning as in (3-23), we can show that

$$p_n(k_1, k_2, \ldots, k_r) = \frac{n!}{k_1!k_2! \cdots k_r!} p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r} \tag{3-62}$$

The proof of the above is similar to the proof of (3-23). It will be given in the next example for $r = 6$.

Example 3-20. A fair die is rolled n times. We shall determine the probability $p_n(k_1,k_2, \ldots, k_6)$ that the ith face shows k_i times, where

$$k_1 + k_2 + k_3 + k_4 + k_5 + k_6 = n$$

The total number of outcomes of our experiment equals 6^n, namely, all possible n element sequences

$$\zeta_1\zeta_2 \ldots \zeta_n \tag{3-63}$$

that one can form with

$$\zeta_i = f_1 \text{ or } f_2 \ldots \text{ or } f_6$$

where $f_1, \ldots, f_6$ are the six faces of the die. An elementary event is a particular sequence (3-63), and its probability equals $1/6^n$ because all elementary events have the same probability. (This follows from the fact that the die is fair and the rollings independent.) Therefore [see (2-16)] the probability of any event $\mathcal{A}$ equals

$$\frac{N_{\mathcal{A}}}{6^n}$$

where $N_{\mathcal{A}}$ is the total number of elements in $\mathcal{A}$. Our event

$$\{i\text{th face shows } k_i \text{ times}\} \qquad i = 1, 2, \ldots, 6$$

consists of all sequences (3-63) containing k_1 times the face $f_1, \ldots, k_6$ times the face f_6. It is well known from combinatorial analysis that the total number of such sequences equals

$$\frac{n!}{k_1!k_2! \cdots k_6!}$$

Therefore the probability of the above event is given by

$$p_n(k_1, k_2, \ldots, k_6) = \frac{n!}{k_1!k_2! \cdots k_6!} \frac{1}{6^n} \tag{3-64}$$

DeMoivre-Laplace Theorem. One can show as in (3-36) that, for $n \to \infty$, the right-hand side of (3-62) is asymptotically given by

$$\frac{n!}{k_1! \cdots k_r!} p_1^{k_1} \cdots p_r^{k_r} \simeq \frac{e^{-\frac{1}{2}\left[\frac{(k_1 - np_1)^2}{np_1} + \cdots + \frac{(k_r - np_r)^2}{np_r}\right]}}{\sqrt{(2\pi n)^{r-1} p_1 \cdots p_r}} \tag{3-65}$$

This formula facilitates the evaluation of $p_n(k_1, \ldots, k_r)$ for large n and for k_i in the $\sqrt{n}$ vicinity of np_i. It can be easily shown that (3-36) is a special case of (3-65) (show it).

Poisson Theorem. If the $r - 1$ events $\mathcal{A}_1, \ldots, \mathcal{A}_{r-1}$ have small probabilities, so that

$$p_1 + p_2 + \cdots + p_{r-1} \ll 1 \tag{3-66}$$

then one can show, as in (3-48), that for large n,

$$\frac{n!}{k_1! \cdots k_r!} p_1^{k_1} \cdots p_r^{k_r} \simeq \frac{e^{-np_1}(np_1)^{k_1}}{k_1!} \cdots \frac{e^{-np_{r-1}}(np_{r-1})^{k_{r-1}}}{k_{r-1}!} \tag{3-67}$$

provided k_i is of the order of np_i. If

$$n \to \infty \qquad \text{and} \qquad np_i \to a_i \qquad i = 1, 2, \ldots, r-1$$

then

$$\frac{n!}{k_1! \cdots k_r!} p_1^{k_1} \cdots p_r^{k_r} \to \frac{e^{-a_1} a_1^{k_1}}{k_1!} \cdots \frac{e^{-a_{r-1}} a_{ar-1}^{k_{r-1}}}{k_{r-1}!} \tag{3-68}$$

The proof is similar to the proof of (3-49).

Random points in time (continued). We place n points at random in the $(0,T)$ interval, and we shall determine the probability

$$P\{k_a \text{ in } t_a, k_b \text{ in } t_b\}$$

that k_a of these points lie in the interval t_a, and k_b in the interval t_b. We assume that the intervals t_a and t_b are nonoverlapping as in Fig. 3-5.

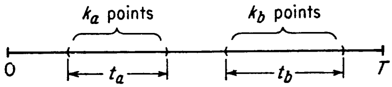

Fig. 3-5

The above can be considered as a problem in repeated trials. We take as our experiment the placing of a single point in the interval $(0,T)$ and denote by

$$\mathcal{A}_1 \qquad \mathcal{A}_2 \qquad \mathcal{A}_3 = \overline{\mathcal{A}_1 + \mathcal{A}_2}$$

the events {the point is in t_a}, {the point is in t_b}, and {the point is outside the intervals t_a and t_b}, respectively. Clearly,

$$P(\mathcal{A}_1) = \frac{t_a}{T} \qquad P(\mathcal{A}_2) = \frac{t_b}{T} \qquad P(\mathcal{A}_3) = 1 - \frac{t_a}{T} - \frac{t_b}{T}$$

The placing of n points can be considered as the n-time repetition of the experiment $\mathfrak{S}$. The event $\{k_a \text{ in } t_a, k_b \text{ in } t_b\}$ is equivalent to the event $\{\mathfrak{A}_1 \text{ occurs } k_a \text{ times}, \mathfrak{A}_2 \text{ occurs } k_b \text{ times}, \mathfrak{A}_3 \text{ occurs } n - (k_a + k_b) \text{ times}\}$. From (3-62) we thus obtain with

$$k_1 = k_a \qquad k_2 = k_b \qquad k_3 = n - (k_a + k_b)$$

$$P\{k_a \text{ in } t_a, k_b \text{ in } t_b\} = \frac{n!}{k_1!k_2!k_3!}\left(\frac{t_a}{T}\right)^{k_1}\left(\frac{t_b}{T}\right)^{k_2}\left(1 - \frac{t_a}{T} - \frac{t_b}{T}\right)^{k_3} \tag{3-69}$$

Notice that the events $\{k_a \text{ in } t_a\}$ and $\{k_b \text{ in } t_b\}$ are not independent because the probability of their product (set product) $\{k_a \text{ in } t_a, k_b \text{ in } t_b\}$, given by (3-69), does not equal the product

$$P\{k_a \text{ in } t_a\}P\{k_b \text{ in } t_b\}$$

[see (3-51)].

Suppose now that

$$n \to \infty \qquad T \to \infty \qquad \frac{n}{T} \to \lambda$$

We have

$$\frac{nt_a}{T} \to \lambda t_a \qquad \frac{nt_b}{T} \to \lambda t_b$$

Hence [see (3-68)]

$$P\{k_a \text{ in } t_a, k_b \text{ in } t_b\} = e^{-\lambda t_a}\frac{(\lambda t_a)^{k_a}}{k_a!}e^{-\lambda t_b}\frac{(\lambda t_b)^{k_b}}{k_b!} \tag{3-70}$$

Since [see (3-54)]

$$P\{k_a \text{ in } t_a\} = e^{-\lambda t_a}\frac{(\lambda t_a)^{k_a}}{k_a!}$$

$$P\{k_b \text{ in } t_b\} = e^{-\lambda t_b}\frac{(\lambda t_b)^{k_b}}{k_b!}$$

we conclude that

$$P\{k_a \text{ in } t_a, k_b \text{ in } t_b\} = P\{k_a \text{ in } t_a\}P\{k_b \text{ in } t_b\}$$

i.e., the events $\{k_a \text{ in } t_a\}$ and $\{k_b \text{ in } t_b\}$ are independent. This can be generalized to any number of nonoverlapping intervals.

Application. Suppose that there are

$$k_c = k_a + k_b$$

points in the interval

$$t_c = t_a + t_b$$

What is the probability that k_a of these will be in the interval t_a, and k_b in the interval t_b?

We want the conditional probability

$$P\{k_a \text{ in } t_a, k_b \text{ in } t_b | k_c \text{ in } t_c\} = \frac{P\{k_a \text{ in } t_a, k_b \text{ in } t_b\}}{P\{k_c \text{ in } t_c\}}$$

From (3-70) and (3-54) we conclude that this probability is given by

$$\frac{[e^{-\lambda t_a}(\lambda t_a)^{k_a}/k_a!][e^{-\lambda t_b}(\lambda t_b)^{k_b}/k_b!]}{e^{-\lambda t_c}(\lambda t_c)^{k_c}/k_c!} = \frac{k_c!}{k_a!k_b!}\left(\frac{t_a}{t_c}\right)^{k_a}\left(\frac{t_b}{t_c}\right)^{k_b} \tag{3-71}$$

The above is the same as (3-26) if we make

$$n = k_c \qquad k = k_a \qquad t_2 - t_1 = t_a \qquad T = t_c$$

Thus: The conditional probability (3-71) equals the probability of finding k_a points in the interval t_a out of a total of k_c points, each uniformly distributed in the interval t_c. In particular, if we know that there is only one point in the interval t_c, then the probability that it will be in the interval $t_a (k_a = 1,\ k_b = 0)$ equals t_a/t_c.

3-5. Bayes' theorem in statistics

As an application of the concepts introduced in this chapter we shall consider the important problem of assigning probabilities to various events of a physical experiment (step 1, page 4). To be concrete, we assume that our experiment is the tossing of a given coin, and we shall attempt to determine the probability

$$P\{h\} = p$$

of the event {heads}. To this end we toss the coin

$$n = 1{,}000$$

times and we observe a sequence

$$\zeta_1\zeta_2 \cdots \zeta_n$$

of $k = 490$ heads and $n - k = 510$ tails. What conclusion can we draw from the above observation about the unknown number p?

We are, at first, tempted to conclude that p must be "close" to 0.49:

$$p \simeq \frac{k}{n} \tag{3-72}$$

However, such a conclusion cannot be drawn within the framework of probability theory as developed in this book.†

† We are not, of course, suggesting that conclusion (3-72) is erroneous; we are merely saying that it is based on inductive reasoning. Such reasoning can be presented in quantitative terms, as some authors claim; however, it falls outside the scope of this book. For related discussion, see Myron Tribus, *Journal of Applied Mechanics*, March, 1961, p. 2; E. T. Jaynes, *Physical Review*, vol. 106, pp. 620–630, and vol. 107, pp. 171–190, 1957; G. Polya, "Mathematics and Plausible Reasoning," vols. 1 and 2, Princeton University Press, Princeton, N.J., 1954.

Since we cannot answer the above real question, we shall consider an artificial but "related" problem that can be solved with the tools of probability theory. We have a pile of r coins numbered from 1 to r. We assume that their probabilities are known and denote by

$$P\{h_i\} = p_i \qquad i = 1, 2, \ldots, r \tag{3-73}$$

the probability of "heads" for the ith coin. We pick at random one coin and toss it n times. We then observe a sequence of $k = 490$ heads and $n - k = 510$ tails. Our problem is to find the probability $p(i)$ that we selected the ith coin.

Using Bayes' theorem we shall show that

$$p(i) = \frac{p_i^k(1-p_i)^{n-k}}{p_1^k(1-p_1)^k + \cdots + p_r^k(1-p_r)^k} = Cp_i^k(1-p_i)^{n-k} \tag{3-74}$$

where C is a constant that does not depend on i. Thus the unknown probability is the value of the function

$$Cx^k(1-x)^{n-k} \tag{3-75}$$

for $x = p_i$. It is easy to see that this function has a maximum for

$$x = \frac{k}{n} \tag{3-76}$$

and for n large the curve drops rapidly away from k/n (Fig. 4-21). Therefore $p(i)$ is maximum for $i = m$ such that p_m is closest to k/n, and the corresponding probability $p(m)$ that we selected the mth coin is almost 1, because all other probabilities have small values.

Proof. To prove (3-74) we must define carefully our experiment $\mathfrak{G}$ (selection of a coin and n tossing of it) and express $p(i)$ as a probability of an event in this experiment.

We denote by $\mathcal{S}^i$ the space of the single tossing of the ith coin, and by $\mathcal{S}_n^i$ the space of the nth tossing of this coin (cartesian product):

$$\mathcal{S}_n^i = \mathcal{S}^i \times \mathcal{S}^i \times \cdots \times \mathcal{S}^i$$

Clearly, $\mathcal{S}^i$ consists of 2 elements and $\mathcal{S}_n^i$ of 2^n elements (see Example 3-6). With $P_i(\ \ )$ the probabilities in this space and

$$\{\zeta_1^i\zeta_2^i \ldots \zeta_n^i\} \tag{3-77}$$

the elementary event $\{k$ heads and $n - k$ tails in a specific order$\}$, we have from (3-11)

$$P_i\{\zeta_1^i\zeta_2^i \ldots \zeta_n^i\} = p_i^k(1-p_i)^{n-k} \tag{3-78}$$

The space $\mathcal{S}$ of the experiment $\mathfrak{G}$ is the sum space (page 55)

$$\mathcal{S} = \mathcal{S}_n^1 + \mathcal{S}_n^2 + \cdots + \mathcal{S}_n^r \tag{3-79}$$

consisting of $r2^n$ elements. In this space, $\mathcal{S}_n{}^i$ is a subset and its probability is given by

$$P(\mathcal{S}_n{}^i) = \frac{1}{r} \tag{3-80}$$

because $\mathcal{S}_n{}^i$ occurs if the ith coin is selected, and this selection is by assumption random. In the space $\mathcal{S}$ the event

$$\mathcal{A} = \{k \text{ heads and } n - k \text{ tails in a specific order}\}$$

is not an elementary event, but it consists of r elements, namely, the r sequences of the form (3-77). We know that

$$P(\mathcal{A}|\mathcal{S}_n{}^i) = p_i{}^k(1 - p_i)^{n-k} \tag{3-81}$$

This is the conditional probability of the event $\mathcal{A}$, assuming that we selected the ith coin. We want the probability $p(i)$ that we selected the ith coin under the assumption that we observed a sequence of k heads and $n - k$ tails; in other words, we want the conditional probability of the event $\mathcal{S}_n{}^i$, assuming $\mathcal{A}$:

$$p(i) = P(\mathcal{S}_n{}^i|\mathcal{A}) \tag{3-82}$$

Since the events

$$\mathcal{S}_n{}^t \qquad t = 1, 2, \ldots, r$$

are mutually exclusive and their sum equals $\mathcal{S}$, we conclude from (2-37) that

$$P(\mathcal{S}_n{}^i|\mathcal{A}) = \frac{P(\mathcal{A}|\mathcal{S}_n{}^i)P(\mathcal{S}_n{}^i)}{\sum_{t=1}^{r} P(\mathcal{A}|\mathcal{S}_n{}^t)P(\mathcal{S}_n{}^t)} \tag{3-83}$$

From the above follows that [see (3-80) to (3-82)]

$$p(i) = \frac{p_i{}^k(1 - p_i)^{n-k}\dfrac{1}{r}}{\sum_{t=1}^{r} p_t{}^k(1 - p_t)^{n-k}\dfrac{1}{r}} \tag{3-84}$$

and (3-74) is thus proved.

The quantities $P(\mathcal{S}_n{}^i)$ and $P(\mathcal{S}_n{}^i|\mathcal{A})$ are known as *a priori* and *a posteriori* probabilities that the ith coin was selected. In our example we assumed $P(\mathcal{S}_n{}^i)$ equal to $1/r$; if this is not the case, one must accordingly modify (3-84). The numbers that one assigns to $P(\mathcal{S}_n{}^i)$ in a real problem are often chosen by inductive reasoning. For small n their values very much affect the value of the a posteriori probabilities $p(i)$. However, for large n, $p(i)$ equals almost 1 for i such that p_i is close to k/n and negligible for any other i [see also (2-39)]. Thus, in the face of overwhelming evidence (large n), we conclude with practical certainty that the selected coin is the one whose probability is closest to the observed ratio k/n. At the end of the next chapter we shall extend these results to an infinite supply of coins.

Problems

3-1. Given two experiments $\mathfrak{E}_1$ and $\mathfrak{E}_2$, we perform the experiment $\mathfrak{E} = \mathfrak{E}_1 + \mathfrak{E}_2$ (page 55). With P_1, P_2, and P their respective probabilities and $\mathcal{A}$ an event of $\mathfrak{E}$, show that $P(\mathcal{A}|\mathcal{S}_1) = P_1(\mathcal{A}\mathcal{S}_1)$, where $\mathcal{S}_1$ is the certain event of $\mathfrak{E}_1$.

3-2. Two fair dice are rolled 10 times. Find the probability p that "seven" will show at least once. *Answer.* $1 - p = (5/6)^{10}$.

3-3. A coin is tossed n times. Show that the probability that the number of heads is even equals $\frac{1}{2}[1 + (q - p)^n]$, where $p = P\{\text{heads}\}$.

3-4. A system has 1,000 components. The probability that a particular component will fail in the time interval (t_1,t_2) is given by

$$\int_{t_1}^{t_2} \alpha(t)\,dt \qquad \text{where} \qquad \alpha(t) = \frac{1}{T} e^{-t/T} U(t)$$

Find the probability that prior to $t = T/4$, no more than 100 components will fail. Evaluate approximately the resulting sum.

3-5. A fair coin is tossed $n = 900$ times. Find the probability p that the number of heads will be between 420 and 465. *Answer.* $p = \operatorname{erf} 2 + \operatorname{erf} 1 \simeq 0.819$.

3-6. A fair coin is tossed n times. How large should n be so that the probability that heads is between $0.49n$ and $0.52n$ is at least 0.9? *Answer.* $\operatorname{erf}(0.04\sqrt{n}) + \operatorname{erf}(0.02\sqrt{n}) \geq 0.9$; hence $n > 4{,}556$.

3-7. The probability that a driver will have an accident in 1 month equals 0.02. Find the probability p that in 100 months he will have 3 accidents. *Answer.* $p \simeq 4e^{-2}/3$.

3-8. We place at random n particles in $m > n$ boxes. Find the probability p that the particles will be found in n preselected boxes, one in each box (see Example 1-2).

Outline. (a) *Maxwell-Bolzmann.* There are m ways of placing one particle in m boxes; hence the total number of alternatives equals $N = m^n$. The favorable alternatives are all permutations of the particles in the preselected boxes; hence $N_{\mathcal{A}} = n!$ and $p = n!/m^n$. (b) *Bose-Einstein.* We place the m boxes and the n particles on a straight line and permute the $n + m - 1$ objects, consisting of the n particles and the $m - 1$ walls separating the boxes. The resulting $(n + m - 1)!$ permutations contain all alternatives; however, each is counted more than once. The $(m - 1)!$ permutations of the walls and the $n!$ permutations of the particles result in the same number of particles in each box; hence $N = (n + m - 1)!/(m - 1)!n!$. But $N_{\mathcal{A}} = 1$; therefore $p = 1/N$. (c) *Fermi-Dirac.* The first particle can be placed in any one of m boxes; the second in any one of $m - 1$ boxes; the last in any one of the remaining $m - n + 1$ boxes. Therefore there are $m(m - 1) \cdots (m - n + 1)$ ways of placing n particles in m boxes having no more than one in each box. Since the n particles are not distinguishable, all their permutations must be considered as one alternative. Hence $N = m(m - 1) \cdots (m - n + 1)/n!$. But $N_{\mathcal{A}} = 1$; therefore $p = 1/N$.

3-9. (a) Show that

$$P(\mathcal{A}) = P(\mathcal{A}|\mathcal{M})P(\mathcal{M}) + P(\mathcal{A}|\bar{\mathcal{M}})P(\bar{\mathcal{M}})$$

(b) Two players roll dice alternately. Whoever rolls "seven" wins. Find the probability p that the first player will win. *Answer.* The probability that he wins at the first rolling (event $\mathfrak{M}$) equals $\frac{6}{36}$; if he does not win, he becomes the second player and the probability for winning equals $1 - p$. Hence, $\frac{1}{6} + 5(1 - p)/6 = p$.

(c) In a game played alternately by two players, the probability of winning at a single try equals p_1. Show that the probability that the first player will win is given by $p = 1/(2 - p_1)$.

(d) Repeat (c) for a game with three players.

4

The Concept of a Random Variable

We dealt so far with outcomes of experiments and their collections, which we called events. These outcomes were various objects that could be identified in the performance of the experiment, e.g., "heads" or "black" or "ace of spades." In some experiments one can consider as outcomes real numbers, e.g., "time of a call" or "six on a die"; however, the numerical character of these outcomes is merely a means of identifying them. In this chapter we shall assign a number $\mathbf{x}(\zeta)$ to every outcome ζ of our experiment. We shall thus define a function $\mathbf{x}(\zeta)$ whose "independent variable" ζ will not be a number, but an element of the set $\mathcal{S}$; this function will be called random variable.

4-1. Random variables; distributions; densities

We are given an experiment $\mathfrak{E}$ whose outcomes ζ are various objects belonging to the certain event $\mathcal{S}$. To every ζ we now assign according to some rule a number

$$\mathbf{x}(\zeta)$$

This number could be the gain or loss in a game of chance, the voltage of a random source, the cost of a manufactured part, or any other numerical quantity that is of interest in the performance of the experiment. We thus have established a relationship between the elements ζ of the set $\mathcal{S}$ and certain numbers. Such a relationship is called random variable (abbreviation: r.v.).

Example 4-1. Our experiment is the rolling of a die. With $f_1, f_2, \ldots, f_6$ its six faces, we take

$$\mathbf{x}(f_1) = 10, \qquad \mathbf{x}(f_2) = 20, \ldots, \qquad \mathbf{x}(f_6) = 60$$

The value

$$\mathbf{x}(f_i) = 10i$$

of the r.v. so constructed could be our gain if f_i shows in a game of dice.

Example 4-2. A telephone call occurs at random in the interval $(0,T)$. The space $\mathcal{S}$ consists of all numbers t in this interval. We define the r.v. $\mathbf{x}(t)$ as follows: With (t_1,t_2) an interval included in $(0,T)$, we take

$$\mathbf{x}(t) = \begin{cases} 1 & \text{if } t_1 \leq t \leq t_2 \\ 0 & \text{otherwise} \end{cases}$$

Thus our r.v. takes only the values 0 and 1.

To clarify further the basic concept of a random variable, we shall review briefly the notion of a function.

The Meaning of a Function

We know from calculus that a function

$$x = x(t)$$

is a rule of correspondence between values of t and x. The "independent variable" t takes certain numerical values, forming a set $\mathcal{S}_t$ on the t axis; to every such t we assign, according to some rule, a number $x(t)$ to the "dependent variable" x. This rule could be expressed by a curve, by a table, by a formula like

$$x = t^2 \tag{4-1}$$

or by some other means. The resulting values of x form a set $\mathcal{S}_x$ on the x axis. The set of values $\mathcal{S}_t$ that t takes is called the *domain* of the function, and the set $\mathcal{S}_x$ of the values of x, its *range*. If the function is specified by a table, then its domain and range are clearly indicated by the two columns in the table. However, if the function is given by a formula, then one must clearly specify its domain. Thus, if x is given by (4-1) and its domain $\mathcal{S}_t$ is the entire t axis, then its range $\mathcal{S}_x$ is the positive x axis. However, it is possible that x is related to t by (4-1), but t takes only integral values $t = 1, 2, \ldots, n, \ldots$. The resulting range of x is then the set $1, 4, \ldots, n^2, \ldots$ of squares of integers.

We shall, finally, make a comment about notation. A function is usually denoted by $x(t)$; this notation is ambiguous. The symbol $x(t)$

could mean the particular number $x(t)$ corresponding to the value t of the independent variable; it could also mean the function $x(t)$, namely, the rule of correspondence between any value of t in $\mathcal{S}_t$ and the resulting value of x. To distinguish between the two interpretations, we shall in general denote the latter merely by x, leaving its dependence on t understood [the notation $x(\cdot)$ is also used to represent the function].

Generalization. The foregoing definition of an ordinary function can be phrased as follows: We are given two sets of numbers $\mathcal{S}_t$ and $\mathcal{S}_x$. To every

$$t \in \mathcal{S}_t$$

we assign a number $x(t)$ belonging to the set $\mathcal{S}_x$:

$$x(t) \in \mathcal{S}_x$$

From this interpretation, the generalization of the concept of a function to arbitrary sets follows readily.

We are given two sets of objects $\mathcal{S}_\zeta$ and $\mathcal{S}_\eta$ consisting of the elements ζ and η, respectively:

$$\zeta \in \mathcal{S}_\zeta \qquad \eta \in \mathcal{S}_\eta$$

We say that η is a function of ζ:

$$\eta = \eta(\zeta)$$

if to every element ζ of the set $\mathcal{S}_\zeta$ we make correspond an element η of the set $\mathcal{S}_\eta$. We assume that $\mathcal{S}_\eta$ is so chosen that all its elements are used in this pairing. The set $\mathcal{S}_\zeta$ is the domain of the function, and the set $\mathcal{S}_\eta$ its range. If, for example, $\mathcal{S}_\zeta$ is the set of children in a community and $\mathcal{S}_\eta$ the set of their fathers, then the pairing of a child with his father is a function. Clearly, to a given ζ there corresponds a single $\eta = \eta(\zeta)$. However, more than one ζ might be paired with the same η (a child has only one father, but one father might have more than one child). Therefore the number of elements of $\mathcal{S}_\eta$ does not exceed the number of elements of $\mathcal{S}_\zeta$. In Example 4-2 the domain of the function consists of all numbers in the $(0,T)$ interval, but its range has only two elements, namely, the numbers 0 and 1.

The Random Variable

We are given an experiment $\mathfrak{S}$ specified by the space $\mathcal{S}$, the field $\mathfrak{F}$ of subsets of $\mathcal{S}$ called events, and the probability P assigned to these events. To every outcome ζ of this experiment we assign a number $\mathbf{x}(\zeta)$. We have thus defined a function $\mathbf{x}$ whose domain is the set $\mathcal{S}$ of all outcomes, and range a set of numbers. This function is called random variable provided it satisfies certain general conditions, to be soon given.

All random variables will be written in boldface letters $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$, ... The symbol $\mathbf{x}(\zeta)$ will indicate the number assigned to the specific outcome ζ, whereas $\mathbf{x}$ will indicate the rule of correspondence between any element of $\mathcal{S}$ and the number assigned to it. In Example 4-1, $\mathbf{x}$ is the entire table pairing the faces of the die with the numbers 10, . . . , 60; the domain of this r.v. is the set of the six faces, and its range the set of the above six numbers. The expression $\mathbf{x}(f_2)$ is the number 20.

If we talk about several r.v., we shall always assume that they are defined on the same experiment.

Notations. In dealing with r.v. one is often interested in questions of the following form: Given a real number x, what is the probability that the r.v. $\mathbf{x}$ is less than x; or given two numbers x_1 and x_2, what is the probability that $\mathbf{x}$ takes values between x_1 and x_2? If, for example, our r.v. is the length of a manufactured product, we might want the probability that this length will not exceed certain bounds. As we know, probabilities are assigned only to events, i.e., to various subsets of the certain event $\mathcal{S}$. Therefore, in order to answer the above questions, we should be able to express the conditions imposed on $\mathbf{x}$ as events. As we mentioned on page 19, if x is a given number, the notation $\{\mathbf{x} \leq x\}$ represents a set consisting of all outcomes ζ such that

$$\mathbf{x}(\zeta) \leq x$$

We elaborate on the meaning of $\{\mathbf{x} \leq x\}$. Suppose that the r.v. $\mathbf{x}$ is specified by a table. At the left column of this table, we list all outcomes ζ_i and at the right the corresponding values (numbers) $\mathbf{x}(\zeta_i)$ of $\mathbf{x}$. Given an arbitrary but fixed number x (this number could be designated by any letter), we find all right column entries $\mathbf{x}(\zeta_i)$ that are less than or equal to x. The corresponding ζ's at the left column form a set. This set is a *collection of experimental outcomes* (not of numbers) and is denoted by $\{\mathbf{x} \leq x\}$. This notation specifies the set $\{\mathbf{x} \leq x\}$ not explicitly by its elements but by their property as in (2-2). The reader should have a very clear understanding of its meaning.

Similarly, with x_1 and x_2 two given numbers, $\{x_1 \leq \mathbf{x} \leq x_2\}$ is the set consisting of all outcomes ζ such that

$$x_1 \leq \mathbf{x}(\zeta) \leq x_2$$

Finally, $\{\mathbf{x}(\zeta) = x\}$ is the set of all ζ's such that

$$\mathbf{x}(\zeta) = x$$

More generally, if $\mathcal{I}$ is any set of numbers, then

$$\{\mathbf{x} \in \mathcal{I}\}$$

is the set of all ζ's such that $\mathbf{x}(\zeta)$ is a number belonging to the set $\mathcal{I}$.

We shall illustrate the above notations with the r.v. introduced in Example 4-1. The set $\{\mathbf{x} \leq 35\}$ consists of the elements f_1, f_2, f_3:

$$\{\mathbf{x} \leq 35\} = \{f_1, f_2, f_3\}$$

because $\mathbf{x}(\zeta) \leq 35$ only for $\zeta = f_1$ or f_2 or f_3. Similarly, $\{\mathbf{x} \leq 5\}$ is the empty set 0:

$$\{\mathbf{x} \leq 5\} = 0$$

because for no outcome we have $\mathbf{x}(\zeta) \leq 5$. The set $\{20 \leq \mathbf{x} \leq 35\}$ consists of the elements f_2 and f_3:

$$\{20 \leq \mathbf{x} \leq 35\} = \{f_2, f_3\}$$

because $\mathbf{x}(\zeta)$ is a number between 20 and 35 only for $\zeta = f_2$ or f_3. Finally,

$$\{\mathbf{x} = 40\} = \{f_4\} \qquad \{\mathbf{x} = 35\} = 0$$

as it is easy to see.

We shall now give the conditions that the function $\mathbf{x}$ must satisfy in order to be a r.v. Clearly, $\{\mathbf{x} \leq x\}$ is a subset of $\mathcal{S}$; we want it to be an event, i.e., to belong to the probability field $\mathfrak{F}$ (see page 28) for every x. Only then can we talk about the probability that the r.v. $\mathbf{x}$ is less than the number x.

From the above condition follows that the set $\{x_1 < \mathbf{x} \leq x_2\}$ is also an event. Indeed,

$$\{x_1 < \mathbf{x} \leq x_2\} = \{\mathbf{x} \leq x_2\} - \{\mathbf{x} \leq x_1\} \tag{4-2}$$

as it is easy to see. And since $\{\mathbf{x} \leq x_1\}$ and $\{\mathbf{x} \leq x_2\}$ are events by assumption, their difference is also an event because events form a field.

One can show similarly that if $\mathcal{I}$ is any set of real numbers that can be written as a countable sum or product of intervals,† then the set

$$\{\mathbf{x} \in \mathcal{I}\}$$

is an event because it can be written as a countable sum or product of events of the form $\{\mathbf{x} \leq x\}$ belonging to a Borel field.

From the above we conclude that $\{\mathbf{x} = x\}$ is an event for any x. In particular, the sets

$$\{\mathbf{x} = +\infty\} \qquad \text{and} \qquad \{\mathbf{x} = -\infty\}$$

are also events. These events are not in general empty; i.e., we allow the r.v. $\mathbf{x}$ to equal $+\infty$ or $-\infty$ for some outcomes ζ. However, we shall demand that the set of such outcomes have zero probability. With this final condition we have completed the specification of a r.v.

† As we have mentioned, any set of numbers that is of interest in applications is of this kind.